MASS

Modular **A**coustic **S**uppression **S**ystem

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June 4th, 2026

Statement of Disclaimer

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Contents of this Report

This report fulfills the Senior Project requirement for the Mechanical Engineering Department. This document contains:

1. Scope of Work (SOW)

2. Preliminary Design Review (PDR) Report

3. Critical Design Review (CDR) Report

4. Final Design Review (FDR) Report

5. Project Expo Poster

Noise Suppression System for Artemis Human Lander

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23 October 2025

Noise Suppression System for NASA Human Landers

Project Overview

As we progress further in space exploration, astronauts are going to need to stay in space for longer durations of time. Currently, the environmental control and life support systems in the human lander are too loud for astronauts to healthily live and communicate aboard the landers. NASA needs a way to suppress noise on the Artemis human lander capsule because the onboard environmental control and life support system makes significant ambient noise that affects the physical and mental health of the astronauts onboard. The project is sponsored by Professor John Chen and will be carried out by a team of mechanical engineering students at California Polytechnic State University, including Emmi Cayer, Gabriel Choi, Johnathan Corvera, Lucas Kaemmerer, Julia Megargle, and Michael Milner.

Key Specifications

A list of specifications and the target values based off of the customer needs, which can be seen in further detail in Table #, can be seen in the table below:

Specification	Target Value
Volume	< 1 m^3
Repairability	With common tools in < 60 min
Weight	< 1000 lb
Power	< 9.5 kW
Manufacturability	Made onsite in < 1 year
Adjustability	Fit into different sized spaces ranging from 60 m^3 to 1000 $m^3$and adjust to decibel levels between 25 – 70 dBa
Long Term Cost	< $1,000
Short Term Cost	< $10,000
Durability	Survive a minimum of 5 missions
Noise Suppression	< 70 dBa

Scope and Deliverables

The following will be delivered to Professor John Chen and NASA:

- A final report and technical paper describing the proposed solution and containing documentation of the design process.
- Analysis of the system's theoretical weight, volume, noise suppression/decibel level and power consumption
- A CAD model and complete drawing illustrating the final design and scaled prototype.
- A scaled prototype of the final solution demonstrating the concept of the noise suppression system.

What's Next

The initial research phase has been completed, and the timeline has been verified with John Chen and the given NASA HuLc competition guidelines. Research will continue as ideation begins, and an initial concept plan will be produced for the first prototype. This will lead into the preliminary design review where Rick Lasko, our senior project coach, will verify our design and changes will be discussed and made for the next reviews.

Figure 1. Four-Panel Chart Summarizing Project Status.

Overview

The proposed project focuses on developing an innovative solution to manage and reduce the noise generated by the environmental control and life support systems (ECLSS) in long-duration space habitats, with the intent of mitigating excess noise that can have a significant impact on crew health and communication.

Directing this task is our problem statement, which outlines our team's objective in the NASA HuLC: NASA needs a way to suppress noise on the Artemis human lander capsule because the onboard environmental control and life support system makes significant ambient noise that affects the physical and mental health of the astronauts onboard.

In addition to our problem statement, competition regulations stipulate the design must fit inside the human lander

dimensions and cannot change the ECLSS functionality; illustrated in the boundary sketch of Figure 2.

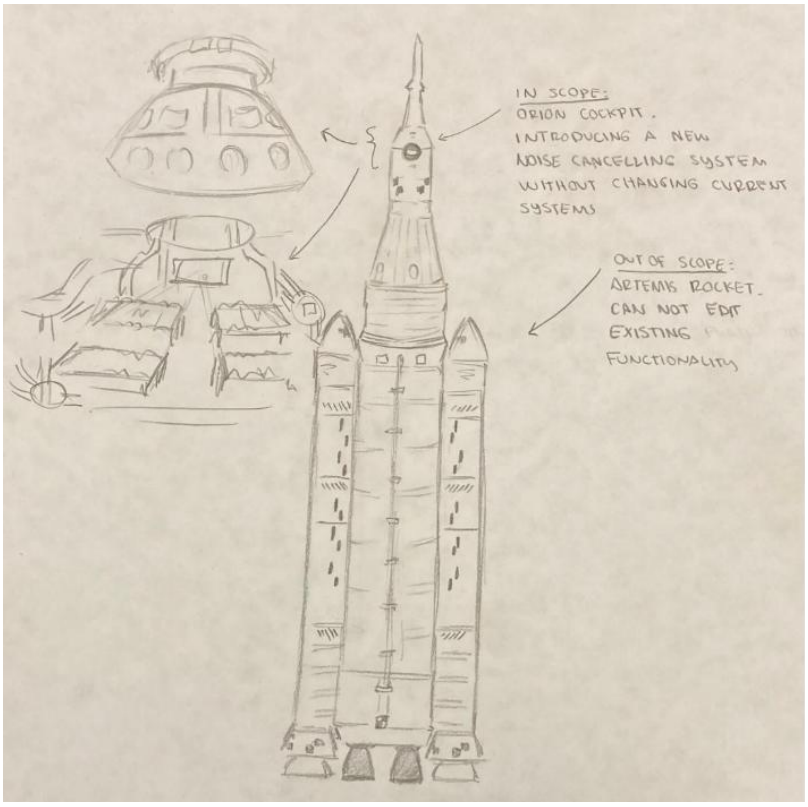


Figure 2. Boundary Sketch Depicting Scope and Boundaries of the Project.

This competition-based project sponsored by Professor John Chen from the Cal Poly Mechanical Engineering Department will follow the guidelines of the NASA Human Lander Competition. As our project sponsor, John Chen will be expected to provide insight into the design and specify areas of improvement. The key stakeholders are NASA and the Artemis astronauts that will be using the system.

The team is made up of six Cal Poly undergraduate mechanical engineering students – Emmi Cayer, Gabriel Choi, Jonathan Corvera, Lucas Kaemmerer, Julia Megargle, and Michael Milner. Lucas Kaemmerer will act as the team's treasurer and manage the teams budget and any material orders, Gabriel Choi will act as the team's secretary and maintain file organization and meeting notes, Emmi Cayer and Jonathan Corvera will lead in planning by taking care of all meeting scheduling and maintaining the Ghantt chart, and Julia Megargle and Michael Milner will maintain communication with the sponsor and any additional experts or faculty. All team members will actively participate in all aspects of the design project including research, ideation, design, prototyping and testing.

This report is structured as follows: The first section presents existing solutions and current market technologies. The next section provides details of customer needs derived from stakeholder interviews and competition requirements. The section following customer needs outlines engineering specifications with measurable targets, and the last section provides an annotated bibliography of 20+ sources including journal articles, patents, and technical reports. The appendix includes the QFD House of Quality analysis in Appendix A and the current Gantt chart in Appendix B.

Researched Topics and Existing Solutions

This section outlines previous and current solutions that have been thought of and developed to address the issue

of unwanted ambient noise. Much can be learned from these solutions, and there are elements that can be built upon and improved. Table 1 shows these solution types, providing a description and examples of each solution type. These examples can be found in further detail in the annotated bibliography.

Table 1. Existing Solutions

This table details the different types of solutions currently used to address similar issues concerning sound levels and provides examples of their usage.

Solution Type	Description	Examples
Removeable headpiece with noise cancellation	This type of design uses removeable headpieces or headphones to decrease ambient noise using a soundwave that is opposite the ambient noise	Apple Noise Cancelling Headphones [1,2]
Soundproof Panels – Acoustic Absorption	This type of design works by absorbing sound waves to reduce the ambient noise within a room. It is a passive system that works by trapping sound energy in the materials which are then converted into a small amount of heat energy	Soundproof Cow [3]
Vibration Damping Panels	This type of design works by layering damping material, usually viscoelastic elastomers, to dampen large ranges of frequency	DYAD [4]
Origami Geometric Panels	This type of design is based on being able to identify sound waves, and the material will respond by changing its length, radius, or material properties. Which will absorb sound and pressure.	Acoustic Materials [5]
Quieter designs of life support system components	The design has multiple elongated propeller elements with tips of each propeller to form a closed structure to add rigidity and reduce the acoustic signature.	Toroidal Propeller [6]
Eliminate undesired acoustic noise via frequency signals	This will be done by producing and implementing an acoustic interference signal in the desired area. Based on analyzing the unwanted acoustic noise.	Generating Acoustic Quiet Zone [7]
Fiberglass or Mineral Wool Structure	This design works by placing optimized structures in order to absorb sound as it travels through walls.	Mineral Wool Insulation [8]
ECLSS Fan noise reduction	This is done by optimizing the blade geometry by analyzing the vortex shredding pattern. Optimize swept blade and tip of blade, which will reduce noise of the fan.	Hybrid-Electric Hybrid [9]
Quiet Fan Incorporating Active Noise Cancellation	An active noise cancelation in a confined space by analyzing the noise made by the fans and reproducing noise from a speaker to reduce the overall noise.	Quiet Fan [10]

Customer Needs

This section outlines the needs of the customer and user, NASA and the Artemis astronauts, respectively. The needs were tabulated and used to create relevant engineering specifications based on qualitative and quantitative observations. These needs, specifications, and justifications can be found in Table 2. The specifications and their further descriptions can be found in Table 3. Derivation of these needs can be found in the QFD House of Quality

in Appendix A.

Table 2: Customer Needs

This table lists the identified customer needs, providing justification and assigning relevant, measurable engineering specifications that will be used to evaluate the adequacy of the prototype in addressing this need.

Customer Needs	Relevant Engineering Specifications (Spec #)	Justification
Survive launch loads	Spec 7: Long term cost Spec 10: Durability	The solution needs to be functional during the long-term mission, which means it must survive the launch. Components must withstand up to 8g of quasi-static and dynamic acceleration; mounts and joints must tolerate torques with 1.5× safety margin, and resonance should be avoided.
Operational life of 30 days for lunar, 1200 days for Mars missions	Spec 2: Repairability Spec 4: Power Consumption Spec 10: Durability Spec 11: Suppresses Noise	The system needs to last until the duration of the mission, or the health and comfort of the astronauts could be jeopardized. If there is a failure, it must be easily repairable.
No added risk to crew	Spec 2: Repairability Spec 6: Adjustability Spec 9: Environmental Sustainability	The crew's safety is a priority. The design cannot interfere with life-support systems.
Simple to implement	Spec 2: Repairability Spec 5: Manufacturability	Simplicity in implementation is reliant on both reliability and manufacturability. An overly complicated system that is difficult to implement is more than likely to be difficult to repair. More often than not, ease of manufacturability also is strongly correlated to its ease of implementation. We need this so that NASA doesn't spend their little time implementing a complicated system, as well as, in case the system breaks; we don't want the astronauts to struggle to repair.
Minimal barriers to NASA adoption	Spec 1: Volume Spec 3: Weight Spec 4: Power Consumption Spec 6: Adjustability Spec 7: Longer Term Cost Spec 8: Initial Cost Spec 11: Suppresses Noise	The system cannot take up unreasonable mission resources. This is for long-term missions, so it must be low in volume, weight, and power consumption so space can be allocated to the astronauts, and power is not taken away from life sustaining equipment.
Fully manufactured and usable within 5-8 years	Spec 5: Manufacturability	This design must be part of the NASA Artemis missions, which means it needs to be completed within the mission timeline. This means it needs to be feasibly manufacturable within the provided time zone.
Reduce, isolate, or manage noise from ECLSS	Spec 10: Durability Spec 11: Suppresses Noise	Per the objective of the Human Lander challenge, our project is to serve the purpose of managing and suppressing noise.

Customer Needs	Relevant Engineering Specifications (Spec #)	Justification
Longevity	Spec 2: Repairability Spec 7: Long Term Cost Spec 10: Durability	Durability is an important consideration for this as, also from the competition objective, our solution is to last 31 days for a moon landing mission, and 1200 days on mars. Durability is what enables this constraint. Extrapolating this constraint further repairability is also important as in the case that our system breaks the astronauts need to be able to fix the problem. Repairability leads to our long-term cost specification, as if our design isn't easily repairable, it would cost more to fix.
Astronaut comfort	Spec 1: Volume Spec 6: Adjustability Spec 11: Suppresses Noise	Astronauts will be living for extended periods of time, and ambient noise from the ECLSS affects their health and comfort, so being able to suppress the noise that is unhealthy is key [14]. This includes being able to adjust for different frequency ranges and situations. Additionally, space is limited, so the solution must be compact, so it does not encroach on living and working quarters.
Environmental Sustainability	Spec 2: Repairability Spec 4: Power Consumption Spec 9: Environmental Sustainability Spec 10: Durability	The design should ideally have minimal environmental impact for long-term sustainability efforts and safety reasons. The more repairable/durable the solution is the less materials waste and manufacturing related emissions are created. This is also the reasoning for the power consumption specification.

Engineering Specifications

This section further details the engineering specifications that were developed based on the needs of the customer. These specifications were composed into measurable standards to provide quantitative target values and tolerances. Methods of testing were evaluated and chosen based on the specification and how compliance could be determined.

It should be noted that these target values and tolerances are subject to change with developments in the NASA human lander program. Derivation from these specifications can be found in QFD House of Quality Appendix A.

Table 3: Engineering Specifications

This table lists the engineering specifications that address each customer's needs, assigning specific target values and tolerances to each specification to be used in evaluating the prototype.

Spec #	Specification	Description	Target	Tolerance	Risk	Compliance *
1	Volume	The volume of the solution must be contained within the lander vehicle while still allowing for astronauts to have space	1 cubic m	± 0.25 cubic m	Medium	A,I

Spec #	Specification	Description	Target	Tolerance	Risk	Compliance *
2	Repairability	Ability to be repaired by astronauts with basic tools aboard Artemis	30 min	± 5 min	Low	A, S
3	Weight	Needs to be light enough to not interfere with launch	1000 lb	±100 lb	Medium	A,I
4	Power Consumption	Take minimal amount of power to operate, and not strain the current electrical system	0.95 kW	±.1 kW	Medium	A,T
5	Manufacturab -ility	Individual construction of each panel triangle	Manufacture in small scale	small scale, so allowed to take more time	Low	A
6	Adjustability/ adaptability	Can adapt to multiple size and specification environments	High modularity	N/A	High	I,A
7	Long Term Cost	Cost to maintain and support our system	$100,000.00	±$10,000.0 0	Medium	A
8	Initial Cost	Initial cost to manufacture and implement into Artemis	$10,000	±$1000.00	Low	A
9	Environmenta l Sustainability	Design has minimal impact on the environment during manufacturing	N/A	N/A	Low	I
10	Durability	Able to withstand multiple missions	At least 5 missions	N/A	High	I
11	Suppresses Noise	Decreases decibel noise in the Artemis environment	<70 dBa	±10 dBa	High	A,I,T

*Compliance Methods: (A) Analysis, (I) Inspection, (S) Similar to Existing Designs, (T) Test

Annotated Bibliography

[1] Apple Support Team, "Active noise cancellation and transparency modes for airpods," Apple Support, https://support.apple.com/en-us/108918 (accessed Oct. 21, 2025).
This document describes Apple's active noise cancellation technology that uses external microphones to detect ambient sound and generates anti-noise to cancel unwanted frequencies. It is relevant as an example of commercial active noise cancellation that could be used to cancel unwanted frequencies.
[2] Method and system for acoustic passthrough, US20230421945A1, Apple Inc., June 22, 2023.
This patent describes a system for selectively allowing certain sounds to pass through while canceling others based on acoustic characteristics and user preferences. It is relevant because astronauts need to suppress ECLSS noise while maintaining communication capability and hearing critical alarms..
[3] "Soundproofing & Acoustic Solutions Online," Soundproof Cow, https://www.soundproofcow.com (accessed Oct. 21, 2025).
This commercial website catalogs passive acoustic materials including mass-loaded vinyl, acoustic foam panels, and sound barriers for industrial and commercial applications. It is relevant for understanding available off-the-shelf soundproofing materials and their performance characteristics, particularly in terms of passive elements.
[4] "Dyad," SOUNDCOAT, https://soundcoat.com/acoustic-products/damping/dyad (accessed Oct. 21, 2025).
This commercial website describes DYAD, a constrained-layer viscoelastic damping material that reduces vibration transmission across broad frequency ranges (50-5000 Hz). It is directly relevant as a passive solution for damping structural-borne noise from ECLSS components, with the ability to achieve 10-20 dB vibration reduction.
[5] Acoustic metamaterials, US10573291B2, Research Foundation of the State University of New York, Feb. 25, 2020.: https://patents.google.com/patent/US10573291B2/en
This patent describes engineered metamaterial structures with modular cavities and tuned masses designed to absorb specific sound frequencies through resonance. It is relevant as an innovative lightweight passive solution that could meet the volume and weight constraints while targeting ECLSS frequency ranges as can be made in a compact manner. The metamaterial can be optimized for specific frequencies and offers a good performance to weight ratio when compared with current foam absorbers.
[6] Toroidal propeller, US20190135410A1, Massachusetts Institute of Technology, May 9, 2019. https://patents.google.com/patent/US20190135410A1/en
This patent presents a toroidal fan blade geometry that reduces aerodynamic noise generation through altered vortex formation and pressure distribution. It is relevant because ECLSS ventilation fans are significant noise sources in spacecraft habitats. This fan design can reduce noise levels by 10-15 dBa without reducing airflow and efficiency.
[7] Acoustic noise control system, US11887575B2, Independent Inventor, Jan. 30, 2024.
This patent covers noise control for aeronautical applications using passive and active hybrid systems. It is relevant for understanding combined approaches.
[8] "Mineral Wool Insulation," Specialty Products Inc. https://www.spi-co.com/162777/category/mechanical-insulation/mineral-wool-insulation
This commercial site describes mineral wool acoustic and thermal insulation used in mechanical systems. It is relevant as a proven aerospace-grade insulation with both thermal and acoustic properties. It suppresses noise and is also non-flammable, which ties into the safety elements of the design.
[9] Active noise control systems, US8855392B2, Silentium Ltd., Oct. 7, 2014.
This patent covers multi-channel active noise control for enclosed spaces. It is relevant for understanding commercial active noise cancellation implementation.
[10] Quiet fan blade design, US11198519B1, B. A. Seeley, Dec. 14, 2020.
This patent describes blade geometries that reduce aerodynamic noise generation. It directly applies to ECLSS fan noise sources. Modifying the blade design reduces noise by altering the vortex shedding patterns.
[11] SAE International, "Acoustical Considerations for Aircraft Environmental Control System Design," SAE Standard AIR1826A, 2018. https://saemobilus.sae.org/standards/air1826a
This aerospace industry standard establishes acoustic design guidelines and best practices for environmental

control systems in aircraft, including noise source identification, propagation path analysis, and mitigation strategies. It is directly applicable to the project as ECLSS systems in spacecraft share similar components and acoustic challenges with aircrafts. Testing methods were outlined for noise measurement, which can be used in evaluating the final solution.
[12] D. J. Maglieri et al., "Interior Noise Source/Path Identification Technology," SAE Technical Paper 2000-01-1709, 2000. doi: 10.4271/2000-01-1709
This paper describes methods for identifying and isolating noise sources in vehicle interiors. It is relevant because ECLSS noise follows similar propagation paths within the lander cabin.
[13] D. J. Mead, "Recent applications of viscoelastic damping for noise control in automobiles and commercial airplanes," J. Sound Vib., vol. 262, no. 3, pp. 457-474, 2003. doi: 10.1016/S0022-460X(03)00106-8
This journal article reviews viscoelastic damping applications in transportation. It is relevant as a passive noise control method suitable for space environments as constrained layer damping can reduce structural vibration transmission.
[14] J. S. Preisser et al., "Acoustics and noise control in space crew compartments," NASA Technical Reports Server, NASA/TM-2016-219173, 2016. https://ntrs.nasa.gov/api/citations/20160000818
This NASA technical memorandum addresses acoustic requirements for crewed spacecraft. It directly shows the customer needs by establishing NASA's 60-70 dBA limits for crew compartments, giving the specific frequency bands that need to be blocked that impact astronaut health.
[15] M. P. Hurtado, "Optimum Design of Compact, Quiet, and Efficient Ventilation Fans," Master's thesis, Dept. Mech. Eng., Virginia Tech, Blacksburg, VA, 2020. https://www.academia.edu/download/89578189/Hurtado_MP_D_2020.pdf
This thesis from Virginia Tech optimizes fan design for noise reduction while maintaining airflow efficiency. It is relevant because ECLSS fans are sources of much noise and geometry modifications like this can reduce noise by up to 15 dBa without impeding performance.
[16] NASA, "Environmental Control & Life Support System (ECLSS): Human-Centered Approach," OCHMO-TB-002, NASA Marshall Space Flight Center, 2019.
This technical brief describes ECLSS architecture and human factors considerations. It is important for understanding the system the solution must integrate with, including component locations, airflow requirements, and access constraints that affect our design boundaries.
[17] N. S. Zawodny et al., "Innovative Approach for Developing Spacecraft Interior Acoustic Requirement Allocation," in NOISE-CON 2016, Providence, RI, 2016, pp. 1-10. https://ntrs.nasa.gov/citations/20160005445
This paper presents a methodology for allocating acoustic requirements across spacecraft subsystems. It is relevant for establishing noise suppression targets and understanding how the system fits within overall vehicle acoustic budget.
[18] Y. Jing et al., "Reconfigurable Origami-Inspired Metamaterials for Controllable Sound Manipulation," Phys. Rev. Applied, vol. 12, no. 3, p. 034029, Sep. 2019. doi: 10.1103/PhysRevApplied.12.034029
This research article describes tunable acoustic metamaterials using origami geometry. It represents an emerging technology for adaptive noise control that uses geometric reconfiguration to shift absorption frequency bands and add adjustability to the design.
[19] R. Singh et al., "Noise reduction in pumping systems," Proc. Inst. Mech. Eng. E, vol. 232, no. 4, pp. 432-445, 2018. doi: 10.1177/0957456518782227
This journal article examines noise reduction techniques for pumps and fluid systems. It is relevant because ECLSS includes pumps for coolant circulation that add to the ambient noise.
[20] J. L. Perry and L. Parrish, "Design and Analysis of the ISS Environmental Control and Life Support System," SAE Technical Paper 2001-01-2345, NASA, 2001. https://www.nasa.gov/wp-content/uploads/2015/05/design_iss_environment_control_and_life_support_systems.pdf
This technical paper details the ISS ECLSS design, including acoustic considerations. It provides precedent for long-duration habitat life support systems, operational noise levels, and crew feedback that informed the durability and noise suppression specifications.
[21] Systems and methods for active noise cancellation, US20220114996A1, Independent Inventor, Apr. 14, 2022.
This patent describes active noise cancellation systems using adaptive filtering. It is relevant as a potential solution

approach, though power requirements may be prohibitive.
[22] N. S. Zawodny and D. D. Boyd Jr., "Geometric Acoustics for Aircraft Noise Scattering," in AIAA Aviation 2022 Forum, Chicago, IL, June 2022, Paper AIAA 2022-3077. doi: 10.2514/6.2022-3077
This paper presents geometric acoustic methods for modeling sound propagation, reflection, and diffraction around aircraft structures including ray tracing and surface creeping waves. It is relevant for understanding how noise propagates within the confined geometry of spacecraft cabins and reflects off interior surfaces.
[23] "Meta-mirror reflects sound waves in any direction," Phys.org, Feb. 2019. https://phys.org/news/2019-02-metasurfaces-unusual.html
This article describes metasurface technology that can redirect sound waves in controlled directions through engineered surface geometries. It is relevant as an emerging passive technology for managing sound propagation without traditional bulky absorbers. Metasurfaces can enable the design to be more compact and lightweight while still providing acoustic control.
[24] K. Kailasanath et al., "Pure Cancellation of Noise in a Supersonic Rectangular Jet," in AIAA Aviation 2024 Forum, Las Vegas, NV, July 2024, Paper AIAA 2024-3594. doi: 10.2514/6.2024-3594
This paper investigates active noise cancellation techniques for high-velocity jet flows using anti-phase acoustic injection. It is relevant for understanding active noise control in high-speed airflows similar to ECLSS ventilation systems.
[25] R. A. Ibrahim, "Recent advances in nonlinear passive vibration isolators," J. Sound Vib., vol. 314, no. 3-5, pp. 371-452, 2008. doi: 10.1016/S0022-460X(08)00043-6
This comprehensive journal review examines nonlinear vibration isolation systems including quasi-zero stiffness mechanisms and nonlinear damping. It is relevant for isolating ECLSS component vibrations before they transmit to cabin structure. Nonlinear isolators can achieve a broader frequency isolation range than linear systems while still being compact.
[26] M. Maciel et al., "Analytical Modeling of Active Acoustic Louvres," in Proc. INTER-NOISE 2016, Hamburg, Germany, Aug. 2016, pp. 1-12. https://www.researchgate.net/publication/289663823
This paper develops analytical models for active acoustic louvers that dynamically adjust aperture geometry to control sound transmission. It is relevant as a potential adjustable/adaptive solution for controlling noise while maintaining necessary airflow. Mechanically adjusting barriers can provide 5-15 dBa variability in sound transmission which adds adjustability to the system.
[27] Systems and methods for active noise cancellation, US8750531B2, Semiconductor Components Industries LLC, June 10, 2014.
This patent describes integrated circuit solutions for active noise cancellation including adaptive filtering algorithms and low-power digital signal processing. It is relevant for understanding power-efficient active noise control implementations.
[28] Acoustic treatment system, EP3091415B1, Kevin Allan Dooley Inc., May 23, 2018. [Online]. Available: https://patents.google.com/patent/EP3091415B1/en
This patent covers modular acoustic treatment panels designed for aerospace applications with quick-mount installation systems. It is relevant for our assembly and repairability specifications. Modular panel designs can enable quicker installation and removal which supports the need for easy implementation.
[29] "Zero Sound Panels," Zero Sound, 2023. [Online]. Available: https://zerosound.com/
This commercial website describes large-format acoustic panels designed to dampen ambient noise in commercial and residential spaces. It is relevant for understanding high-performance passive absorption materials and their installation methods.
[30] "Active Noise Cancellation Systems," Silentium, 2022. https://www.silentium.com/ecomotion-2022/
This product website describes commercial active noise cancellation systems for vehicles and equipment enclosures using distributed microphone/speaker arrays. It is relevant for understanding state-of-the-art active noise control implementation.
[31] "Denoize Active Sound Cancellation," Denoize, 2023. https://www.denoize.com/
This commercial website describes active sound cancellation technology for automotive and industrial applications using adaptive algorithms. It is relevant as a commercial example of active noise control.

Appendix A: QFD House of Quality

The Quality Function Deployment (QFD) is a process that is used to determine the specifications of a new design that is based on the needs of the customer and what is currently on the market. It is a visual representation of these needs and specifications that outlines the importance and relationship between each requirement.

Correlations
Positive +
Negative –
No Correlation

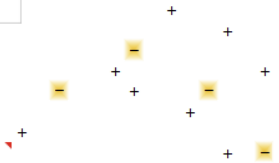
Relationships
Strong ●
Moderate ○
Weak ▽

Direction of Improvement
Maximize ▲
Target ◇
Minimize ▼

QFD House of Quality

Project: NASA HuLc Challenge: Noise Suppression

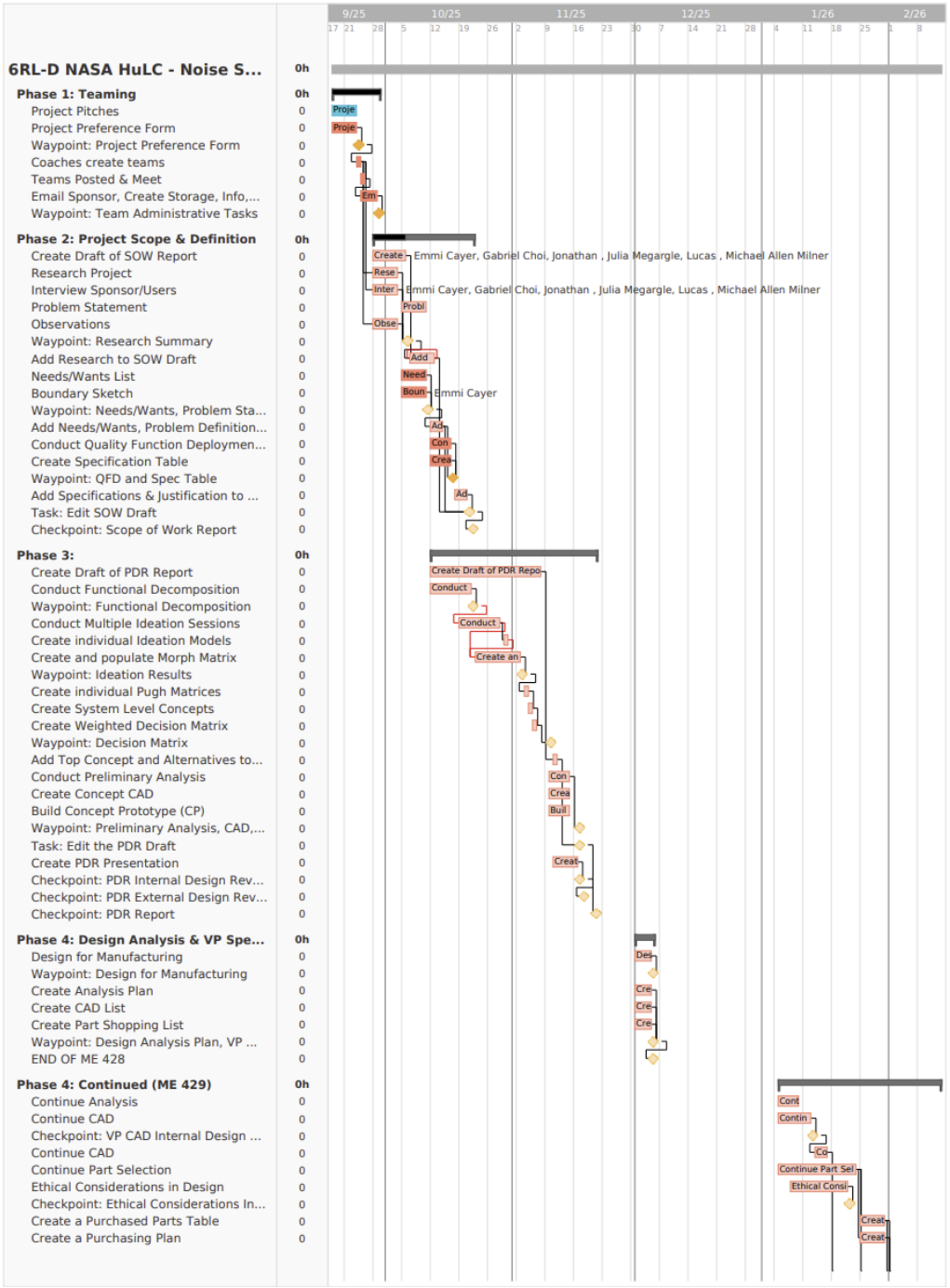
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		WHO: Customers																		NOW: Curr. Products						
	Column #							1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16					
	Direction of Improvement							▼	▲	◇	▼	▲	▲	▼	◇	▲												
Row #	Weight Chart	Relative Weight	NASA	Artemis Astronauts			Maximum Relationship	How: Engineering Specifications (Tests) / WHAT: Customer Requirements (Needs/Wants)	Volume	Repairability	Weight	Power Consumption	Manufacturability	Adjustability/adaptability	Long Term Cost	Initial Cost	Environmental Sustainability	Durability	Suppresses Noise						Our Current Product	Competitor #2: *Apple Headphones*	Competitor #2: Soundproof Cow	1 cubic	Competitor #4: DYAD	0 1 2 3 4 5	Row #
1		14%	10	10			9	Survive Launch Load		▽	▽	▽	▽	▽	●	○	▽	●												1
2		14%	10	10			9	Operational Life of 30-days for lunar, 1200 days for Mars missions	▽	●		●	▽	○	▽	▽	▽	●	●							1	5	3	4		2
3		10%	6	9			3	No additional risks to crew	○	▽	○	○	▽	▽	▽	▽	▽	○								2	5	2	5		3
4		9%	8	5			9	Minimal barriers to NASA adoption (low power, low weight)	●		●	●	○	●	●	●		○	●							0	4	1	1		4
5		10%	6	9			9	Simple to implement	▽	●	▽	○	●	▽	▽	○			○							5	4	1	4		5
6		10%	9	6			9	Ready to use within 5-8 years					●						▽							5	5	2	5		6
7		14%	10	10			9	Reduce, isolate, or manage noise from ECLSS systems	▽		▽	▽		○				●	●							3	2	2	3		7
8		9%	8	5			9	Longevity		●		○		▽	●	▽	○	●	○							3	4	2	4	Our Product Competitor #1 Competitor #2 Competitor #3 Competitor #4	8
9		10%	6	9			9	Astronaut comfort	●	○	○			○			▽		●							2	3	3	4		9
								HOW MUCH: Target Values	<1 cubic meter	Repaired with tools in 60 min	<1000 lb	<.95 kW	Team, made onsite, ≤ 1 yr	Adjustable?	under $10000	under $100000	Is it environmentally safe?	lasts for at least 5 missions	<70 dBa											
								Max Relationship	9	9	9	9	9	9	9	9	3	9	9											
								Technical Importance Rating	241.1	350.7	179.5	319.2	249	236.3	317.8	184.9	74.66	507.5	487	0	0	0	0	0						
								Relative Weight	8%	11%	6%	10%	8%	8%	10%	6%	2%	16%	15%	0%	0%	0%	0%	0%						
Curr. Products								Our Current Product																						
								Competitor #2: Apple Headphones	5	1	5	5	5	3	5	5	1	0	0											
								Competitor #2: Soundproof Cow	3	1	5	5	5	2	3	3	4	3	4											
								1 cubic	5	2	5	3	5	4	5	5	3	4	5											
								Competitor #4: DYAD	5	1	5	5	5	1	3	3	3	4	4											
								Column #	1	2	3	4	5	6	7	8	9	5432	11	12	kcx	14	15	16						

Appendix B: Gantt Chart

The Gantt Chart displays the project schedule illustrated as a horizontal bar chart. Individual tasks are outlined for each phase then each task is visualized as a timeline, with tasks that rely on each other emphasized.




9/25
10/25
11/25
12/25
1/26
2/26
17 21 28 5 12 19 26 2 9 16 23 30 7 14 21 28 4 11 18 25 1 8
Create a FP Build Plan 0
Waypoint: FB Plan, iBOM, Parts Table... 0
Create an Indented Bill of Materials 0
Create a Drawing Package 0
Checkpoint: Drawing Internal Design... 0
Create a Design Verification Plan 0
Create a Feasibility Prototype (FP) 0
Create a Manufacturing Plan 0
Checkpoint: Manufacturing Internal ... 0
Checkpoint: CDR Internal Design Rev... 0
Checkpoint: CDR External Design Rev... 0
Checkpoint: CDR Report 0
Problem Statement 0h
Creat
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Noise Suppression System for Artemis Human Lander
Preliminary Design Review

by
Emmi Cayer
Gabriel Choi
Jonathan Corvera
Lucas Kaemmerer
Julia Megargle
Michael Milner

Senior Project
Sponsor: Dr. John Chen
California Polytechnic State University,
San Luis Obispo
College of Engineering



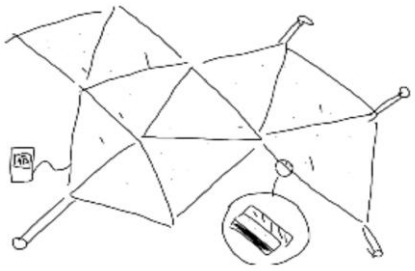
17 November 2025

NASA Human Lander Challenge – Noise Suppression Preliminary Design Review

Project Overview

As humanity progresses further in space exploration, astronauts are going to need to stay in space for longer durations of time. Currently, the Environmental Control and Life Support Systems (ECLSS) in the human lander are too loud for astronauts to healthily live and communicate aboard the lander. NASA (National Aeronautics and Space Administration) needs a way to suppress this noise on the Artemis human lander capsule. This means a solution must be designed that dampens and eliminates a wide range of frequencies within the lander to provide astronauts with a safe and comfortable living environment. This includes meeting volume, weight, sound, and power requirements within the human lander vehicle. The design will include a barrier with a variety of features to combat the full range of frequencies generated by the ECLSS. The project is being carried out by a team of mechanical engineering students at California Polytechnic State University, San Luis Obispo (Cal Poly), including Emmi Cayer, Gabriel Choi, Jonathan Corvera, Lucas Kaemmerer, Julia Megargle, and Michael Milner. This project is sponsored by Dr. John Chen with funding provided by Cal Poly.

Concept description



The barrier concept designed to block noise generated by the ECLSS features diamond-shaped panels that can fold in two directions, with adjustment angles ranging from 45 to 90 degrees. These adjustable positions allow the design to alter the angle of the reflected frequency and redirect it to a specific area on lander and away from the astronauts in the space. This adjustable frame can also hold layers of acoustic damping material. Each layer and location on the structure can be optimized to absorb different frequency ranges, particularly high frequencies.

Concept Justification

Aboard the ISS, the realest example of an extended duration space habitat, ambient ECLSS noise is a large inhibitor of communication. Under the scrutiny of a weighted decision matrix [Appendix B] comparison to other solutions like sound absorption reservoirs and active noise cancelation, the hinge system stood out as the most flexible and efficient option to mitigate that noise and improve astronaut communication. The modular structure of the panels allows the system to fit irregular mounting configurations, and the adjustable pitch of the absorptive ridges allow the panels to meet a broader range of noise frequencies. Modularity also brings the added benefit of improved reparability.

What's Next

This document acts as a proposal for an initial design which will be refined based on sponsor feedback. The sponsor, John Chen, will be requested to review the proposal and approve or reject aspects of the design. If the design is approved, the concept will continue to be developed and analyzed in anticipation of the Critical Design Review (CDR). The CDR report will describe the details of the final prototype and justify the manufacturing of the design. Justification will include acoustic analysis, durability testing, safety evaluation, structural prototype, and budget plan.

Figure 1. Four-Panel Chart for Preliminary Design Review

Overview

As humanity progresses further in space exploration, astronauts are going to need to stay in space for longer durations of time. The proposed project focuses on developing an innovative solution to manage and reduce the noise generated by the Environmental Control and Life Support Systems (ECLSS) in long-duration space habitats. Currently, the ECLSS in the human lander is too loud for astronauts to healthily live and communicate aboard the lander. The project's intent is to mitigate excess noise that can have a significant impact on the crew's health and communication. A solution must be designed that dampens and eliminates a wide range of frequencies within the lander to provide astronauts with a safe and comfortable living environment. This includes meeting volume, weight, sound, and power requirements within the human lander vehicle. The design will include a barrier with a variety of features to combat the full range of frequencies generated by the ECLSS.

Funding for the prototype will be provided by California Polytechnic State University for up to $500 for material costs. The project will be carried out by a team of Cal Poly engineering students including Emmi Cayer, Gabriel Choi, Jonathan Corvera, Lucas Kaemmerer, Julia Megargle, and Michael Milner. This team is responsible for the design, manufacturing, and testing of the acoustic barrier and optimizing it to meet the customer's needs.
The team has done extensive research into the needs of the customer, existing solutions, and the necessary specifications. Much ideation was conducted using the ISS as a point of reference for the design, and at this point in the design process the team has narrowed down the results from ideation to a single leading concept design, as shown in Figure 2.

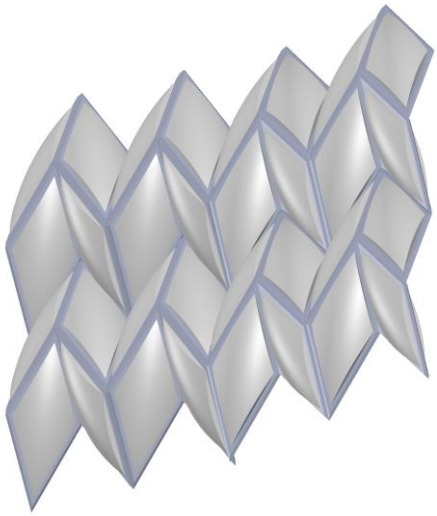


Figure 2. Foldable barrier panel concept CAD.

The goal of this report is to present the selected solution concept and outline its key design features. The selection process will be justified through comparison with other designs and how each of these meets the customer's design criteria. The report is ordered as follows:

- Concept Description and Justification: Describes the leading concept and how it meets the competition and customer requirements
- Alternative Solutions: Summarizes other concepts that were discussed during ideation, explaining how they worked with the given constraints and why they can act as alternatives.

- Appendices: Includes supporting materials, such as the Gantt chart and ideation tools including the weighted decision matrix, results of concept model testing, and preliminary analysis.

If the project is approved by the sponsor, the team will continue further analysis, optimization, and testing in anticipation for the Critical Design Review (CDR) at the end of Cal Poly's winter quarter. The CDR will present the finalized prototype design and justify its readiness for fabrication. Additionally, by March 4, 2026, the team will submit a proposal package to the NASA Human Lander Challenge competition consisting of a written report and video presentation.

Concept Description and Justification

This section explains how the leading concept was selected for the noise-mitigation system and the technical details relevant to its design and fabrication.

After substantial ideation, the team developed a concept prototype that combines the most effective aspects of previous designs and alternatives into a single solution. According to the ISS Crewmembers' Noise Exposures [1], the frequency range to be attenuated is 63–1000 Hz. Acoustic panels were chosen as a starting point for their ability to attenuate frequencies in the range of 125–1000 Hz, which covers most of the target band. Additional foams with varying properties can be used with the adjustable panels to attenuate the remaining lower frequencies. Each tile is a rhombus with hinges on all sides, allowing it to bend independently and reduce coupling with neighboring tiles. This hinge system creates groups of rhombi that act as deflection shapes when moving together. The rhombi are hollowed-out frames that are to be filled with acoustic damping material in the final design. The chosen design has been fully modeled and can be seen in Figure 2. Following this section, a brief comparison to hex-shell and linear-folding alternatives is provided in the Alternative Solutions section, with the weighted decision matrix in Appendix B.

Motivation

For the acoustic suppression panel to perform well for the human lander crewed habitat, it must attenuate broadband life support system noise while remaining lightweight, compact, and straightforward to integrate. The first of these requires effective control of both low to mid frequency scattering and general sound absorption. This is fairly straightforward: a surface that presents only a single, hard, fixed geometry will tend to reflect sound energy, while a panel that cannot vary its standoff relies on permanent thickness for low-band absorption performance.

In this way, this concept with variable facet orientation is suited to disperse a range of frequencies with the angle able to be varied to strengthen the range of frequency absorption. The lightweight nature of the design stems from material selection and minimal hardware, which will be further detailed in the Concept section.

The merit of this acoustic panel system is also tied to its serviceability, a metric at which the acoustic panel design functions well. Many other solutions already in implementation are passive sound suppression because their simplicity supports years of service with minimal maintenance. This approach adds controlled articulation to achieve a more capable reverberation and sonic suppression function while maintaining simplicity through the modular tiles. The fundamental compliance of the hinge network allows the surface to reconfigure without sliders that can rattle or wear. Each panel is replaceable at the module level, preserving the maintainability of passive systems while enabling precise tuning not available in current alternatives. In this sense, expected service life remains comparable to conventional panels, with improved adaptability to evolving cabin layouts and noise fields.

Concept

The chosen design is based on prior research regarding the functionality of acoustic panels and their deflection shapes. The angle of attack of the deflection shapes can be optimized for the frequency or size of the sound wave that is being deflected. In an American Institute of Aeronautics Astronautics article on noise dispersion [2], the science behind the angle of attack on acoustic damping technology and the size of the panel used directly affects how effectively noise reverberation is reduced. High frequencies are more effectively dispersed by deflection shapes with a sharp angle of attack, while lower frequencies are less dependent on the angle of attack and more dependent on the distance between the panel and the wall as well as the thickness of the panel itself. When the angle of attack on the deflecting shapes is increased, the overall thickness of the air gap shrinks due to compression of the surface area. Because of this, the angle of attack of the panels on the concept prototype changes to prioritize either deflection shape angle or panel thickness based on the target frequency so that the sound can be most effectively dispersed.

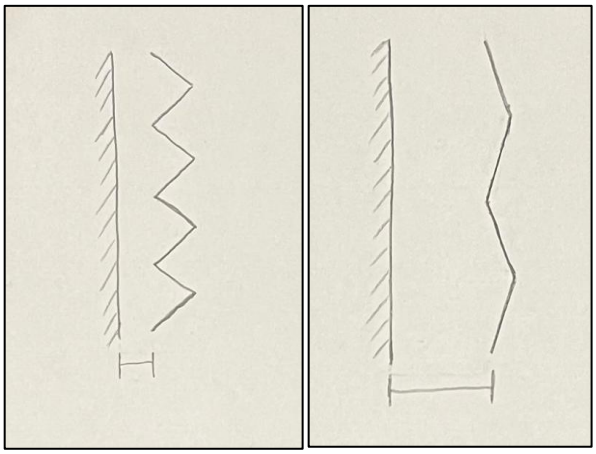


Figure 3. Sketch showing Design Configuration prioritizing high frequencies (left) and low frequencies (right).

As seen in Figure 3, in a "high frequency" configuration, the height of the deflection shapes is prioritized over the thickness of the air gap between the panel and the wall to disperse high frequencies more effectively. In the "low frequency" configuration, the deflection shapes are almost negligible to increase the thickness of the air gap, which more effectively reduces low frequency reverberation.

Because the overall size of the panels shrinks when the angle of attack is increased, the design of the panel will need to be very large to accommodate deflection at the highest target frequency, as shown in calculations done in Appendix C. This design will be able to attach to copies of itself in a lattice-like structure so that panels of any shape and size can be created. Due to hinges in two dimensions, the design will also be able to be placed on surfaces of any shape if they do not contain sharp angles of less than 60 degrees, which is very uncommon for rocket interiors that normally incorporate cylindrical or square-like space design. In a rare case where angles of less than 60 degrees are involved, panels can be placed on opposing walls, and acoustic material can be placed in the corner, creating little to no efficiency losses by the sharp corners.

Considerations

While this design effectively meets the project's objectives, there are a few unknowns and challenges that must be considered. Actuator reliability must be heavily considered. To move a large sheet of angled acoustic material, many actuators must be in place. This will increase the number of potential electrical failure points. Additionally,

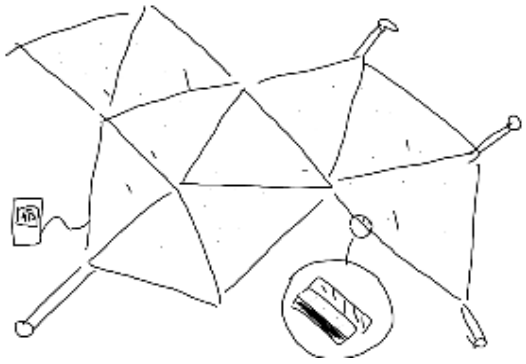
material fatigue must also be considered. The panels' repeated folding and angle adjustments could lead to wear at the hinges and points of material interaction, reducing noise suppression performance over time and potentially affecting astronaut health and communication.

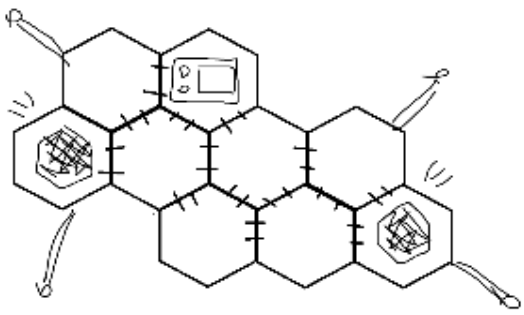
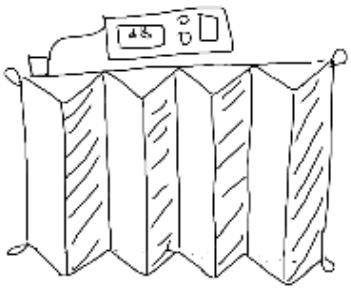
Alternative Solutions

During ideation, design solutions were ranked based on eleven engineering specification criteria: volume, repairability, weight, power consumption, manufacturability, adjustability, adaptability, long term cost, short term cost, environmental sustainability, durability, and noise suppression. These criteria are shown in Appendix B in the weighted decision matrix.

Based on these criteria, three solutions ranked much higher than the others. These three design solutions are shown below in Table 1. If the customer does not prefer our chosen solution, the other two solutions shown can be further worked and designed to meet their specifications. These three design solutions were chosen specifically due to their adjustability and ability to absorb a range of frequencies, which were weighted as the two most important criteria.

Table 1. Alternative Solutions Considered.

Solution Name	Description/Sketch	Justification
Triangular Pattern Panels	 The sketch illustrates a modular system composed of interconnected triangular panels in a geometric pattern. Each panel is mounted on adjustable supports that allow variation in orientation and angular displacement. The configuration emphasizes the use of lightweight structural members and origami to inspire geometry. There is inclusion of measurement and control components to respond to changes in frequency and environmental conditions.	This design focuses on the theme of origami shapes and the formation of complex, 3D angular displacements of the panels. This design has the highest weighted adjustability and adaptability to the environment score, as well as an equal noise suppression score to the other designs. Since it ranked highest overall in the Weighted Decision Matrix, it is the design of choice when it comes to projected customer satisfaction.

Hex Shell Panels	 The sketch depicts a modular array of hexagonal panels connected in a honeycomb configuration. Each hexagonal unit is designed to interlock with adjacent panels, forcing a continuous and rigid structure. The system incorporates embedded components such as control modules and acoustic devices positioned within select panels. Support rods are illustrated extending from the structure to attach to the surroundings.	This design focuses on rigid modularity, meaning that the panels have very little adjustability but can be fit together in such a way to slightly angle their overall path. They can, therefore, be used on walls and machines with different geometry, but do not adjust to dampen specific frequencies with a respective panel angle.
Linear Folding Panels	 The sketch illustrates a series of vertically aligned foldable panels connected along linear hinges, forming an accordion-like structure. The design incorporates a control module positioned at the top to adjust the panel angles.	This design focuses on ease of actuation and angular displacement. Due to its adjustable nature this design ranked high on the adjustability and adaptability scale but wasn't the best option since this adjustability was in only one direction.

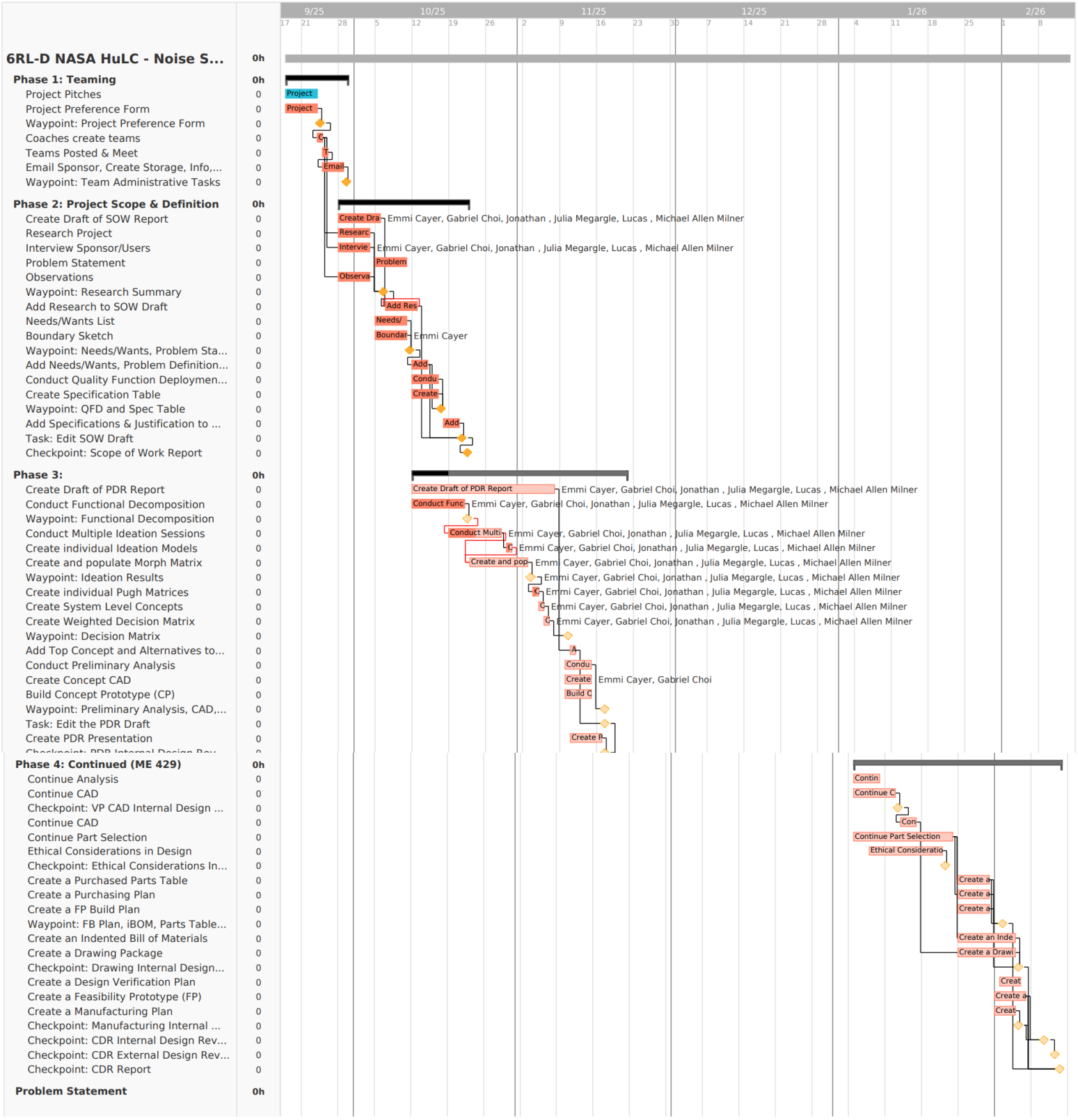
Works Cited

[1] J. Limardo, C. S. Allen, R. W. Danielson, and A. J. Boone, "Status - International Space Station (ISS) Crewmembers' Noise Exposures," *INTER-NOISE and NOISE-CON Congress and Conference Proceedings*, vol. 263, no. 4, pp. 2740–2754, Aug. 2021, doi: https://doi.org/10.3397/IN-2021-2219.

[2] "On the Aerodynamic-Noise Sources in a Circular Cylinder Coated with Porous Materials | Aeroacoustics Conferences," *Aeroacoustics Conferences*, 2022, doi: https://doi.org/10.2514/MAERO22;page:string:Article.

Appendix A: Gantt Chart

For organization and planning, the team uses a Gantt chart through a website called “teamgantt”. This gives a clear timeline on a horizontal bar chart that visually maps the project and the tasks that need to be done. The bars on the chart indicate when the tasks should be started, how long they will take, and when they are due. Team members are assigned to each task, with names appearing next to the tasks that are assigned.


9/25
10/25
11/25
12/25
1/26
2/26
6RL-D NASA HuLC - Noise S... 0h
Phase 1: Teaming 0h
Project Pitches 0
Project Preference Form 0
Waypoint: Project Preference Form 0
Coaches create teams 0
Teams Posted & Meet 0
Email Sponsor, Create Storage, Info,... 0
Waypoint: Team Administrative Tasks 0
Phase 2: Project Scope & Definition 0h
Create Draft of SOW Report 0
Research Project 0
Interview Sponsor/Users 0
Problem Statement 0
Observations 0
Waypoint: Research Summary 0
Add Research to SOW Draft 0
Needs/Wants List 0
Boundary Sketch 0
Waypoint: Needs/Wants, Problem Sta... 0
Add Needs/Wants, Problem Definition... 0
Conduct Quality Function Deploymen... 0
Create Specification Table 0
Waypoint: QFD and Spec Table 0
Add Specifications & Justification to ... 0
Task: Edit SOW Draft 0
Checkpoint: Scope of Work Report 0
Phase 3: 0h
Create Draft of PDR Report 0
Conduct Functional Decomposition 0
Waypoint: Functional Decomposition 0
Conduct Multiple Ideation Sessions 0
Create individual Ideation Models 0
Create and populate Morph Matrix 0
Waypoint: Ideation Results 0
Create individual Pugh Matrices 0
Create System Level Concepts 0
Create Weighted Decision Matrix 0
Waypoint: Decision Matrix 0
Add Top Concept and Alternatives to... 0
Conduct Preliminary Analysis 0
Create Concept CAD 0
Build Concept Prototype (CP) 0
Waypoint: Preliminary Analysis, CAD,... 0
Task: Edit the PDR Draft 0
Create PDR Presentation 0
Phase 4: Continued (ME 429) 0h
Continue Analysis 0
Continue CAD 0
Checkpoint: VP CAD Internal Design ... 0
Continue CAD 0
Continue Part Selection 0
Ethical Considerations in Design 0
Checkpoint: Ethical Considerations In... 0
Create a Purchased Parts Table 0
Create a Purchasing Plan 0
Create a FP Build Plan 0
Waypoint: FB Plan, iBOM, Parts Table... 0
Create an Indented Bill of Materials 0
Create a Drawing Package 0
Checkpoint: Drawing Internal Design... 0
Create a Design Verification Plan 0
Create a Feasibility Prototype (FP) 0
Create a Manufacturing Plan 0
Checkpoint: Manufacturing Internal ... 0
Checkpoint: CDR Internal Design Rev... 0
Checkpoint: CDR External Design Rev... 0
Checkpoint: CDR Report 0
Problem Statement 0h
Emmi Cayer, Gabriel Choi, Jonathan , Julia Megargle, Lucas , Michael Allen Milner
Emmi Cayer
Emmi Cayer, Gabriel Choi

Appendix B: Ideation Process

This appendix consists of the tools that were used in the ideation process leading to the selected design concept. The functional decomposition tree is depicted in Figure B.1. This was the start of the ideation process and was used to determine the key functions of the design that are needed to satisfy the customer. These functions were used in the brainstorming process, where multiple solutions to these functions were proposed by team members as depicted in Figure B.2. Sketches and preliminary prototypes were done and collected into a morphological matrix that was used to map out different combinations of attributes. This is shown in Figure B.3. These combinations were then sketched and used in a weighted decision matrix, as shown in Figure B.4., that lead to the final selection of the design discussed in the preliminary design report.

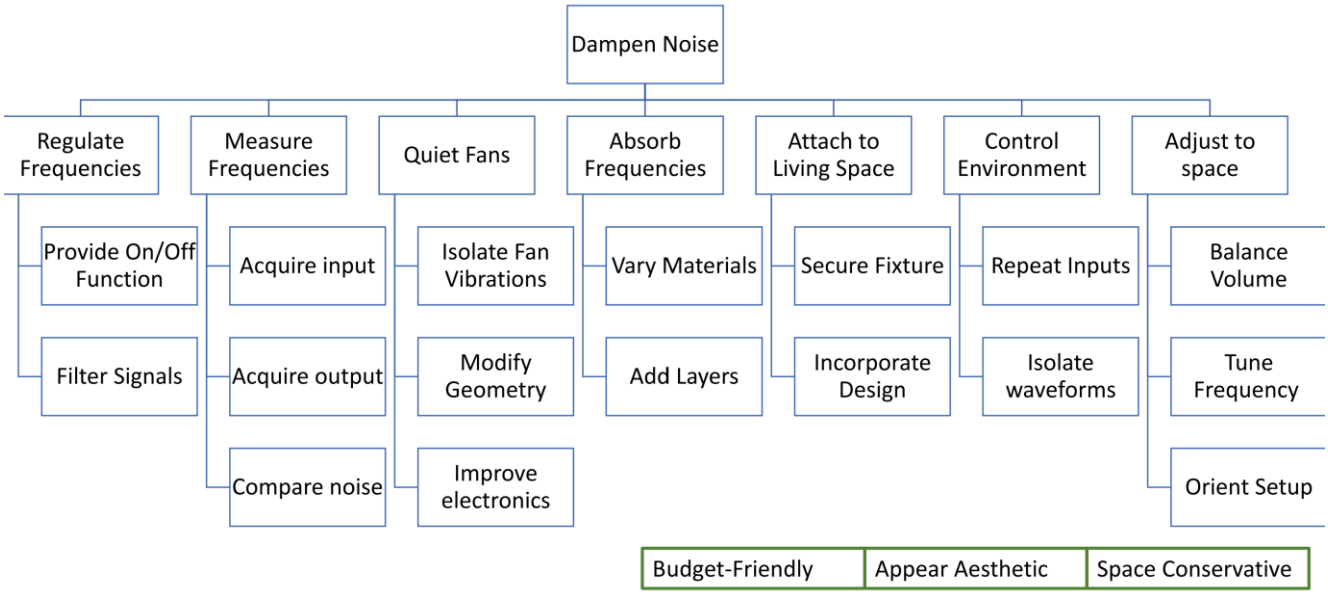


Figure B.1. Function tree outlining main goal and relating constraints.

In Figure B.1. presents a function tree breaking down the primary goal of dampening noise. The diagram systematically breaks down the overarching function into seven key branches. The constraints listed at the base figure as budget-friendling, aesthetic appearance, and space conservatism, which represent guiding design limitations that influence all functional decisions. Overall, the function tree provides a structured framework for identifying and prioritizing essential design requirements during the ideation phase.

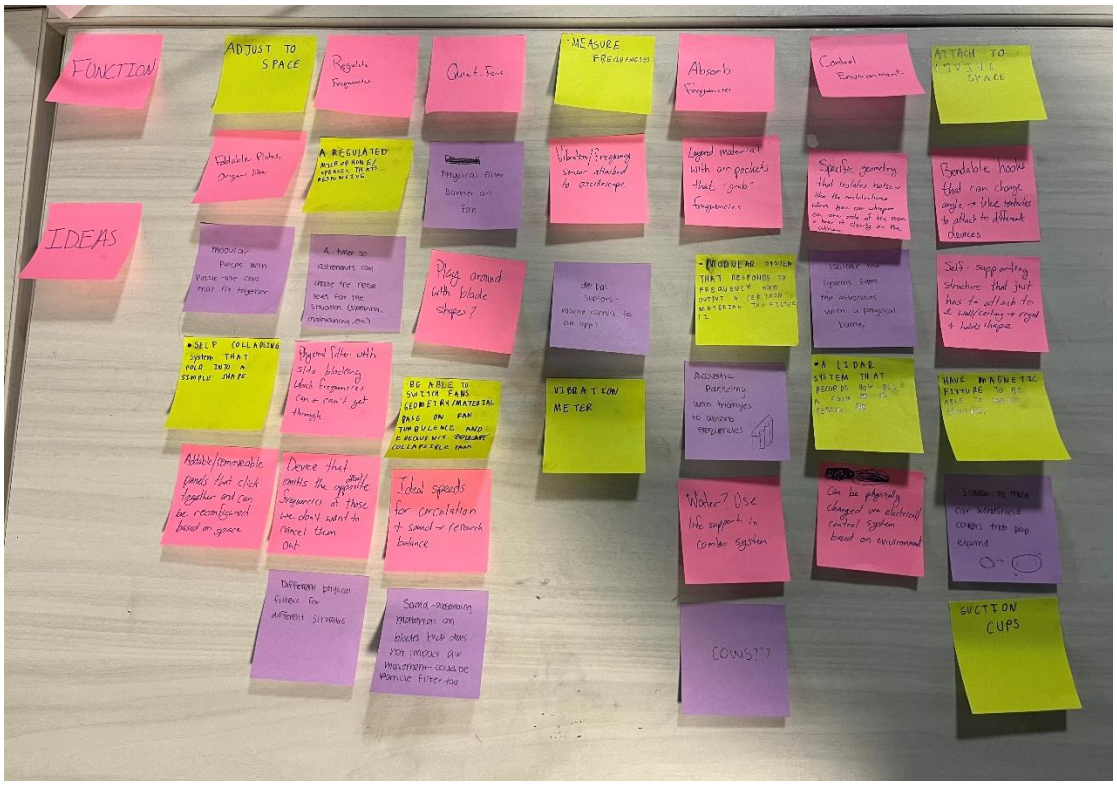


Figure B.2. Brainstorming process – sticky notes with proposed solutions to each function.

This illustrates the brainstorming process conducted during the ideation phase, in which a potential solution was developed for each sub function from the function tree shown above in Figure B.2. The figure displays a collection of color coded sticky notes organized by the functional category. Each note represents an individual idea generated by team members to address functions. The visual organization facilitated collaborative discussion, allowing the team to compare, refine, and group similar concepts.

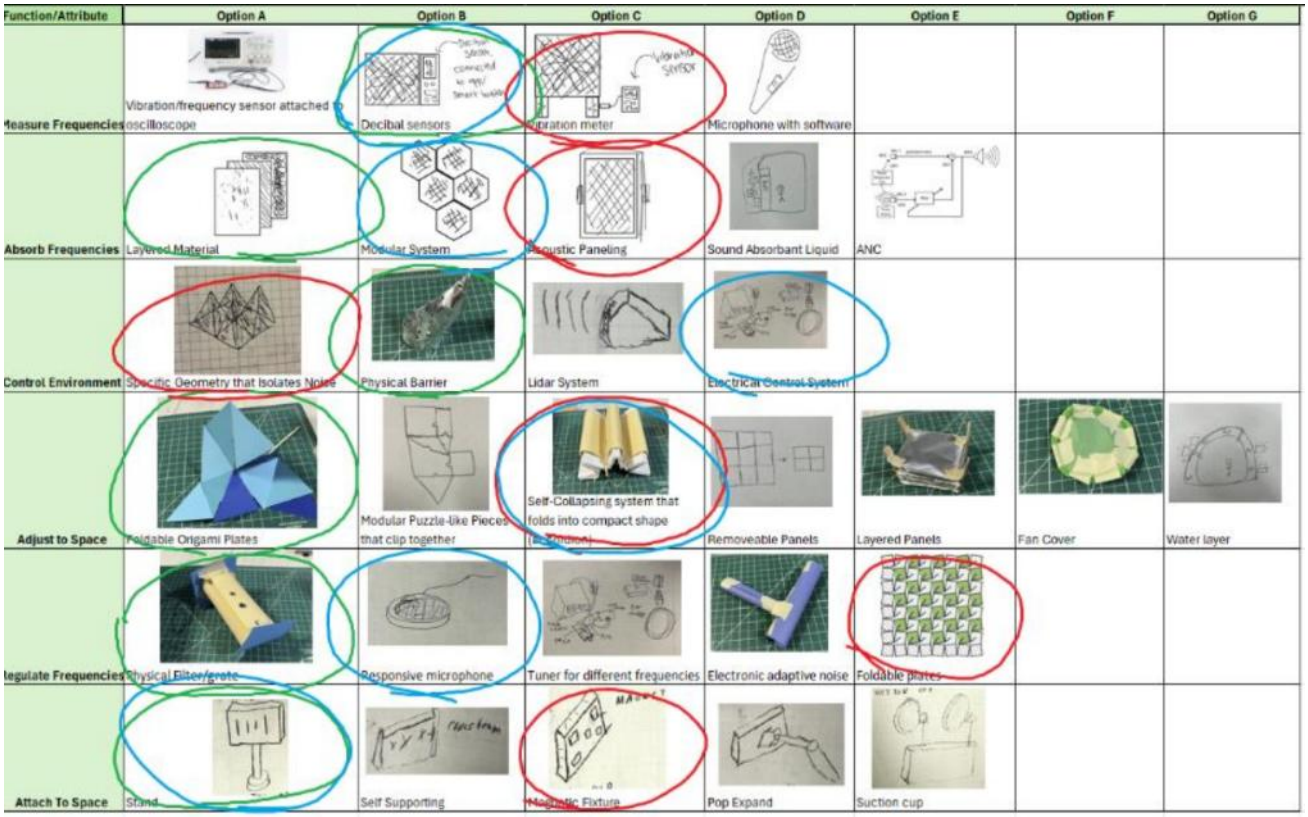


Figure B.3. Morphological matrix with different attributes mapped out in color-coded circles.

The morphological matrix shown above in Figure B.3. was used to organize and compare potential design attributes for each main function. The table includes conceptual sketches of alternative solutions arranged by functional category, with color coded circles identifying the most promising option. This method allowed the design team to systematically evaluate the combination of features and narrow down viable concepts for further development.

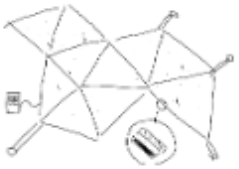
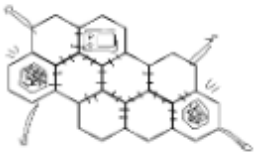
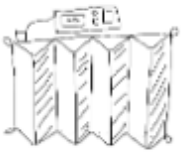
		Design 1		Design 2		Design 3	
		 Green		 Blue		 Red	
Specification	**Weight**	**Score**	**Total**	**Score**	**Total**	**Score**	**Total**
Volume	4	3	12	4	16	3	12
Repairability	3	3	9	2	6	2	6
Weight	3	4	12	3	9	2	6
Power Consumption	4	4	16	1	4	4	16
Manufacturability	2	2	4	3	6	4	8
Adjustability/Adaptability	5	5	25	4	20	3	15
Long Term Cost	2	4	8	4	8	4	8
Short Term Cost	2	3	6	4	8	3	6
Environmental Sustainabilty	1	1	1	1	1	1	1
Durability	4	4	16	3	12	3	12
Suppresses Noise	5	5	25	5	25	5	25
SUM			134		115		115

Figure B.4. Weighted decision matrix showing leading design that was selected as the primary concept prototype.

The weighted decision matrix shown above in Figure B.4. was used to evaluate and compare the three final design concepts based on key performance criteria. Each specification was assigned a weight according to its importance, and individual designs were scored accordingly. The Triangular Pattern Panels (Design 1) achieved the highest total score, indicating superior performance in adaptability, noise suppression, and environmental sustainability. This outcome led to its selection as the primary concept prototype for further development.

Appendix C: Preliminary Analysis

Appendix C consists of the work done to justify the solution concept. As the design consists of panels that change angles relative to each other to block off different frequencies, calculations were done to prove that a certain thickness of panel and panel angle would block off the target frequencies generated by NASA's Environmental Control and Life Support Systems. Calculations shown in Figure C.1 prove that the chosen thickness of the absorbent material can in fact absorb high frequencies. Calculations shown in Figure C.2 determine the panel size and angle at which low frequencies can be reflected. Figure C.3 shows the preliminary 3D printed prototype of the selected concept.

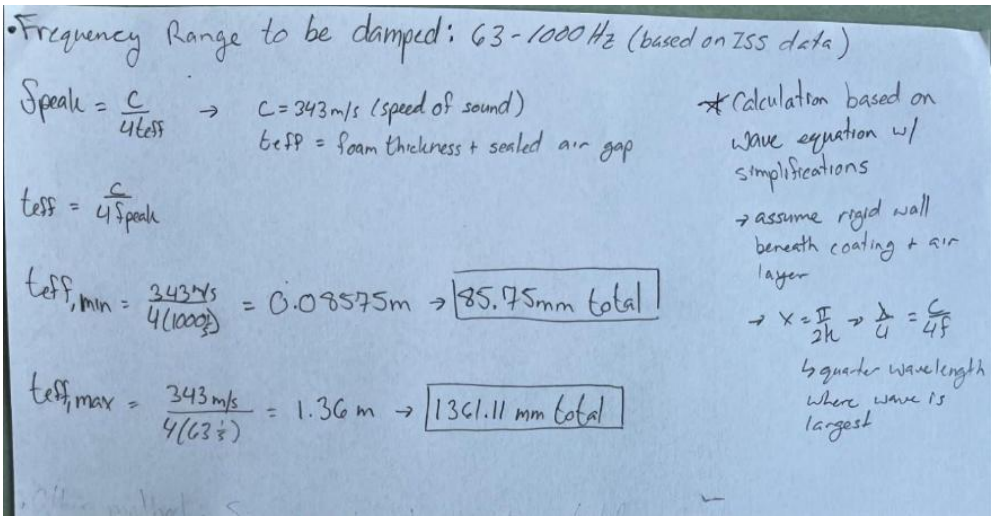


Figure C.1. Calculations for chosen thickness

The preliminary equation of hand calculations is shown above in Figure C.1. It is used to determine the effective panel thickness required to damp frequencies within the target range of 63-100 Hz. The computation applies to the quarter wavelength formula, incorporating the speed of sound and material properties to estimate both minimum and maximum effective thickness values. These results guided the dimensional parameters selected for prototype fabrication.

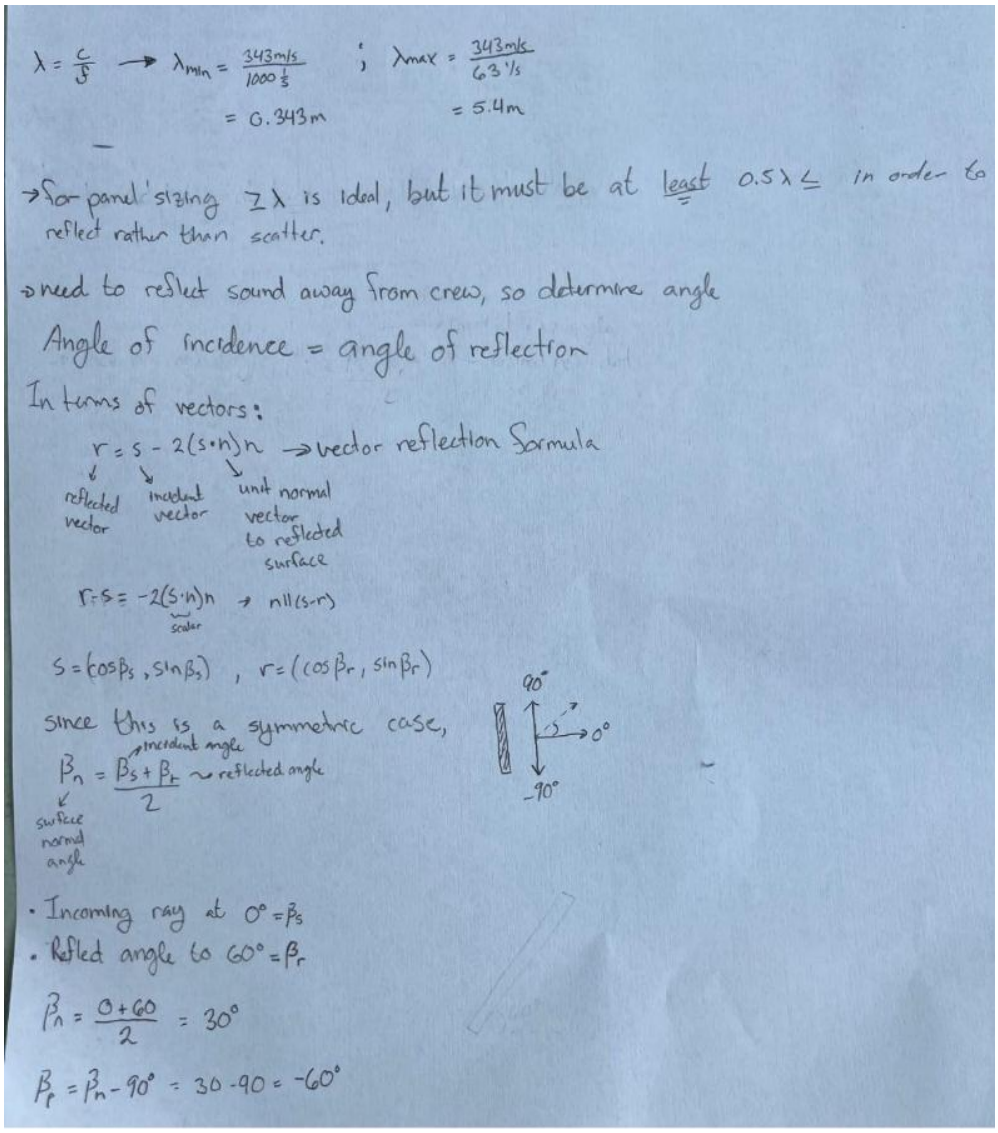


Figure C.2. Calculation for angles to dampen certain frequencies.

The analytical calculations used to determine the optimal panel angles for sound reflection and frequency dampening are shown in Figure C.2. The analysis applies vector reflection principles and the relationship between wavelength and angle of incidence to identify configurations that effectively redirect and minimize unwanted sound waves within the target frequency range.

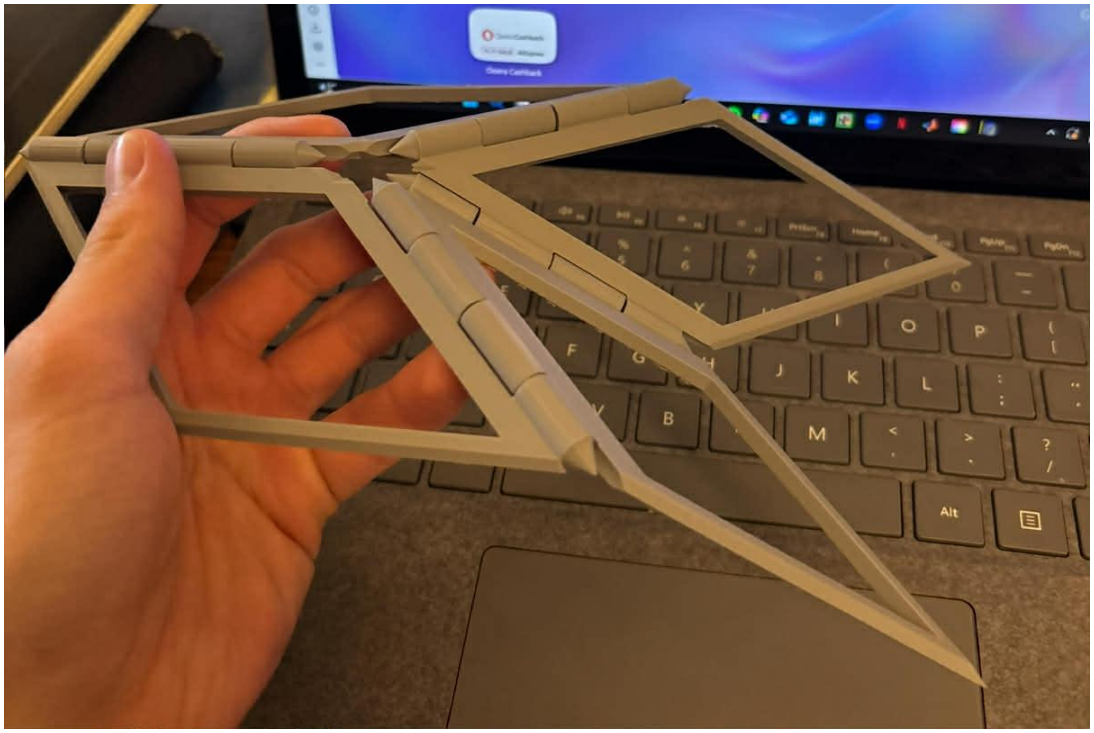


Figure C.3. 3-D printed Prototype Concept

The Figure C.3 displays the 3D-printed prototype of the selected concept, illustrating the structural framework and joint configuration of the adjustable triangular panel system. The prototype demonstrates the intended range of motion and modular assembly, validating the feasibility of the design folding mechanism and geometric adaptability prior to full scale fabrication. This physical prototype allowed the team to see the adjustability of the angles and better figure out how to make it more adaptable. The team learned that they could add larger panels as long as they were the same geometry to reflect the smaller frequencies.

Noise Suppression System for Artemis Human Lander

Critical Design Review

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Senior Project

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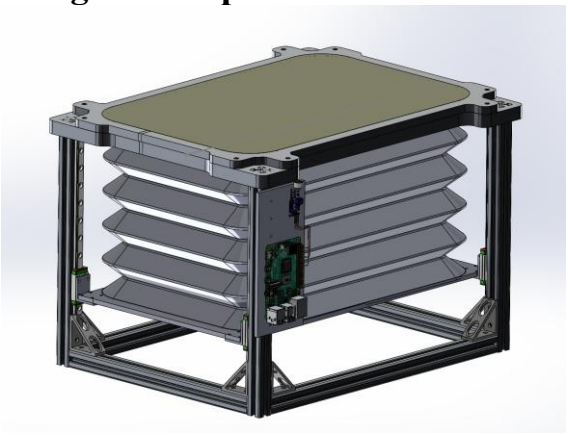


February 11, 2026

Project Overview

As humanity progresses in space exploration, astronauts are going to need to stay in space for longer durations of time. Currently, the Environmental Control and Life Support Systems (ECLSS) in the human lander are too loud for astronauts to healthily live and communicate aboard the lander. NASA (National Aeronautics and Space Administration) requires a method to suppress this noise within the Artemis human lander capsule. Therefore, a solution must be designed that dampens and eliminates a wide range of frequencies within the lander to provide astronauts with a safe and comfortable living environment. This solution must meet volume, weight, sound, and power requirements within the human lander vehicle. The design will include an adjustable acoustic membrane dampener that will target the lower frequency ranges generated by the ECLSS. The project is being carried out by a team of mechanical engineering students at California Polytechnic State University, San Luis Obispo (Cal Poly), including Emmi Cayer, Gabriel Choi, Jonathan Corvera, Lucas Kaemmerer, Julia Megargle, and Michael Milner. This project is sponsored by Dr. John Chen, with funding provided by Cal Poly.

Design Description



The design features a neoprene membrane that is stretched over a thermoplastic polyurethane (TPU) reverberation chamber with an adjustable depth controlled by a linear actuator. The bottom is sealed with a TPU gasket and mounted onto an aluminum plate to provide support. Aluminum T-slot framing was used to support and mount the design on the walls of the human lander. The TPU used to make the reverberation chamber, seal, and gasket connecting the membrane cost approximately $42 USD. The metal creating the base and frame for the design costs about $100 USD. Electronics used to control the design include a linear actuator, acoustic sensors, a microcontroller, and battery for power cost about $460 USD bringing the total cost to approximately $600 USD for the final design.

Design Specifications

The key design specifications include meeting the acoustic, weight, volumetric, and power criteria. Based on detailed calculations that can be found in Appendix A, there is convincing evidence that the design meets the key acoustic criteria required in NASA's Noise Criterion specifications [1]. A detailed CAD model provided evidence that the dimensions and volume would meet the established criteria. Reviewing the volume and materials with the CAD model provided convincing evidence that the weight of this design was within the established criteria. Finally, power was determined based on motor and electrical component needs.

Manufacturing and Test Plans

The reverberation chamber, which acts as the main housing, and the gasket will be 3D-printed with TPU filament using Cal Poly's Makerspace Bambu 3D printers. All aluminum parts for the base and frame will be bought from McMaster-Carr and will be machined to the dimensions specified by the design. The nylon membrane, all connectors, and all electronics will be bought from an online vendor and assembled in the Cal Poly shops. Testing will be comprised of visual inspection and acoustic testing to verify design criteria.

Figure 0. Four-Panel Chart for Critical Design Review.

Overview

As humanity progresses further in space exploration, astronauts will need to stay in space for longer durations of time. The proposed project focuses on developing an innovative solution to manage and reduce the noise generated by the Environmental Control and Life Support Systems (ECLSS) in long-duration space habitats. Currently, the ECLSS in the human lander generates noise levels that are too high for astronauts to live and communicate safely aboard. The project's intent is to mitigate excess noise that can have a significant impact on the crew's health and communication. The solution must dampen a wide range of frequencies within the lander to provide a safe and comfortable living environment. This includes meeting volume, weight, acoustic, and power requirements within the human lander vehicle. The design will include an adjustable acoustic membrane dampener that will target the lower frequency ranges generated by the ECLSS.

Funding for the prototype will be provided by California Polytechnic State University for up to $500 for material costs. The project will be carried out by a team of Cal Poly engineering students including Emmi Cayer, Gabriel Choi, Jonathan Corvera, Lucas Kaemmerer, Julia Megargle, and Michael Milner. This team is responsible for the design, manufacturing, and testing of the acoustic barrier and optimizing it to meet the customer's needs.

At the presentation of this Critical Design Review (CDR), the team has done extensive research into the necessary specifications as seen in the previously reviewed Scope of Work [2] and current acoustic technologies. Ideation was conducted on possible solutions to the proposed project and prototyping based on initial analysis was done to narrow these ideas into one proposed solution that was presented in the Preliminary Design Review (PDR). Continued analysis revealed several issues with the selected design, and with sponsor support, a redesign was conducted. During the following informal design reviews, the team received positive feedback on the redesign and with the approval of the sponsor, the design direction was verified.

Following the verification of the design direction, the team has worked on a detailed CAD model and calculations based on selected materials to verify that the design meets the critical specifications. A manufacturing plan and drawing package was created to prepare for the construction of a prototype that will continue to be tested. This final design includes key features that meet the required specifications and allow for a seamless manufacturing process, and is shown below in Figure 1.

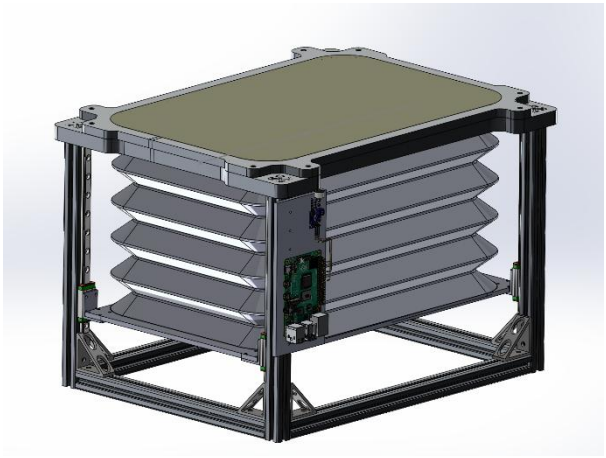


Figure 1. Final Acoustic Membrane Damper Assembly Render.

The purpose of this report is to present the final design solution to the proposed problem and outline its key features, justify how it meets the design specifications through analysis, and describe the future plans for verifying the prototype and manufacturing. With this in mind, the report has key sections including Concept Description, Design Justification, Manufacturing Plan, Design Verification, and Budget. Additional appendices include design analysis, a Gantt chart detailing the project timeline, information on the structural prototype, a detailed manufacturing and assembly plan, a design hazard checklist, the complete project budget, and failure modes and effects analysis (FMEA). These detail and justify the methods by which progress on this project has been made and outline future work.

This team seeks final approval from Dr. John Chen to proceed with prototype manufacturing and testing under the current design direction. If this design is approved, development and analysis will continue in anticipation of the Final Design Review (FDR) on May 29. The final design report will present a verification prototype and results from testing in detail. The estimated timeline for the project can be found in the Gantt Chart displayed in Appendix F.

Concept Description

The concept proposed is a tuned membrane absorber that will be able to absorb and reduce specific frequencies within a targeted range. Tuned membrane absorbers function by using a flexible membrane, acting as a spring at ambient pressure, that effectively reduces sound by achieving resonance with incoming sound waves, usually at low frequencies. To target a range of frequencies, existing versions of the tuned membrane absorber must be modified by adding an adjustable plate, allowing it to target multiple frequencies by changing its configuration.

The functionality of the tuned membrane absorber is mathematically supported by a lumped-parameter acoustic model that treats the membrane-cavity system as a mass-spring resonator. This model provides an analytical method of determining the necessary cavity depth to suppress a given incident frequency in the target range of 60-200 Hz.

Lumped Mass-Spring Representation

The system consists of:

- A flexible membrane with a surface mass density, $m_s(\frac{\text{kg}}{\text{m}^2})$
- A sealed air cavity of depth, $d_m(m)$
- Ambient air at pressure, P_0 (Pa)

When incident sound waves excite the membrane, it behaves like an oscillating mass. The sealed air cavity provides a restoring force due to compression of the trapped air. Under small oscillations, which can be assumed for sound waves, the pressure change in the cavity follows the relation:

$$\Delta P = -\gamma P_0 \frac{\Delta V}{V} \tag{1}$$

Where:

- $\gamma =$ ratio of specific heats for air (≈ 1.4)
- $V =$ cavity volume
- $\Delta V =$ change in cavity volume due to membrane displacement

For a membrane moving with displacement x over area A:

$$\Delta V = Ax \tag{2}$$

This results in an effective acoustic spring constant per unit area:

$$k' = \frac{\gamma P_0}{d} \tag{3}$$

The membrane surface mass density m_s acts as the oscillation mass per unit area. The system therefore behaves as a simple harmonic oscillator with natural frequency:

$$\omega_0 = \sqrt{\frac{k'}{m_s}} \tag{4}$$

Substituting for k':

$$\omega_0 = \sqrt{\frac{\gamma P_0}{m_s d}} \tag{5}$$

Converting frequency in Hz:

$$f_0 = \frac{1}{2\pi}\sqrt{\frac{\gamma P_0}{m_s d}} \tag{6}$$

This expression defines the resonance frequency of the membrane absorber.

Required Cavity Depth for Target Frequency and Material Selection

Rearranging Equation (6) provides a direct design relationship for cavity depth

$$d = \frac{\gamma P_0}{(2\pi f_0)^2 m_s} \tag{7}$$

This relationship shows that increasing membrane surface mass, which is the product of material density and thickness, decreases the required cavity depth. Due to the inverse square relationship, it can be ascertained that lower target frequencies require significantly larger cavity depths as depth varies proportionally with $\frac{1}{f^2}$.

Using the cavity depth expression, the MATLAB script, seen in Appendix A, calculates the required depth for a suppressible frequency within the frequency operating range. This enables active tuning of the absorber by adjusting the mounting plate position.

Based on the selected membrane material, the required depth to achieve optimal decibel reduction at a specific frequency varies. A denser material requires less depth to reduce a specific frequency, while a less dense material requires more depth. The bounds on the adjustability of the design have been set to a minimum of 0.05 meters when the accordion structure is fully compressed, and a maximum of 0.305 meters when the structure is fully outstretched. 3mm neoprene was chosen as the current membrane material, with the relationship between cavity depth and suppressible frequency range shown in the plot in Figure 2. According to the simulation, the design can suppress a range of 60-130 Hz at depths ranging from 0.05 m to 0.25 m. While research was done that supports the effectiveness of this material for the membrane, further material research is currently in progress in hopes that a more optimized material may be identified to utilize the full range of the design and reach the entire target range of frequencies.

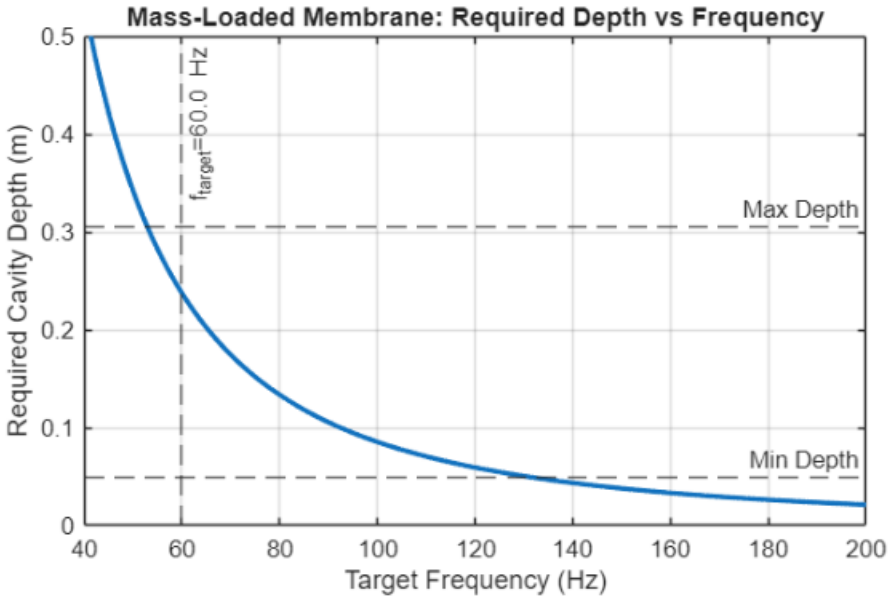


Figure 2. Plot of required depth in meters versus target frequency in Hz for 3mm Neoprene.

Acoustic Impedance and Absorption Coefficient

To estimate the percentage of incident sound energy absorbed, the system can be modeled using the surface acoustic impedance. The surface impedance of the membrane-cavity system is:

$$Z_s(\omega) = R_{loss} + j\omega m_s - j\frac{\gamma P_0}{\omega d} \tag{9}$$

where:

- R_{loss}= effective damping term
- $\omega = 2\pi f$

The normalized impedance is:

$$z = \frac{Z_s}{Z_0} \tag{10}$$

where:

- $Z_0 = \rho_0 c$ is the characteristic impedance of air.

The reflection coefficient at normal incidence is:

$$R = \frac{z-1}{z+1} \tag{11}$$

The absorption coefficient is then:

$$\alpha = 1 - |R|^2 \tag{12}$$

At resonance, the reactive terms cancel, and if the resistive term is tuned near Z_0, the absorber approaches maximum energy absorption.

Relevance to Adjustable Design

Because resonance depends on cavity depth d, adjusting the mounting plate directly changes the tuned frequency according to:

$$f \propto \frac{1}{\sqrt{d}} \tag{13}$$

Thus, by mechanically varying cavity depth, the system can shift its peak absorption across the desired frequency range. The mathematical relationship implemented in the control subsystem allows the actuator to position the mounting plate based on real-time frequency measurements.

This analytical framework confirms that the proposed adjustable membrane absorber can theoretically target and suppress the low-frequency bands through controlled geometric reconfiguration.

Physical Design Description

The system consists of three main subsystems: an acoustic barrier, a reverberation chamber, and a control system. The final assembly of this acoustic damper system is shown in Figure 3.

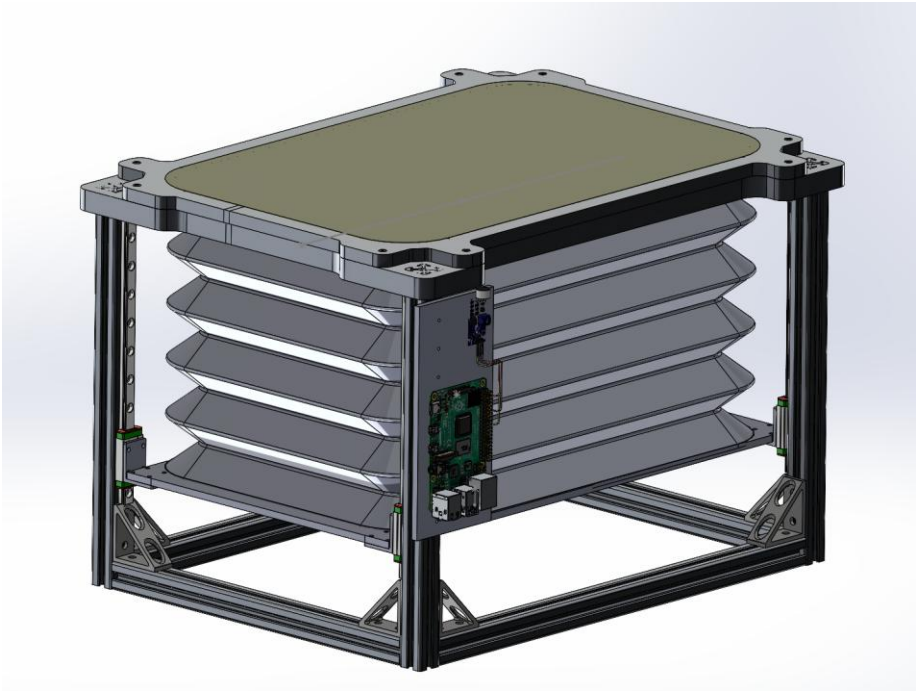
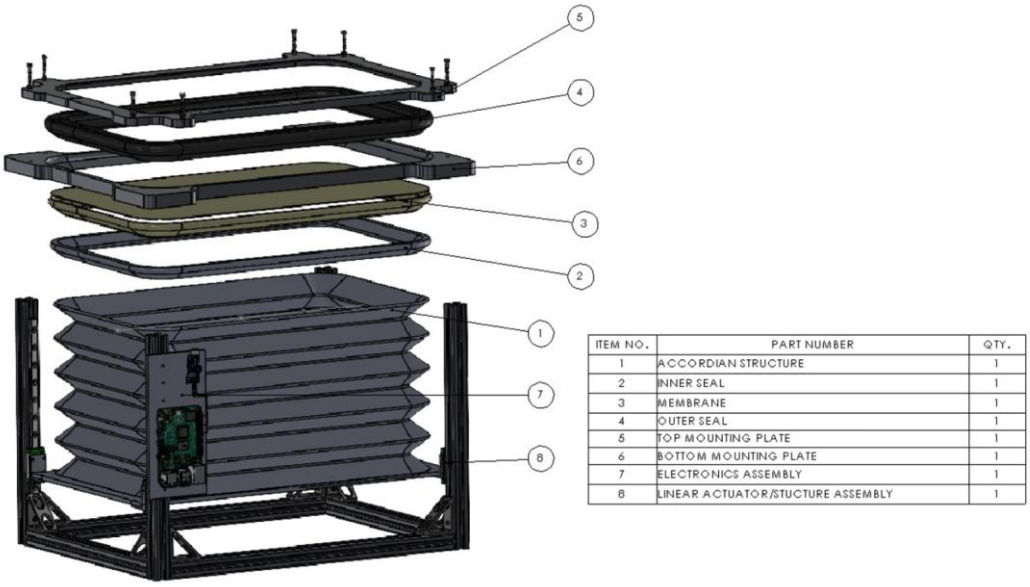


Figure 3. Acoustic damper CAD final assembly.

The acoustic barrier consists of a membrane material that will act in tandem with sealed air in the cavity to create a spring-like system that will absorb the vibration of emitted sound waves. The reverberation chamber houses the sealed air cavity and the mounting plate, which completes the pressure seal on the enclosed air cavity. The control system will facilitate movement of the mounting plate, which will be discussed in further detail below. An exploded view of the system with each subsystem separated is shown in Figure 4.



ITEM NO.	PART NUMBER	QTY.
1	ACCORDIAN STRUCTURE	1
2	INNER SEAL	1
3	MEMBRANE	1
4	OUTER SEAL	1
5	TOP MOUNTING PLATE	1
6	BOTTOM MOUNTING PLATE	1
7	ELECTRONICS ASSEMBLY	1
8	LINEAR ACTUATOR/STUCTURE ASSEMBLY	1

Figure 4. Acoustic damper assembly exploded view.

Based on the depth of the sealed air cavity behind the membrane, the membrane absorber will be able to suppress a specific select frequency. The target frequency range for this solution is 60-200 Hz as previously discussed. The proposed solution will have an adjustability factor for the mounting plate controlling the depth of the sealed air cavity behind the membrane, meaning that the solution will be able to target a range of frequencies based on its configuration. The mounting plate will be adjusted using a linear actuator in one corner of the frame, whereas the other corners will have linear rails to ensure a uniform movement of the entire plate. Additionally, the mounting plate will have a small enough orifice to allow the air cavity to slowly return to ambient pressure upon movement of the mounting plate but will maintain constant pressure during vibration.

To actuate the design, first the frequency with the highest decibel rating will be determined by using a frequency sensor in the control subsystem. A MATLAB calculation script to determine the corresponding reverberation chamber depth and linear actuator voltage will be coded and run on the Arduino based on the measured frequency. A plot of the required depth for each corresponding frequency is shown above in Figure 2.

A TPU printed housing was decided on for the reverberation chamber as it is most optimal in terms of balancing weight, cost, and effectiveness, shown in Figure 5. Additionally, since TPU is inherently flexible, slightly thinner material at the edges of the accordion structure will allow the design to bend in the desired way when the mounting plate is actuated.

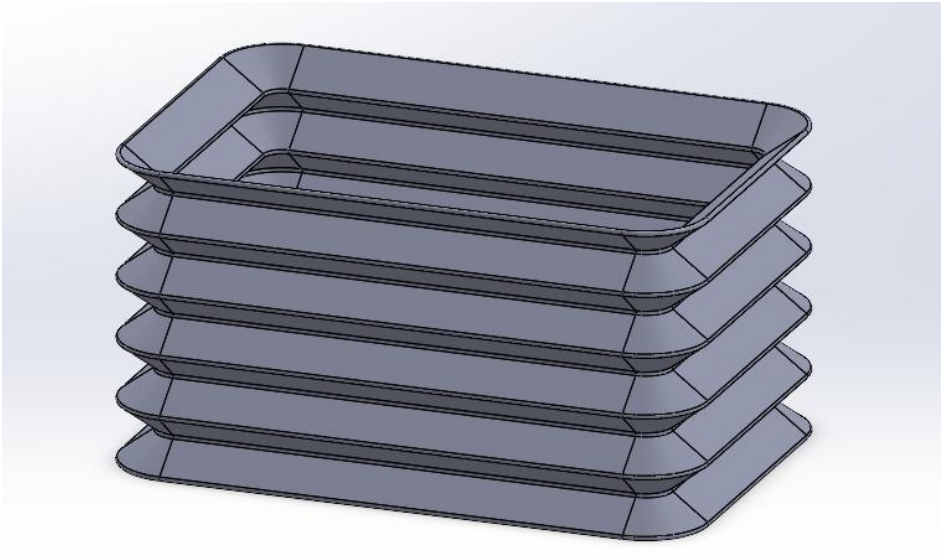


Figure 5. CAD Design of Accordion Housing Structure

In addition, the gasket sealing the acoustic barrier to the housing will is an accordion-like structure printed out of TPU, as its flexible nature combined with mechanical fasteners attaching the subsystem components together will be crucial for creating an effective pressure seal.

In terms of actuation, a linear actuator was selected that provides enough holding torque to fully actuate the mounting plate, while also staying under power consumption limits. The selected actuator has a holding torque of 270 oz-in, which would be more than enough to actuate the mounting plate, and an input power of 126 W, which is well beneath our consumption limit of 950 W. Because a single actuator had enough torque to fully actuate the plate, the team decided to use one actuator paired with three linear rails in the remaining corners of the housing to minimize cost, complexity, and ambient noise. The linear rails selected would be tall enough to span the actuation range, while also making sure they could handle the load of the mounting plate.

Many potential hazards with the tuned membrane absorber design have been considered, as listed in Appendix C. Identified hazards have corresponding corrective actions that are currently being implemented. Some of these corrective procedures will be with regards to the design, such as changing the material of the frame, whereas others will concern new features external to the design, such as providing additional covering around the module for astronaut protection. These procedures can also be found in Appendix C. Additionally, we have considered possible causes and solutions regarding maintenance of the design. The most likely cause of failure for the tuned membrane absorber would be due to fatigue in the linear actuator or exceeding the duty cycle for the part. To adapt to this potential failure, the linear actuator and control system are near the outside of the part, providing easy access to make replacements for worn-out parts. Since the membrane and frame move very minimally during operation, they will most likely not experience any short-term fatigue that is to be considered.

Design Justification

To ensure that the acoustic membrane damper meets sponsor requirements, five primary specifications were established. These specifications are summarized in Table 1, which outlines each requirement and the supporting justification. Specification 1 pertains to the main function of the device in terms of frequency damping. A frequency range of 60-200 Hz was selected based on analysis of noise conditions within the ISS living quarters [3]. This specification ensures that these low frequencies are mitigated to support astronaut health and communication. Specifications 2 and 3 apply to the integration of the design into NASA Artemis Human Landers. The total system weight must not exceed 1000 pounds so it does not interfere with the payload, and it cannot have a volume larger than 35 cubic feet so as not to interfere with the space astronauts need to live and work. Specification 4 is based on cost limits determined by the Mechanical Engineering Department for the senior project series. The project cannot exceed the allotted $1000 budget, which will be tracked using a bill of materials (Appendix D) and detailed bookkeeping.

To justify Specification 1 and ensure the design mitigated the proper frequency range, calculations and simulations were done in MATLAB using the depth of the reverberation chamber and the density of the membrane, as shown in Appendix A. This analysis predicts that this design with variable depth will dampen the specified frequency range, providing support for the selected design direction. However, prototype testing within this frequency range is required to experimentally verify the efficacy of the design. To justify the weight and volume specifications, the detailed CAD model and drawings were used. Drawing dimensions confirmed that the design was within the volume, and material assignment and mass estimations were used in SolidWorks to ensure the weight. Both specifications were well below their respective limits, providing confidence in the design meeting the integration requirements. After manufacturing, the design will need to be weighed and measured to provide physical values, but there is enough margin that it will meet the specifications if there is a slight discrepancy.

Table 1. Justification that our design will meet each of our specifications.

#	Spec	Specification Description	Justification
1	Acoustic	The acoustic membrane damper must decrease frequencies within the range of 60 to 200 Hz	Calculated frequency damping of 60 and 200 Hz. See frequency calculations in Appendix A.
2	Weight	The acoustic membrane damper's weight must not interfere with payload and should be <1000 lb.	Weight estimated from CAD model is predicted to be 5 lb. It is estimated that 10 panels will be used in the living space, bringing the final weight to 50 lb. The breakdown can be found further detail in Appendix E.

#	Spec	Specification Description	Justification
3	Volume	The acoustic membrane dampener must not take up more than 35 cubic feet of space throughout the lander.	Each acoustic membrane is estimated to be 1.35 cubic feet from the CAD model. 10 panels bring the final volume to 15.3 cubic feet of space.
4	Power	The acoustic membrane damper must not put a strain on the lander's electrical system and should not draw more than 0.95 kW of power.	The motor controlling the actuator will draw a peak of 126 W and the microcontroller will draw approximately 1 W depending on the current mode of operation.
5	Cost	The project must not exceed the designated budget of $1000.	Estimated project cost is $547.09. See the Project Budget in Appendix D for further breakdown of project.

Manufacturing Plan

To keep the project on track in both cost and schedule, the fabrication approach and supplier choices for every part of the prototype was mapped out. These details are grouped by subsystem and summarized in Table 2. For each component, the table indicates whether it will be purchased (P) and used as is, purchased then altered by the team (M), or fabricated in house (B). When a part requires modification or in-house fabrication, Table 2 also notes the necessary raw materials, where they will be obtained, and the key equipment and operations required for manufacturing and assembly.

Table 2. Manufacturing summary.

Subsystem	Component	Purchase (P) Modify (M) Build (B)	Raw Materials Needed	Where/ how procured?	Equipment and Operations
Acoustic Barrier	1.1	P	N/A	Amazon	N/A
	1.2	B	TPU	Bambu Lab Store	Bambu 3D Printer – STL to print
	1.3	B	N/A	McMaster	N/A
Reverberation Chamber	2.1	B	TPU	Bambu Lab Store	Bambu 3D Printer – STL to printer

Subsystem	Component	Purchase (P) Modify (M) Build (B)	Raw Materials Needed	Where/ how procured?	Equipment and Operations
	2.2	B	N/A	McMaster	All other parts assemble onto this plate
	2.3	B	N/A	McMaster	Rubber mounts are used to assemble most other parts onto the mounting plate
Control and Actuation	3.1	P	N/A	Amazon	N/A
	3.2	P	N/A	Amazon	N/A
	3.3	P	N/A	Amazon	N/A
	3.4	P	N/A	Amazon	N/A
	3.5	P	N/A	Amazon	N/A
	3.6	P	N/A	Amazon	N/A
	3.7	P	N/A	Amazon	N/A

The protype was intentionally developed using low complexity manufacturing processes to minimize fabrication risk and schedule uncertainty. As outlined in the Manufacturing Plan, custom components are fabricated using TPU 3D printing to enable use of complex geometry and flexibility, while standardized components and hardware are sourced through Amazon and McMaster to minimize lead time and complexity. Fit and sealing performance are verified through dimensional tolerance checks between the gasket, main frame, and mounting plate prior to final assembly. Post processing operations ensure proper mating surface quality and sealing integrity. Final integration is completed by mounting chambers and control components to the plate using rubber mechanical fastener, then sealing the acoustic barrier to the top of the chamber to form the complete system. See Appendix G for greater details.

Design Verification

Testing of the membrane acoustic dampener will primarily evaluate acoustic performance metrics regarding tonal reduction at key frequency bands, general sound pressure reduction, and mechanical durability. Verification testing will begin with static sized dampers as proof of concept, followed by the testing of the adjustable prototype to determine optimal membrane absorber size. As detailed in Table 3, the Design Verification Plan (DVP) includes functional acoustic testing as well as membrane durability and mounting lifespan validation to ensure engineering rigor. Table 3 summarizes the formal

DVP, including test descriptions, required equipment, responsible team members, and scheduled start dates. These start dates can be found in the Gantt chart shown in Appendix F.

Table 3. Design Verification Plan

Project:		Sound Suppression	Sponsor:	Dr. Chen			Date:	2/5/26
TEST PLAN								
Test #	Spec #	Test Description	Measurements	Acceptance Criteria	Facilities/ Equipment	Parts Needed	Member	TIMING Start date
1	1	Low Frequency Reduction	dB, Hz	NC40, dB < 60, 60-200 Hz	SLM, quiet room, testing fan set up	Fan, Motor, Prototype	Lucas	2/17
2	1,2,3	Pressure	(kilo)Pascal	Pressure differential from ATM in cavity	Digital Media Gauge, quiet room	Prototype	Jonathan	2/19
3	2,3,5	Mechanical Durability	Temperature, cycles, dB, Hz	Low detuning under stress	Quiet room, heat lamp, COMSOL	Prototype	Julia	2/17
4	2	Mounting Safety	Cycles	Mounting designed for infinite life	COMSOL	CAD	Michael	2/14

Budget

The team was given a budget of $1,000 to create a verification prototype of the full assembly. This assembly consists of three subsystems, as shown in Table 2: Acoustic Barrier, Reverberation Chamber, and Controls and Actuation. The Acoustic Barrier subsystem includes three components – the neoprene membrane, the gasket, and ¼-20 nylon fasteners, all of which require purchasing components or materials for manufacturing. The Reverberation Chamber subsystem has three components as well – the housing walls, housing base, and the rubber mounting system. The walls and base will be manufactured by the team using TPU filament, and the rubber mounting system uses commercially available parts designed for minimizing acoustic vibrations on personal computer fan systems. The Controls and Actuation subsystem has seven components, all of which are purchased – the linear actuator, the acoustic sensor, the microcontroller, the PCB, the T slot framing, the linear rails, and a 12 V battery. All component vendors, internal and external part numbers, and total costs are shown in a budget summary below in Table 4.

Table 4. Budget summary.

Item Description	Vendor	Vendor's Part #	iBOM #	Total Cost	Purchase Status
Neoprene membrane	Amazon	B0D81ZMWTV	1.1	$10.99	Not purchased
TPU for the gasket, housing walls, and housing base	Bambu Labs	U00-K0-1.75-1000-SPL	1.2, 2.1, 2.2	$41.99	Not purchased
¼-20 nylon fasteners	McMaster-Carr	94613A542	1.3	$12.59	Not purchased
Rubber mounting fasteners	QuietPCUSA	AFM02B	2.3	$8.88	Not purchased
Linear actuators	Amazon	B0BYDDJMF7	3.1	$402.40	Not purchased
Acoustic sensors	Amazon	B0CCF5R7CR	3.2	$7.99	Not purchased
Microcontroller	Amazon	B008GRTSV6	3.3	$20.70	Not purchased
PCB	Amazon	B008GRTSV6	3.4	$8.99	Not purchased
T slot framing	McMaster-Carr	47065T101	3.5	$11.67	Not purchased
Linear rails	Amazon	H21DSN9EWJ	3.6	$7.99	Not purchased
12 V battery	Amazon	B0D9D5L3B5	3.7	$20.89	Not purchased
Total Budget: **Amount Spent:** **Planned Spending:**	$1,000.00 $0.00 $547.09				
Remainder:	$452.91				

References

[1] *Noise Criterion (NC) Levels: Definitions, Standards & Calculator*, www.engineeringtoolbox.com/nc-noise-criterion-d_725.html. Accessed 11 Feb. 2026.

[2] Cayer, Emmi, et al. Noise Suppression System for Artemis Human Lander. Mechanical Engineering Department California Polytechnic State University, Nov. 2025.

[3] Senior Member, AIAA; ISS Acoustics Lead *Boeing Environments, J. Huntsville, and Goodman, "International Space Station Acoustics - A Status Report," 2003. Available: https://ntrs.nasa.gov/api/citations/20100039608/downloads/20100039608.pdf

Appendix A: Design Analysis

A MATLAB script was developed to analytical model and size the tunable membrane absorber system. The script implements the lumped mass-spring resonance equation for a membrane backed by a sealed air cavity to determine the required cavity depth for a specified target frequency within the 60-200 Hz range. Using material properties (neoprene density and selected thickness), the code calculates membrane surface mass density and corresponding cavity depth limits based on the physical constraints of the adjustable mounting plate. The script also generates a depth versus frequency plot to visualize actuator requirements across the operating band. In addition to geometric sizing, the model incorporates an acoustic impedance formula to estimate the theoretical absorption coefficient $\alpha(f)$. This is computed using the surface impedance of the membrane-cavity system and the reflection coefficient relative to the air's characteristic impedance. The resulting absorption curve provides a prediction of resonance behavior and validates that depth adjustment shifts the peak suppression frequency as intended. The script serves as a parametric design tool to support actuator sizing, material selection, and theoretical performance prior to fabrication and testing.

```
%% Tunable Neoprene Membrane Absorber (60–200 Hz) + alpha(f)
% Model: lumped, pistonic membrane backed by a sealed air cavity.
%
% Tuning relation:
%   f0 = (1/(2*pi)) * sqrt( (gamma*P0) / (m_s*d) )
%   d(f) = (gamma*P0) / ( (2*pi*f)^2 * m_s )
%
% Surface impedance (normal incidence):
%   Zs(omega) = R_loss + j*omega*m_s  -  j*(gamma*P0)/(omega*d)
% Reflection coefficient:
%   R = (z - 1)/(z + 1),  z = Zs/Z0,  Z0 = rho0*c
% Absorption coefficient:
%   alpha = 1 - |R|^2
%
% Notes:
% - This predicts resonance placement well for a limp, well-sealed membrane.
% - Peak absorption requires losses; R_loss approximates damping/leakage/resistive
layers.

clear; clc;

%% ---------------- DESIGN TARGETS ----------------

f_lo = 60;                      % Hz (lowest frequency of interest)
f_hi = 200;                     % Hz (highest frequency of interest)

%% ---------------- MEMBRANE (NEOPRENE) ----------------
% Use thickness as the main input since material availability is driving the choice.

rho_material = 1300;            % kg/m^3 (typical solid neoprene sheet)
t = 0.006;                      % m  (example: 6 mm neoprene). Change to match what
you can buy.
m_s = rho_material * t;         % kg/m^2 surface mass

%% ---------------- AIR / ACOUSTIC PROPERTIES ----------------

gamma = 1.4;
P0    = 101325;                 % Pa

rho0  = 1.21;                   % kg/m^3
c     = 343;                    % m/s
Z0    = rho0 * c;               % Pa*s/m

%% ---------------- GEOMETRY / ACTUATION LIMITS ----------------

d_min = 0.05;                   % m
d_max = 0.305;                  % m

A = 0.36;                       % m^2 (example build area)
aspect_ratio = 1.0;             % X/Y

X = sqrt(A * aspect_ratio);
```

```
Y = A / X;

total_mass = m_s * A;

%% ---------------- DEPTH REQUIRED AT ENDPOINT FREQUENCIES ----------------

d_req_lo = (gamma * P0) / ( (2*pi*f_lo)^2 * m_s );
d_req_hi = (gamma * P0) / ( (2*pi*f_hi)^2 * m_s );

% Simple guidance messages (direction is important!)
if d_req_lo > d_max
    warning(['At %.0f Hz, required depth is %.3f m (> d_max). ' ...
             'To reduce required depth: increase membrane surface mass (thicker /
add mass) or raise the minimum frequency.'], ...
            f_lo, d_req_lo);
elseif d_req_lo < d_min
    warning(['At %.0f Hz, required depth is %.3f m (< d_min). ' ...
             'To increase required depth: reduce membrane surface mass or lower the
minimum frequency.'], ...
            f_lo, d_req_lo);
end

%% ---------------- SUMMARY OUTPUT ----------------

clc;

fprintf('\n');
fprintf('================================================\n');
```

```
================================================
```

```
fprintf('  Tunable Neoprene Membrane Absorber Summary\n');
```

```
  Tunable Neoprene Membrane Absorber Summary
```

```
fprintf('================================================\n');
```

```
================================================
```

```
fprintf('Frequency band:          %.0f - %.0f Hz\n', f_lo, f_hi);
```

```
Frequency band:          60 - 200 Hz
```

```
fprintf('Neoprene density:        %.0f kg/m^3\n', rho_material);
```

```
Neoprene density:        1300 kg/m^3
```

```
fprintf('Membrane thickness:      %.2f mm\n', 1000*t);
```

```
Membrane thickness:      6.00 mm
```

```
fprintf('Surface mass (m_s):      %.3f kg/m^2\n', m_s);
```

```
Surface mass (m_s):      7.800 kg/m^2
```

```
fprintf('\nPanel area:              %.3f m^2\n', A);
```

```
Panel area:             0.360 m^2
```

```
fprintf('Panel dimensions:        %.3f m x %.3f m\n', X, Y);
```

```
Panel dimensions:       0.600 m x 0.600 m
```

```
fprintf('Total membrane mass:     %.3f kg\n', total_mass);
```

```
Total membrane mass:    2.808 kg
```

```
fprintf('\nRequired depth @ %.0f Hz: %.3f m\n', f_lo, d_req_lo);
```

```
Required depth @ 60 Hz: 0.128 m
```

```
fprintf('Required depth @ %.0f Hz: %.3f m\n', f_hi, d_req_hi);
```

```
Required depth @ 200 Hz: 0.012 m
```

```
fprintf('Available depth range:    %.3f – %.3f m\n', d_min, d_max);
```

```
Available depth range:    0.050 – 0.305 m
```

```
fprintf('================================================\n\n');
```

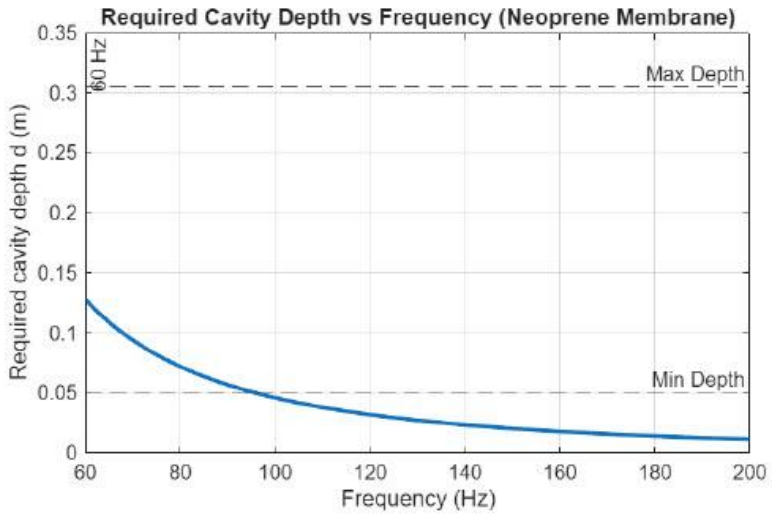
```
================================================
```

```
%% ---------------- DEPTH VS FREQUENCY CURVE ----------------

N = 500;
f = linspace(f_lo, f_hi, N);
omega = 2*pi*f;

d_vs_f = (gamma * P0) ./ (omega.^2 * m_s);

figure;
plot(f, d_vs_f, 'LineWidth', 2);
grid on;
xlabel('Frequency (Hz)');
ylabel('Required cavity depth d (m)');
title('Required Cavity Depth vs Frequency (Neoprene Membrane)');
hold on;
yline(d_min, '--', 'Min Depth');
yline(d_max, '--', 'Max Depth');
xline(f_lo, '--', sprintf('%.0f Hz', f_lo));
xline(f_hi, '--', sprintf('%.0f Hz', f_hi));
hold off;
```




```
%% ---------------- ABSORPTION CURVE alpha(f) ----------------
% Choose a design depth to evaluate alpha(f).
% Common choices:
%   - "tuned at the low end": use d_design = d_req_lo (if feasible)
%   - "max depth available": use d_design = d_max (best shot at low frequency)
%
% Here: evaluate at the depth that would tune exactly to f_lo (and also clamped).

d_design = d_req_lo;                    % depth that tunes to the low-end frequency
d_design_clamped = min(max(d_design, d_min), d_max);

% Loss term: tune this to match expected damping (edge leakage, resistive layer,
etc.)
% R_loss ~ Z0 gives a reasonable "critically coupled" starting point near resonance.
R_loss = Z0;

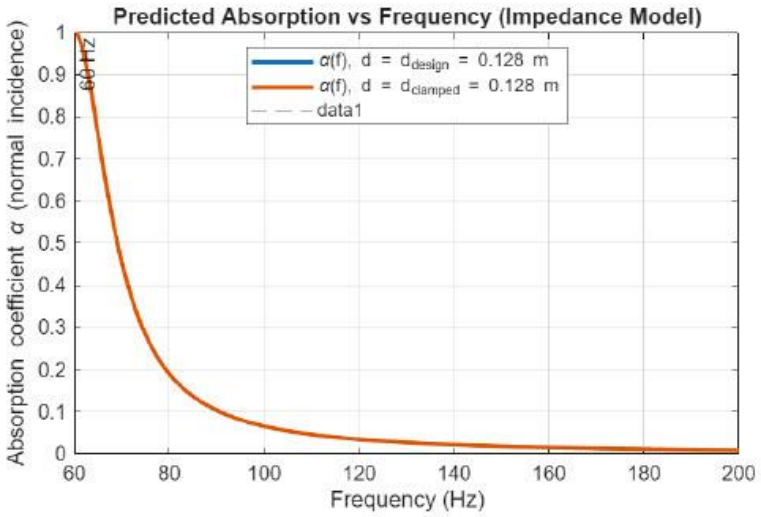
Zs_design = R_loss + 1i*omega*m_s - 1i*(gamma*P0)./(omega*d_design);
z_design  = Zs_design / Z0;
R_design  = (z_design - 1) ./ (z_design + 1);
alpha_design = 1 - abs(R_design).^2;

Zs_clamp = R_loss + 1i*omega*m_s - 1i*(gamma*P0)./(omega*d_design_clamped);
z_clamp  = Zs_clamp / Z0;
R_clamp  = (z_clamp - 1) ./ (z_clamp + 1);
alpha_clamped = 1 - abs(R_clamp).^2;

% Keep alpha within [0,1] (numerical roundoff protection)
alpha_design  = min(max(real(alpha_design),  0), 1);
alpha_clamped = min(max(real(alpha_clamped), 0), 1);

figure;
```

```
plot(f, alpha_design, 'LineWidth', 2); hold on;
plot(f, alpha_clamped, 'LineWidth', 2);
grid on;
xlabel('Frequency (Hz)');
ylabel('Absorption coefficient \alpha (normal incidence)');
title('Predicted Absorption vs Frequency (Impedance Model)');
legend(sprintf('\\alpha(f), d = d_{design} = %.3f m', d_design), ...
       sprintf('\\alpha(f), d = d_{clamped} = %.3f m', d_design_clamped), ...
       'Location','best');
xline(f_lo, '--', sprintf('%.0f Hz', f_lo));
hold off;
```




```
fprintf('\nAbsorption model settings:\n');
```

```
Absorption model settings:
```

```
fprintf('  R_loss = %.1f Pa*s/m (Z0 = %.1f Pa*s/m)\n', R_loss, Z0);
```

```
  R_loss = 415.0 Pa*s/m (Z0 = 415.0 Pa*s/m)
```

```
fprintf('  d_design = %.3f m, d_clamped = %.3f m\n', d_design, d_design_clamped);
```

```
  d_design = 0.128 m, d_clamped = 0.128 m
```

Appendix B: FMEA

FMEA stands for Failure Mode and Effects Analysis. It is a tool used to predict what might break and how severe it would be. It connects the cause of the problem with the plan to catch it. The final score is called the RPN or Risk Priority Number, which tells what risks are the most dangerous and need to be fixed first. Table B.1 outlines the FMEA done on the design at present.

Table B.1 FMEA Table

System / Function	Potential Failure Mode	Potential Effects of the Failure Mode	Severity	Potential Causes of the Failure Mode	Current Preventative Activities	Occurence	Current Detection Activities	Detection	RPN	Recommended Action(s)
Acoustic Barrier	Membrane fails to resonate at target frequency	a.Noise levels remains unsafe ♦. crew discomfort and communication issues	8	a.incorrect membrane desnity b. membrane tension changes over time	a.MATLAB modeling to verify depth and desnity relationship b. Material property research	3	Acoucstic testing with microphone in quiet room	4	96	a. conduct fatigue analysis b. Verify mebrane material properties before assembly
Maintained Sealed Air Cavity	loss of pressure Seal	a.inability to tune b. acoustic damping fails completely	7	a.TPU 3D print defects b. Gasket misalignment c. Fasteners loosening due to vibrations	a.Tolerance check between gaskets and frame b.Flexible TPU gasket for better sealing	4	Pressure differential testing	3	84	Apply post-proccesing selant to TPU parts
Frames and Mounts	Unit detaches from wall	a.Injury to astronauts b. Damage to other lander equipments	9	a.Rubber mount shear failure under launch loads b. Fastener fatigue from vibratrions	a. Design mounting for infinite life b. use of rubber vibration damping mounts	2	Pull test	2	36	Perform mechanical durability testing
Controls and Actuation	Actuator fails to move or adjust to wrong depth	a.System cannot tuned to the dominant noise frequency b. suboptimal suppression	5	a.Battery power loss b. Microntroller error code c. Sensor malfunction d. Wiring short circuit	a.sleeve and reroute wiring to prevent breakage b.code simulation and debugging	4	Functional testing	3	60	a. Implement battery health monitoring b. secure all wire connections with strain relief
User safety	Sharp edges and protruding parts	a.cuts to astronauts b. snagging on space suits	6	a. rough 3D print edges b. expose aluminum extrusion ends	a. provide covering b. file down 3d prints	3	Visual and tactile inspections	1	18	add edge guards

Appendix C: Design Hazard Checklist

A design hazard checklist documents all anticipated hazards and potential safety risks of the product during the design phase. By evaluating the materials, manufacturing processes, and design configurations, it can be guaranteed that the final design is safe for use. This checklist serves as a record of the hazards that were identified during this design phase as shown in Table C.1. Corrective actions have been identified to adapt to these potential hazards, which can be seen in Table C.2 along with their planned integration dates.

Table C.1. Design Hazard Checklist.

Y	N	Category
Y		1. Will any part of the design create hazardous revolving, reciprocating, running, shearing, punching, pressing, squeezing, drawing, cutting, rolling, mixing or similar action, including pinch points and sheer points?
	N	2. Can any part of the design undergo high accelerations/decelerations?
	N	3. Will the system have any large moving masses or large forces?
	N	4. Will the system produce a projectile?
Y		5. Would it be possible for the system to fall under gravity creating injury?
Y		6. Will a user be exposed to overhanging weights as part of the design?
Y		7. Will the system have any sharp edges?
	N	8. Will any part of the electrical systems not be grounded?
	N	9. Will there be any large batteries or electrical voltage in the system above 40 V?
Y		10. Will there be any stored energy in the system such as batteries, flywheels, hanging weights or pressurized fluids?
	N	11. Will there be any explosive or flammable liquids, gases, or dust fuel as part of the system?
	N	12. Will the user of the design be required to exert any abnormal effort or physical posture during the use of the design?
	N	13. Will there be any materials known to be hazardous to humans involved in either the design or the manufacturing of the design?
	N	14. Can the system generate high levels of noise?
	N	15. Will the device/system be exposed to extreme environmental conditions such as fog, humidity, cold, high temperatures, etc.?
	N	16. Is it possible for the system to be used in an unsafe manner?
	N	17. Will there be any other potential hazards not listed above? If yes, please explain on reverse.

Table C.2. Design Hazard and Corresponding Corrective Action Timeline

Description of Hazard	**Planned Corrective Action**	**Planned Date**	**Actual Date**
Mounted system has potential to fall and cause injury	Design mounting for longer life using materials with higher modulus of elasticity. Suggest safer mounting arrangements.	2/15	-
Mounted system protrudes from wall creating overhang	Moldable mounting to allow for out-of-the-way mounting arrangements.	2/19	-
Sharp edges present on aluminum frame	Sand extrusions and provide covering around module	2/19	-
Stored energy in battery powering system	Sleeve and route wiring to prevent accidental breakages or expose live wires.	2/21	-

Appendix D: Drawing & Specification Package

The detailed budget table is shown in Table D.1. It specifies all costs associated with the project. Included is the cost of all material expenses for the parts that will be used in prototype manufacturing. The expenses on the detailed budget table remain below the allotted $1000. A detailed parts table can be seen in Table D.2. with particular part numbers and vendors outlined as well as descriptions of the key features of the selected materials. Figures D.1 – D.6 display drawings of our damper prototype.

Table D.1. Indented Bill of Materials

Level	Number	Descriptive Part Name	Qty	Part Cost	Source	URL	More Info
		Lvl0 Lvl1 Lvl2 Lvl3 Lvl4					
0	100000	Final Assy			------		
1	1	Acoustic Barrier			------		
1	1.1	Neoprene Membrane	1	$ 10.99	Amazon	Nylon_Roll_3mm	
1	1.2	Gasket	1	$ 41.99	Bambu Labs	Bambu-Labs-TPU	
1	1.3	Fasteners -1/4-20 Nylon	8	$ 12.59	McMaster	Nylon_Screws	1" length
2	2	Reverberation Chamber			-------		
2	2.1	Housing Walls	1		Bambu Labs	Bambu-Labs-TPU	Can use roll bought for gasket
2	2.2	Housing Base	1	$ 12.99	Amazon	Mounting PLate	
2	2.3	Rubber Mounting	4	$ 8.88	QuietPCUSA	Quiet_Rubber_Mounts	
3	3	Controls and Actuation			-------		
3	3.1	Linear Actuator	1	$402.38	Amazon	Linear_Actuator	
3	3.2	Acoustic Sensors	1	$ 7.99	Amazon	Acoustic Sensor	
3	3.3	Microcontroller	1	$ 20.70	Amazon	Microcontoller	Arduino for PID control
3	3.4	PCB	1	$ 8.99	Amazon	PCB	Will predesign then order
3	3.5	Vertical T-Slot Framing	4	$45.60	McMaster	Vertical Framing	
3	3.6	Short T-Slot Framing	2	$22.80	McMaster	Short Framing	
3	3.7	Long T-Slot Framing	2	$22.80	McMaster	Long Framing	
3	3.8	Framing Triangle Brackets	8	$74.40	McMaster	Framing Triangle Brackets	
3	3.9	Sliders and Linear Rails	4	$70.76	Amazon	Sliders and Linear Rails	
3	3.10	Socket Head Screws	8	$17.78	McMaster	Screws	
	3.11	Hex Nuts	8	$5.49	McMaster	Hex Nuts	
3	3.12	12 V Battery	1	$ 20.89	Amazon	Battery	
	Total Parts		**57**	**$ 808.02**			

Table D.2. Design Parts Table

Part Number	Part Name	Vendor	Vendor Part Number	Price	Manuf.	Manuf. Part Number	Key Features/Specs	Webpage
1.1	Neoprene	Amazon	-	10.99	Storystore	B0D81ZMWTV	Compression sealing	Nylon_Roll_3mm
1.2	Gasket	Bambu Labs	U00-K0-1.75-1000-SPL	41.99	Tuozhu Technology Limited	U00-K0-1.75-1000-SPL	Breaking elongation rate of > 650	Bambu-Labs-TPU
1.3	Fasteners -1/4-20 Nylon	McMaster	94613A542	12.59	McMaster-Carr		1" Length	Nylon_Screws
2.1	Housing Walls	Bambu Labs	U00-K0-1.75-1000-SPL	41.99	Tuozhu Technology Limited	U00-K0-1.75-1000-SPL	Breaking elongation rate of > 650	Bambu-Labs-TPU
2.2	Housing Base	Bambu Labs	U00-K0-1.75-1000-SPL	41.99	Tuozhu Technology Limited	U00-K0-1.75-1000-SPL	Breaking elongation rate of > 650	Bambu-Labs-TPU
2.3	Rubber Mounting	QuietPCUSA	AFM02B	8.88	Quiet PC USA	AFM02B	Vibration Dampening	Quiet_Rubber_Mounts
3.1	Linear Actuator	Amazon	-	402.38	OITTo	B0BYDDJMF7	Smooth operation, low noise	Linear_Actuator
3.2	Acoustic Sensors	Amazon	-	7.99	WWZMDiB	B0CCF5R7CR	Compatible with Arduino,	Acoustic Sensor
3.3	Microcontroller	Amazon	-	20.7	ARDUINO	B008GRTSV6	Reliable, widely used	Microcontoller
3.4	PCB	Amazon	-	8.99	EPZLON	B09X1D6NPP	3.5x2.05x2.05", sturdy for our desig	PCB
3.5	Vertical T-Slot Framin	McMaster	47065T101	$45.60	McMaster-Carr	47065T101	1" width, comes in varying heights for our assembly. 6063 Aluminum for strength and rigitity, smooth rail texture	Vertical Framing
3.6	Short T-Slot Framing	McMaster	5537T101	$22.80	McMaster-Carr	5537T101	1" width, comes in varying heights for our assembly. 6063 Aluminum for strength and rigitity, smooth rail texture	Short Framing
3.7	Long T-Slot Framing	McMaster	5537T101	$22.80	McMaster-Carr	5537T101	1" width, comes in varying heights for our assembly. 6063 Aluminum for strength and rigitity, smooth rail texture	Long Framing
3.8	Framing Triangle Bra	McMaster	5537T658	$74.40	McMaster-Carr	5537T658	6063 Aluminum	Framing Triangle Brackets
3.9	Sliders and Linear Ra	Amazon	MUNEFEq74fkt0gzy-03	$70.76	MUNEFE	MUNEFEq74fkt0gzy-03	Linear Sliding Rail, 200 mm	Sliders and Linear Rails
3.10	Socket Head Screws	McMaster	91290A125	$17.78	McMaster-Carr	91290A125	Hex shape fastener screws	Screws
3.11	Hex Nuts	McMaster	90695A033	$5.49	McMaster-Carr	90695A033	Hex shape fastener nuts	Hex Nuts
3.12	12 V Battery	Amazon	B0D9D5L3B5	$ 20.89	Rapthor	B0D9D5L3B5	2600 mAh with DC port 5521	Battery

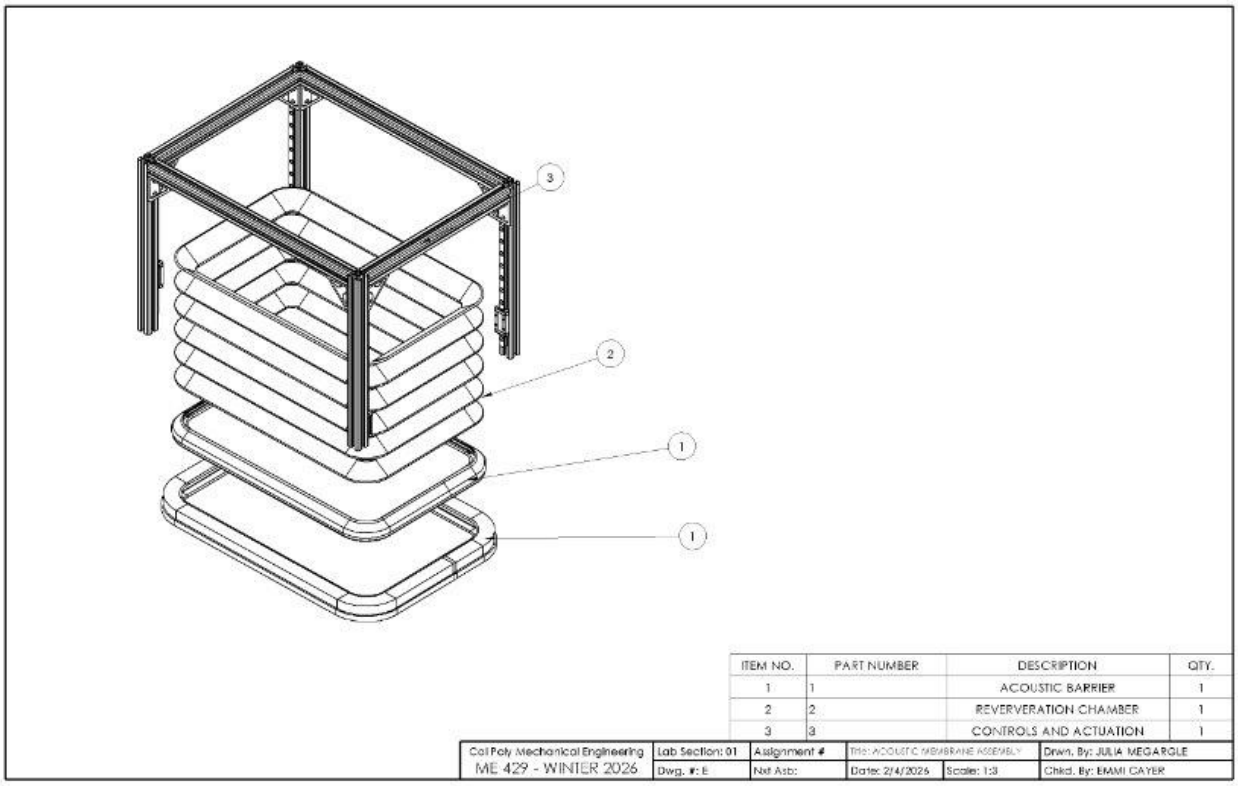


Figure D.1 Acoustic Membrane Assembly

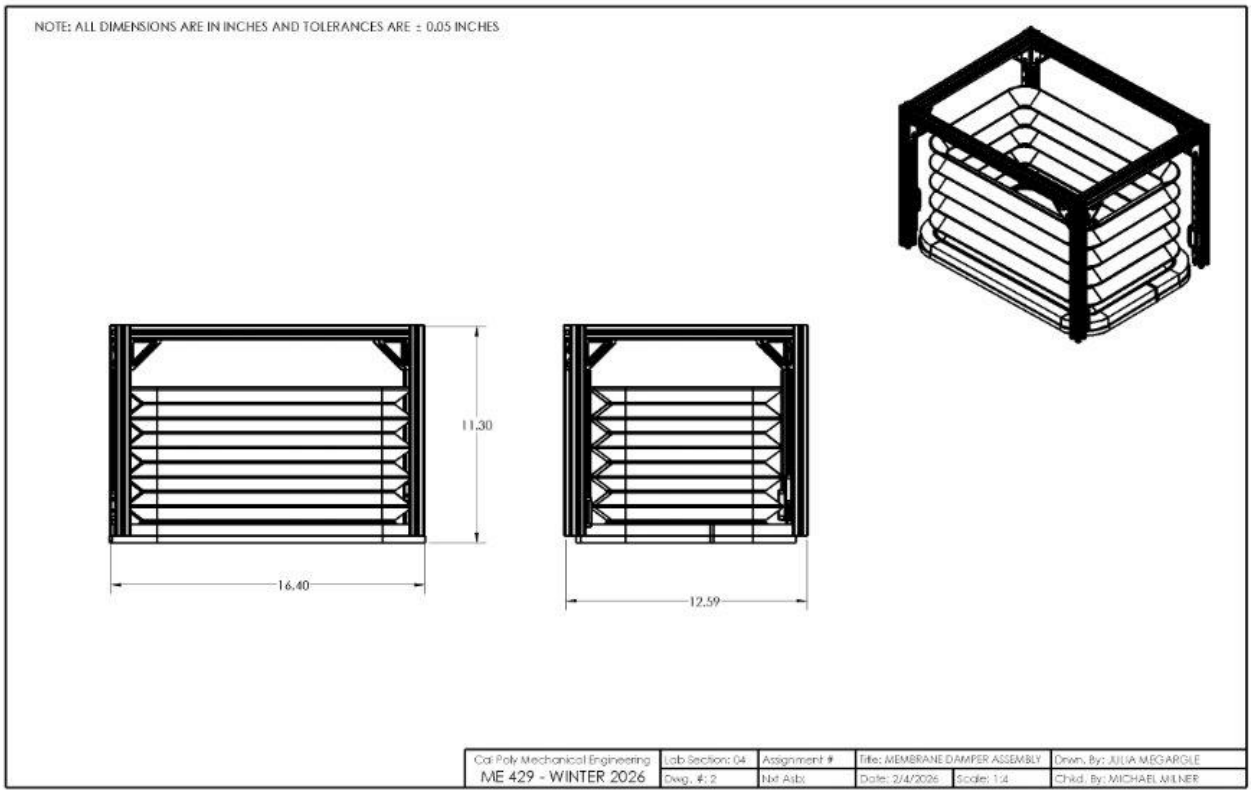


Figure D.2 Damper Assembly

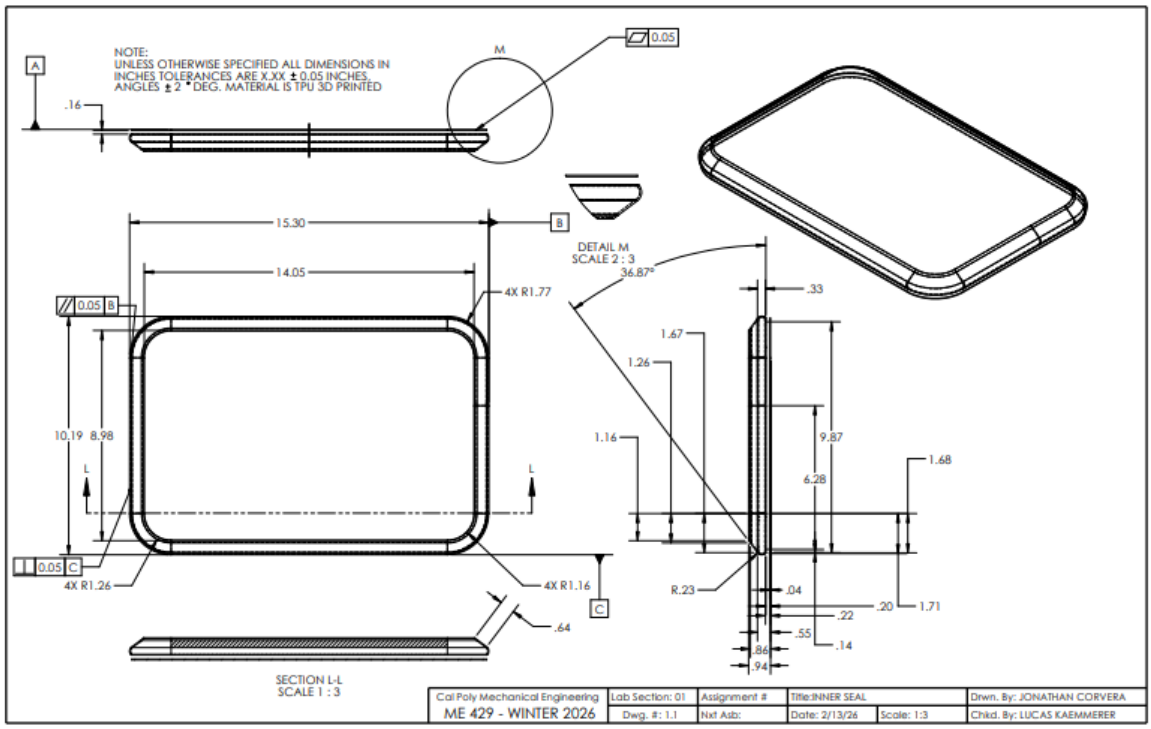


Figure D.3 Inner Seal

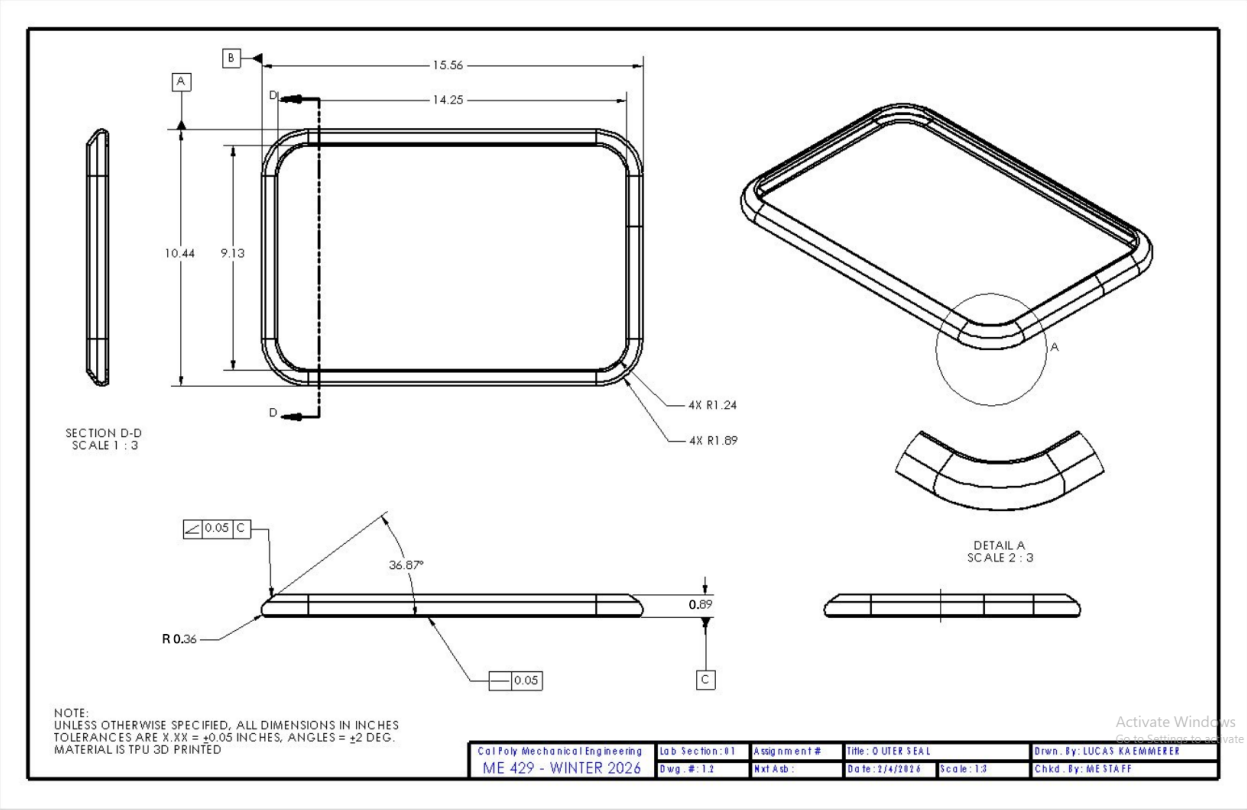


Figure D.4 Outer Seal

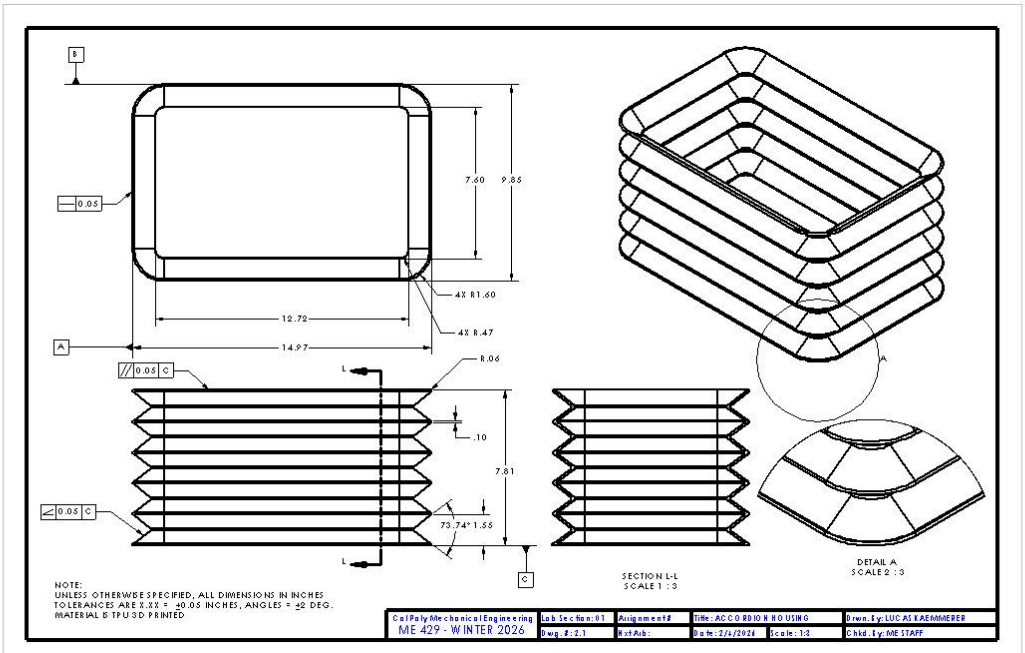


Figure D.5 Accordion Housing

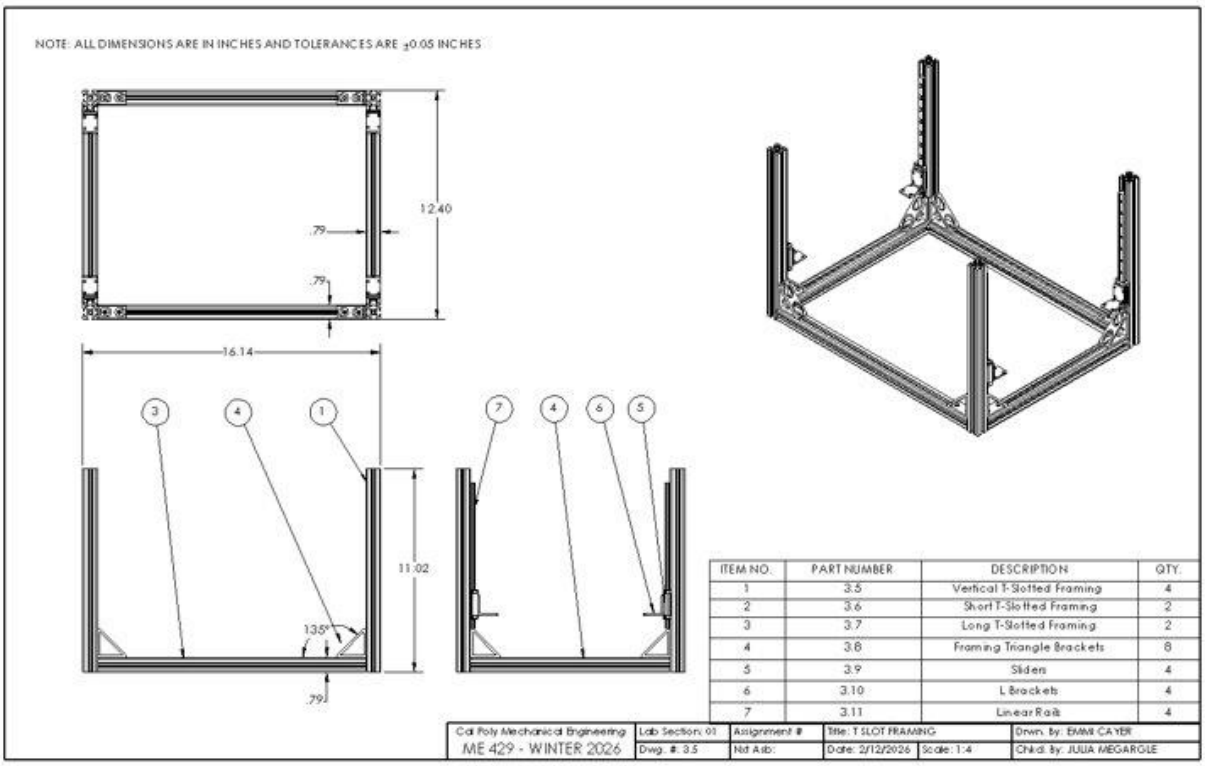


Figure D.6 T Slot Framing

Appendix E: Weight Estimation Table

The weight estimation table, Table E.1, lists subsystem weights, allowing for an accurate estimate of total device weight. By obtaining an accurate device weight, changes can be made to optimize other design criteria while still maintaining a device weight within our design constraints.

Table E.1. Weight Estimation table.

Subsystem	Subsystem Description	Subsystem Weight
1	**Acoustic barrier**, consists of the membrane, gasket and fasteners	.25 lbs
2	**Reverberation chamber,** consists of housing walls and base with rubber mounting	1.75 lb
3	**Controls and actuation,** consisting of structural components, linear actuation, a microcontroller, a PCB, and a battery power source	3 lb

WEIGHT OF 1 DEVICE	5 lb
WEIGHT OF SYSTEM (10 DEVICES)	50 lb

Appendix F: Project Timeline

For organization and planning, the team uses a Gantt chart through a website called “teamgantt”, shown in Figure F.1. This gives a clear timeline on a horizontal bar chart that visually maps the project and the tasks that need to be done. The bars on the chart indicate when the tasks should be started, how long they will take, and when they are due. Team members are assigned to each task, with names appearing next to the tasks that are assigned.

Gantt chart

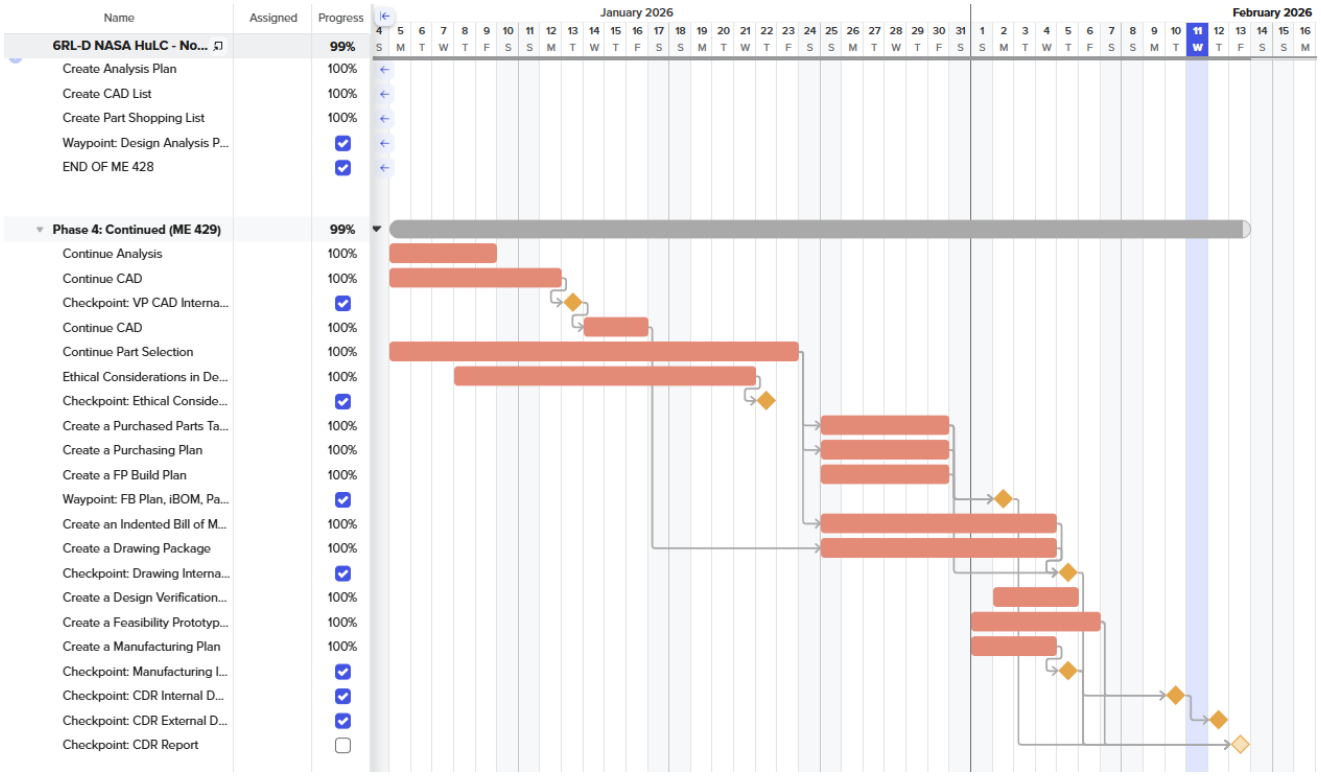


Figure F.1. Team Gantt chart, showing progress towards CDR.

Appendix G: Manufacturing and Assembly Steps

This Manufacturing and Assembly Plan serves as a build guide for the verification prototype, laying out how each part will be obtained, fabricated, and brought together inro the final system. It explains which will be ordered directly from the vendors and which items will be produced by the team using TPU 3D printing on Bambu Lab printers, progressing from STL files to finished prints. The plans also highlight the practical steps that make this assembly work smoothly, basic print cleanup. Checking fits between printed parts and mounting plate, and the order of operations for integrating the acoustic barrier and reverberation chamber. When Final installation is completed on the mounting plate using standard hardware and rubber mounting methos to secure component, while keeping the process simple and repeatable.

MANUFACTURING STEPS:

Subsystem 1: Acoustic Barrier

Membrane (1.1)

1. The membrane will be a 3mm neoprene rubber sheet that will be ordered online through Amazon.
2. This sheet will have an adhesive side that will allow it to be directly attached to the frame.
3. The sheet will be cut with scissors into a 14.05 in x 8.93 in rectangle.

Gasket (1.2)

1. The gasket will be made of TPU and 3D printed using Cal Poly's Makerspace Bambu 3D printers (***whereas the design sent to NASA will use fluorosilicone and will be printed on a resin printer).

a. An STL will be created using the CAD model and designated GD&T.

b. It will be exported and printed using the Cal Poly shop printers.

c. The tolerancing will be checked between the gasket (1.2) and main body frame (2.1) as a go-no-go test.

2. Trees and rough edges of our 3D printed part will be removed using files.
3. The gasket will be secured to the frame using mechanical fasteners (x8 1/4-20 nylon screws [1.3]).

Subsystem 2: Reverberation Chamber

Accordion Housing (2.1)

1. The main body frame accordion housing will be 3D printed with TPU using the Cal Poly Makerspace Bambu 3D printers (***whereas the design sent to NASA will use fluorosilicone and will be printed on a resin printer).

a. An STL will be created using the CAD model.

b. It will be exported and printed using the Cal Poly shop printers.

c. The tolerancing will be checked between the gasket (1.2), main body frame (2.1), and mounting plate (2.2).

2. Tree supports and rough edges of our 3D printed part will be removed using files.
3. The frame will be secured to the mounting plate using mechanical fasteners (rubber mounting).

Mounting plate (2.2)

1. The mounting plate will be purchased through Amazon to act as a base/pressure seal for the frame.
2. The plate will be attached to the frame using rubber mechanical fasteners (2.3).

Rubber mechanical fasteners (2.3)

1. The rubber mechanical fasteners will be used to mount the housing (2.1) and the actuators (3.1) onto the mounting plate.

Subsystem 3: Controls and Actuation:

Actuators

1. A linear actuator (3.1) will be purchased online through Amazon.
2. The actuator will be mounted to part (2.2) shown above with rubber mechanical fasteners (2.3).

Control System and Frame

1. Acoustic sensors (3.2) will be attached to the mounting plate (2.2) using rubber mechanical fasteners (2.3).
2. A simple microcontroller (3.3) will be utilized to turn the input from the sensors (3.3) into a voltage for the actuators.
3. A pre-designed PCB board (3.4) will be ordered and integrated into our controller design.
4. The controller will be mounted to the system using the mounting plate (2.2) and the mechanical fasteners (2.3).
5. Framing will be assembled together (3.5, 3.6, 3.7) using framing triangle brackets (3.8), socket head screws (3.10), and hex nuts (3.11).
6. Linear rails and sliders (3.9) will hold the linear actuators in place on the mounting plate (2.2) using mechanical fasteners (2.3).

ASSEMBLY STEPS

1. With a fully printed and cleaned gasket, the membrane will be attached to the gasket to fully assemble the Acoustic Barrier.
2. The frame and mounting plate will be attached using rubber mounts to fully assemble the Reverberation Chamber.
3. The actuator will be integrated with the control system to fully assemble the Controls and Actuation subsystem.
4. The control system and actuator will be attached to the mounting plate using rubber mounting, allowing adjustability of the sealed air pocket.

5. The acoustic barrier will be attached to the top of the Reverberation chamber, creating a pressure seal and forming the complete system.

MASS

Modular **A**coustic **S**uppression **S**ystem

Final Design Review

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June 4th, 2026

Project Overview

As humanity progresses in space exploration, astronauts are going to need to stay in space for longer durations of time. Currently, the Environmental Control and Life Support Systems (ECLSS) in the human lander are too loud for astronauts to healthily live and communicate aboard the lander. NASA (National Aeronautics and Space Administration) requires a method to suppress this noise within the Artemis human lander capsule. Therefore, a solution must be designed that dampens and eliminates a wide range of frequencies within the lander to provide astronauts with a safe and comfortable living environment. This solution must meet volume, weight, sound, and power requirements within the human lander vehicle. The design will include an adjustable acoustic membrane dampener that will target the lower frequency ranges generated by the ECLSS. The project is being carried out by a team of mechanical engineering students at California Polytechnic State University, San Luis Obispo (Cal Poly), including Emmi Cayer, Gabriel Choi, Jonathan Corvera, Lucas Kaemmerer, Julia Megargle, and Michael Milner. This project is sponsored by Dr. John Chen, with funding provided by Cal Poly.

Design Description

The system consists of three main subsystems: a membrane cartridge, a reverberation chamber, and a control system. The membrane will act in tandem with sealed air in the cavity to create a spring-like system that will absorb the vibration of emitted sound waves [3]. The control system will facilitate vertical movement of the bottom mounting plate, which is attached to the bottom of the accordion pressure cavity, thus enabling the ability to change the distance between the membrane and the bottom of the accordion. Based on the depth of the sealed air cavity behind the membrane, the membrane absorber will be able to suppress a specific select frequency.

This verification prototype represents the CDR design by demonstrating electronic verification, pressure equalization, and structural integrity though launch vibration. These criteria are described in detail int eh design verification section.

Design Image

The MASS team created two designs: one for NASA's HuLC and one that could be completed as a verification prototype with the resources the team had on hand. The first two figures are of the design for NASA, and feature membrane cartridges with a sealed cavity. The final figure is the verification prototype design, which has a similar membrane and cavity design, but its focus was to test for linear actuator and electronic verification, pressure equalization verification,

Design Verification

Three tests were conducted on a feasibility prototype of MASS: a linear actuator and electronic verification test, a pressure equalization verification test, and a vibration test. The linear actuator and electronic verification test performed a sweep for frequencies ranging from 60-250 Hz and was proven to accurately actuate the system. The pressure equalization verification test shifted the chamber depth three inches to verify pressure

and vibration during a launch.

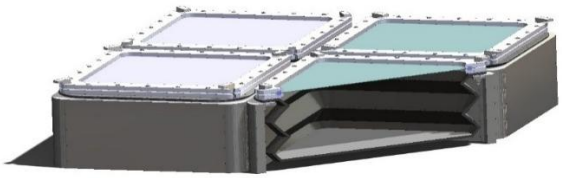


Figure 1. Acoustic Damper CAD Final Assembly.

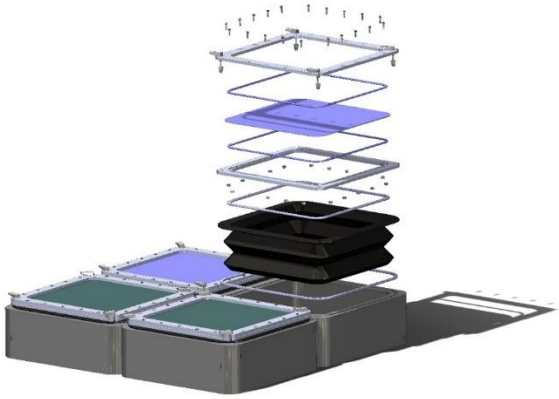


Figure 2. Acoustic damper assembly exploded view.



Figure 6. Feasibility prototype CAD model.

equalization at varying depths throughout five separate tests. The chamber pressure successfully equalized after an average of 4.49 seconds. The vibration test was conducted by mounting the system to a shake table and subjecting it to a random vibration profile sweep from 20-2000 Hz. The final vibration profile reached an RMS acceleration of approximately 10.3 g, and the schedules test completed successfully without premature shutdown.

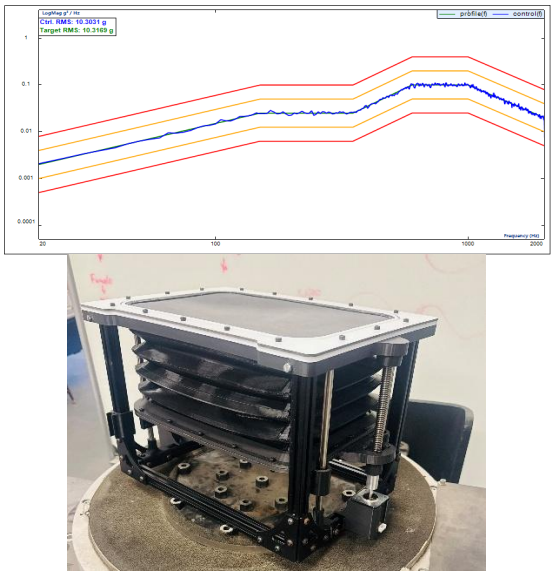


Figure 7. Testing apparatus and decibels versus frequency graph for the vibration test.

The test was used to evaluate structural stability rather than acoustic performance. The prototype held stable during the duration of the test, with the only issue being loose hex nuts after the vibration test. It was determined that lock-nuts would solve this issue; this change was reflected in our budget. Future testing should repeat the procedure using final flight material and include posttest inspection of loosening, deformation, actuator misalignment, membrane damage, and seal degradation.

Figure 0. Four-Panel Chart for Final Design Review

Overview

MASS (Modular Acoustic Suppression System) is a mass tuned damper system that reverberates a membrane atop a cavity to absorb select frequencies produced by the Environmental Control and Life Support Systems (ECLSS) upon NASA's Artemis Human Lander. The system is adjusted via lead screws to target specific frequencies corresponding to different chamber heights.

Under the classical membrane-absorber model, incident acoustic waves coming into contact with the sealed membrane chamber create small oscillations similar to a spring-mass damper system that act as an acoustic spring to dissipate sonic energy [3]. For a constant cross-sectional area, modulating the chamber depth changes the corresponding tuned frequency according to the relationship derived in Eq. (7), where it was found that lower target frequencies require larger cavity depths. This enables active tuning of the absorber by adjusting the vertical position of the mounting plate.

As humanity progresses further in space exploration, astronauts will need to stay in space for longer durations of time [13, 15]. Currently, the ECLSS in the human lander produces sound outside the specified Noise Criterion levels [1,8] that inhibits astronauts' ability to safely live and communicate aboard the lander, with low frequencies affecting concentration and sleep [2, 8, 11]. NASA needs a way to suppress this noise on the Artemis Human Lander capsule.

The California Polytechnic State University (Cal Poly) NASA HuLC team proposes the Modular Acoustic Suppression System (MASS), an adjustable tuned membrane absorber that effectively reduces sound by achieving resonance with incoming sound waves [3]. Using principles of Helmholtz resonance, the system's control scheme takes measured frequency as an input and outputs a required height value to the motor system to change the depth of the acoustic membrane chamber, adjusting to absorb a range of frequency inputs. This design is intended to passively decrease background noise with minimal maintenance, enabling astronauts to more efficiently focus and communicate [8, 9, 11].

The goal of this report is to give an overview of the MASS team's design for NASA's HuLC as well as the verification prototype used for testing and design verification.

The Concept Description and Justification section describes the MASS concept and gives NASA design constraints and analysis to justify design decisions. The Verification section describes the models and simulations utilized to prove the MASS design. The Implementation section describes the tests performed on the MASS verification prototype, the risks involved for NASA, and the budget and timeline information for implementation.

Concept Description and Justification

Current solutions to dampen noise in space and commercial applications are often purely mechanical, bulky, and do not adapt to the environment [8, 10, 14]. MASS is an adjustable system that can adapt to any sound environment within the range of 60-200 Hz throughout the working day and during nighttime [3]. It can, in real time and without user input, adapt to frequency outputs of different research equipment, ECLSS shifts, and vocal levels depending on the time of day. This ability to adjust is also

applicable for long duration shifts as life support equipment wears down, which could lead to adjusted frequences in fans and pumps as dust and wear accumulates. In addition to this, NASA's current acoustic standards and quiet-fan work put emphasis on higher frequency, tonal fan-noise which operates in acoustic bands above MASS's 60-200 Hz target range [8, 10, 14].

Physical Design Description

The system consists of three main subsystems: a membrane cartridge, a reverberation chamber, and a control system. The final assembly of this acoustic damper system is shown in Figure 1.

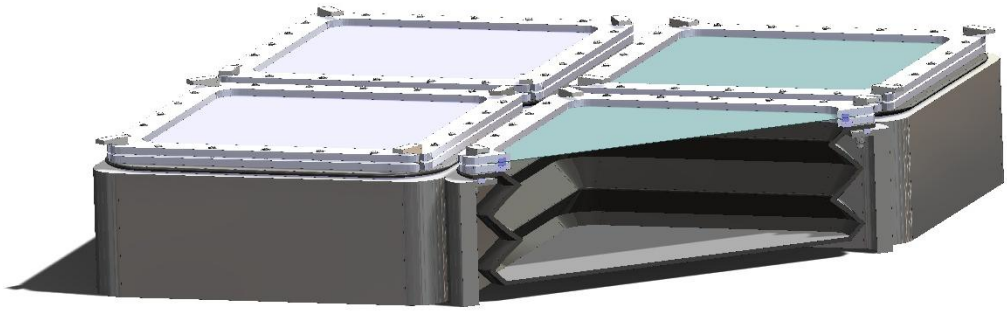


Figure 1. Acoustic Damper CAD Final Assembly.

The membrane cartridge consists of a dense membrane material, a top and bottom plate and gasket, and compression screws. The membrane will act in tandem with sealed air in the cavity to create a spring-like system that will absorb the vibration of emitted sound waves [3]. The reverberation chamber includes another gasket, which completes the pressure seal on the enclosed air cavity, and the accordion pressure cavity. The control system will facilitate vertical movement of the bottom mounting plate, which is attached to the bottom of the accordion pressure cavity, thus enabling the ability to change the distance between the membrane and the bottom of the accordion.

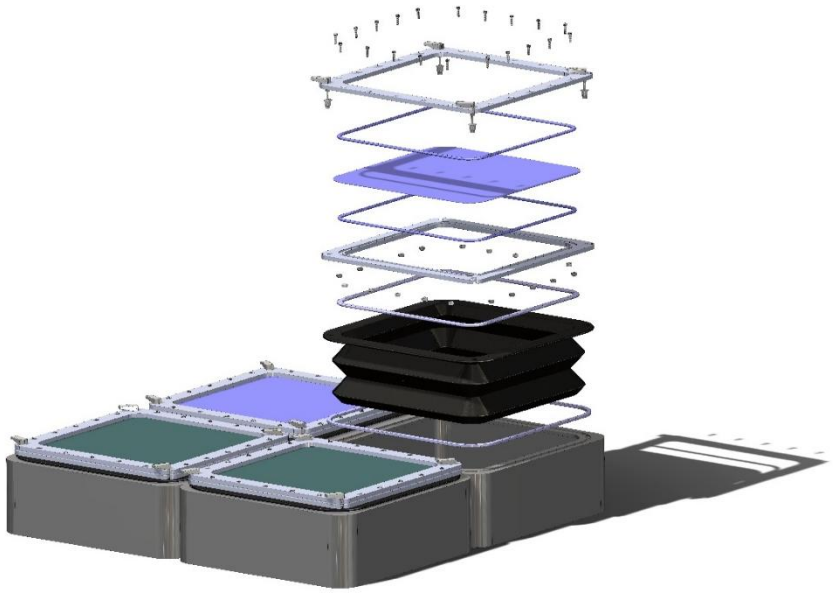


Figure 2. Acoustic Damper Assembly Exploded View.

Based on the depth of the sealed air cavity behind the membrane, the membrane absorber will be able to suppress a specific select frequency. To actuate the design, first the frequency with the highest decibel rating will be determined by using a frequency sensor in the control subsystem.

A fluorosilicone housing will be used for the reverberation chamber as it is most optimal in terms of balancing weight, cost, and effectiveness. Additionally, since fluorosilicone is inherently flexible, slightly thinner material at the edges of the accordion structure will allow the design to bend in the desired way when the mounting plate is actuated. Fluorosilicone is known for its durability and flexibility in aerospace applications [4], making it the perfect candidate for the housing material [4].



Figure 3. CAD Design of Accordion Housing Structure.

In addition, the gasket sealing the acoustic barrier to the housing will also be manufactured with fluorosilicone, as its flexible nature combined with mechanical fasteners attaching the subsystem components together will be crucial for creating an effective pressure seal. Fluorosilicone also has the desired range in density that is ideal for targeting the desired low frequencies

The MASS control system is based on tuning the chamber's geometry to match the dominant acoustic conditions in the environment. Since the membrane resonant behavior depends on chamber depth, pressure, and membrane properties, the controller uses the measured dominant frequency as the input for determining the desired resonant configuration. The goal is to shift the system's natural response so that it more effectively targets low- frequency noise.

The system can be modeled as a finite state machine due to its operations divided into control states. These include initialization, sampling, frequency identification, target depth calculation, adjustment, hold, and fault handling. This structure helps prevent unstable behavior by ensuring that the system only moves when a meaningful change in acoustic input is detected. For potential future improvement a deadband range can also be used so small frequency fluctuation do not cause continuous actuator motion.

The current control approach is primarily feedforward, meaning the measure of frequency is used to predict the chamber for the depth needed for suppression. A possible improvement would be to add closed loop feedback, where the system compares sound levels before and after adjustment and continues tuning until noise reduction is achieved. Additional improvement could include FFT averaging, adaptative frequency to depth mapping, and position feedback to account for backlash, missed steps, or changing environmental conditions.

Design Constraints for NASA Adoption

The implementation of MASS aboard the Artemis Lander is constrained by strict limitations on volume, mass, power consumption, and reliability. Due to minimal habitable volume inside the Artemis Lander, the protrusion of the design into crew space has been minimized so that it can be stowed efficiently near ECLSS hardware [13]. When dealing with implementation of multiple units in a larger system, compact geometry is very favorable. This directly motivated the reduction of maximum cavity depth from 0.305m to 0.205m. The total system weight has also been constrained to remain below 100 lb and below 1.5 kW of power consumption to remain feasible for long duration spaceflight and implementation.

Another large constraint for NASA adoption is long-term system reliability and maintainability. In alignment with other spacecraft noise reduction efforts, the MASS system must function quietly, efficiently, and reliably for extended durations on future Artemis missions while exposed to thermal cycling, vibration, and continuous operation [12]. To mitigate potential failure modes, such as actuator fatigue, membrane degradation, and loss of cavity seal integrity, the linear actuators are positioned externally to allow ease of replacement and maintenance. Additionally, the membrane and mounting plate displace very minimally during operation, which reduces cyclic fatigue of the system and improves the expected operational lifetime.

The tuned membrane absorber depends on maintaining a sealed air cavity during operation to generate a restoring force necessary for resonance-based noise suppression. Since cavity depth varies during operation, maintaining a constant pressure seal, while also allowing cavity pressure to equalize to ambient pressure during adjustment, are equally critical design challenges. Fluorosilicone was chosen as the membrane and gasket material, as its ability to deform while maintaining a pressure seal makes it well suited for the adjustable reverberation chamber used in the MASS design. The material's ability to resist thermal and environmental degradation also makes it compatible for the harsh conditions of space. To allow cavity pressure to equalize, the base plate contains a small central aperture that is sufficiently small to minimize acoustic leakage during operation while still allowing slow pressure equalization during chamber adjustment.

Analysis

When incident sound waves excite the membrane, it behaves like an oscillating mass. The sealed air cavity provides a restoring force due to compression of the trapped air [3]. Under small oscillations, which can be assumed for sound waves, the pressure change in the cavity follows the relation:

$$\Delta P = -\gamma P_0 \frac{\Delta V}{V} \tag{1}$$

where γ is the ratio of specific heats for air (≈ 1.4), V is the cavity volume, P_0 is the ambient air pressure (Pa), and ΔV is the change in cavity volume due to membrane displacement. For a membrane moving with displacement x over area A:

$$\Delta V = Ax. \tag{2}$$

This results in an effective acoustic spring constant per unit area, where d is the sealed air cavity depth (m):

$$k' = \frac{\gamma P_0}{d}. \tag{3}$$

The membrane surface mass density m_s ($\frac{\text{kg}}{\text{m}^2}$) acts as the oscillation mass per unit area. The system therefore behaves as a simple harmonic oscillator with natural frequency:

$$\omega_0 = \sqrt{\frac{k'}{m_s}}. \tag{4}$$

Substituting for k', the equation becomes:

$$\omega_0 = \sqrt{\frac{\gamma P_0}{m_s d}}. \tag{5}$$

Converting frequency in Hz, the final equation is:

$$f_0 = \frac{1}{2\pi}\sqrt{\frac{\gamma P_0}{m_s d}}. \tag{6}$$

This expression defines the resonance frequency of the membrane absorber.

Next, the required cavity depth was calculated, and the corresponding material was selected. Rearranging Equation (6) provides a direct design relationship for cavity depth:

$$d = \frac{\gamma P_0}{(2\pi f_0)^2 m_s}. \tag{7}$$

This relationship shows that increasing membrane surface mass, which is the product of material density and thickness, decreases the required cavity depth. Due to the inverse square relationship, it can be ascertained that lower target frequencies require significantly larger cavity depths as depth varies proportionally with $\frac{1}{f^2}$. Using the cavity depth expression in Equation 7, the required depth for a suppressible frequency can be calculated within the frequency operating range. This enables active

tuning of the absorber by adjusting the mounting plate position.

Next, the acoustic impedance and absorption coefficient were calculated. To estimate the percentage of incident sound energy absorbed, the system can be modeled using the surface acoustic impedance. The surface impedance of the membrane-cavity system is:

$$Z_s(\omega) = R_{loss} + j\omega m_s - j\frac{\gamma P_0}{\omega d} \tag{9}$$

where R_{loss} is the effective damping term, and $\omega = 2\pi f$. The normalized impedance is:

$$z = \frac{Z_s}{Z_0} \tag{10}$$

where $Z_0 = \rho_0 c$ is the characteristic impedance of air. The reflection coefficient at normal incidence is:

$$R = \frac{z - 1}{z + 1} \tag{11}$$

The absorption coefficient is then:

$$\alpha = 1 - |R|^2 \tag{12}$$

At resonance, the reactive terms cancel, and if the resistive term is tuned near Z_0, the absorber approaches maximum energy absorption [7].

Since resonance depends on cavity depth *d*, adjusting the mounting plate directly changes the tuned frequency according to:

$$f \propto \frac{1}{\sqrt{d}} \tag{13}$$

Thus, by mechanically varying cavity depth, the system can shift its peak absorption across the desired frequency range. The mathematical relationship implemented in the control subsystem allows the actuator to position the mounting plate based on real-time frequency measurements.

This analytical framework confirms that the proposed adjustable membrane absorber can theoretically target and suppress low-frequency bands through controlled geometric reconfiguration. To facilitate this reconfiguration, a linear actuator/linear rail system was implemented that will be attached to the mounting plate. The system will include two linear actuators and four linear rails, with each actuator placed on opposite sides of the mounting plate and two linear rails additionally placed on each of these sides. In an ideal scenario an electromechanical actuator that can reduce ambient noise production by

the actuators would be used. A future iteration of the design might include a second membrane or a base plate with a lower effective stiffness. While this does significantly increase weight and system complexity, this would further reduce the required cavity depth to suppress frequencies in the target range.

Verification

Model and Simulation

Fans on the ISS currently generate a range of frequencies [5, 10, 11, 12]. Many implemented sound suppression solutions engineered to maintain the general NC-50 acoustic levels prioritize higher frequency noise, but low-frequency noise from ECLSS hardware, specifically fans and rotating components, have the potential to propagate into the habitable volumes of the lunar lander and result in a chronic stress factor for crew [6, 8, 9]. That background noise both increases fatigue risk and has the capacity to directly degrade mission performance by masking speech cues, reducing intelligibility, and making comms more error prone. These factors are exacerbated for the small reverberation prone environment of the lunar lander. NASA's crew acoustics guidance ties acceptable continuous background noise to NC limits but explicitly does not allow hearing protection as the method of compliance, which is why passive, built-in noise control presents a compelling solution [8].

Based on the selected membrane material, the required depth to achieve optimal decibel reduction at a specific frequency varies. A denser material requires less depth to reduce a specific frequency, but also suppresses a narrower band of frequencies, while a less dense material requires more depth and suppresses a wider band [3]. The bounds on the adjustability of the design have been set to a minimum of 0.05 meters when the accordion structure is fully compressed, and a maximum of 0.205 meters when the structure is fully outstretched. The maximum depth of the reverberation chamber was reduced from .305 meters to .205 meters since our initial proposal in order to reduce empty cavity space. This smaller required cavity depth additionally decreases the amount the wall mounted panel protrudes into the room, thus saving cabin space and increasing the feasibility of implementation. 3 mm fluorosilicone was chosen as the membrane material for the chamber, with the relationship between cavity depth and suppressible frequency range shown in the plot in Figure 4. According to the simulation, the design can suppress a range of 60-170 Hz at depths ranging from 0.05 m to 0.205 m.

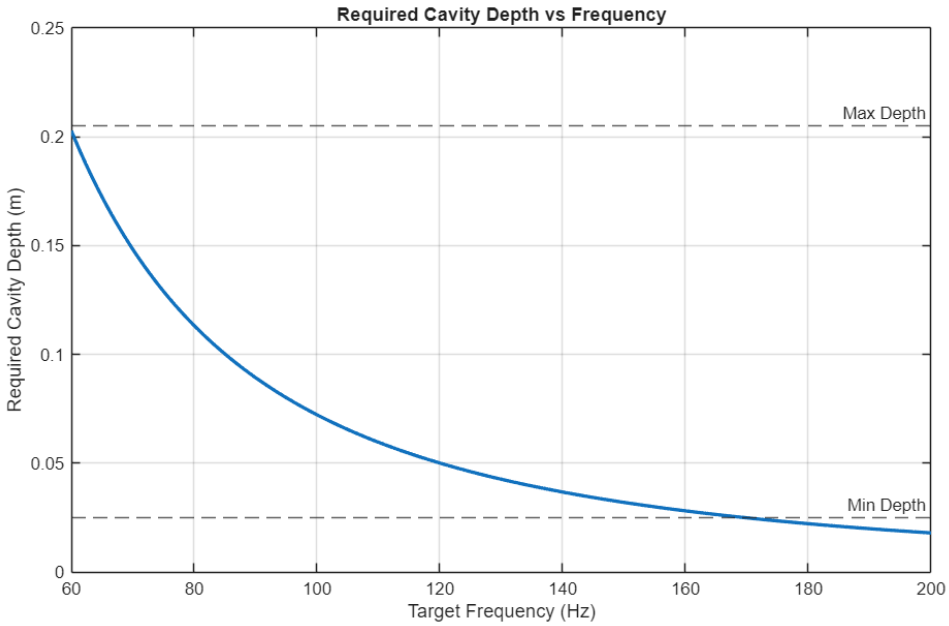


Figure 4. Plot of Required Cavity Depths for 3 mm Fluorosilicone Membrane to Suppress the Target Frequency Range.

Even though fluorosilicone does not cover the entire target range of 60-200 Hz, two different membrane materials or material thicknesses can be used in the larger system of absorbers to encompass the entire range of frequencies. In this case, 2 mm and 3 mm fluorosilicone can be used as the two materials needed to cover the target range. The corresponding range of suppressible frequencies for each material can be seen in Figure 5 below.

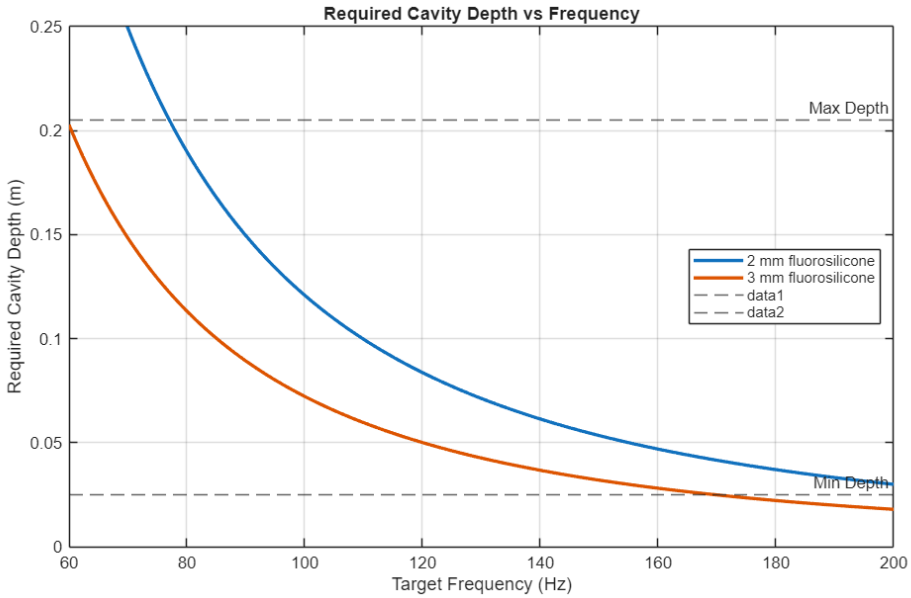


Figure 5. Plot of Required Cavity Depths for 2 mm and 3 mm Fluorosilicone Membrane to Suppress the Target Frequency Range.

3 mm fluorosilicone spans a range of 60-170 Hz, while 2 mm fluorosilicone spans a range of 76-200 Hz. Different materials that cover the target range can be used interchangeably.

Implementation

Feasibility Prototype Validation

Three tests were conducted on a feasibility prototype of MASS: a linear actuator and electronic verification test, a pressure equalization verification test, and a vibration test. This feasibility prototype was manufactured using slightly different materials, the most significant changes being TPU for the housing and mass loaded vinyl for the membrane, but the tests performed showcase the functionality and feasibility of MASS on a larger scale.



Figure 6. Feasibility Prototype CAD Model.

The linear actuator and electronic verification test performed a sweep for frequencies ranging from 60-250 Hz. This ensured that the actuators adjusted to changing frequencies. The developed control system successfully smoothed frequencies to enable consistent transitions despite highly variable sound conditions and could successfully actuate in both directions for the whole height range. Photos of the testing setup procedure are shown in Appendix A.

The pressure equalization verification test shifted the chamber depth three inches to verify pressure equalization at varying depths throughout five separate tests. The chamber pressure successfully equalized after an average of 4.49 seconds. Chamber pressure equalization was obtained with a slight gap in the seal for the prototype, but for NASA's purposes, the seal would be completely enclosed. In

this case, chamber pressure equalization would be obtained through a small orifice at the base of the chamber. Through calculations shown in Appendix B, it was found that a hole diameter of 13 mm would be sufficient.

The vibration test was conducted to verify that the MASS prototype could withstand structural integrity under launches like dynamic vibration conditions. Since launch vehicles subject payload to broadband random vibration, the protype was mounted to a shake table and subjected to a random vibration profile sweep from 20-2000 Hz. This range was selected to represent the dynamic loading that the system could experience during launch and to identify whether any structural loosening, failure, or instability which could occur during a low, mid, and high frequency excitation. The test was performed in stages, increasing from 25% to 50%, 75%, and finally 100%. After the ramp up sequence, the prototype was held at the full test level for 5 minutes. The final vibration profile reached an RMS acceleration of approximately 10.3 g, and the schedules test completed successfully without premature shutdown.

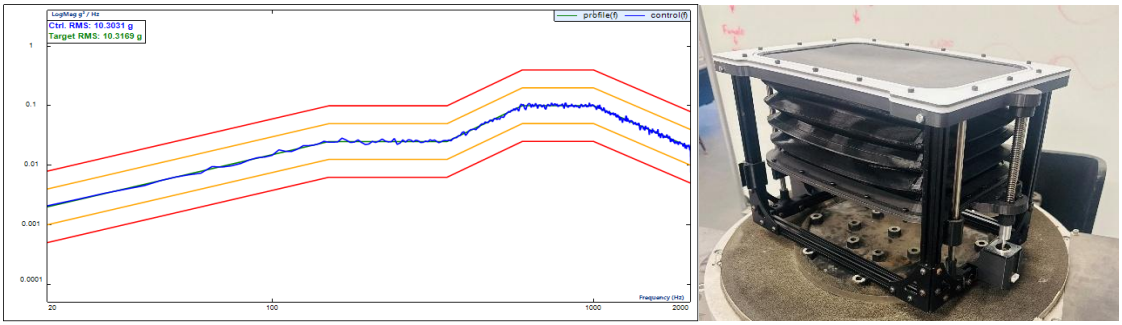


Figure 7. Testing Apparatus and Decibels versus Frequency Graph for the Vibration Test.

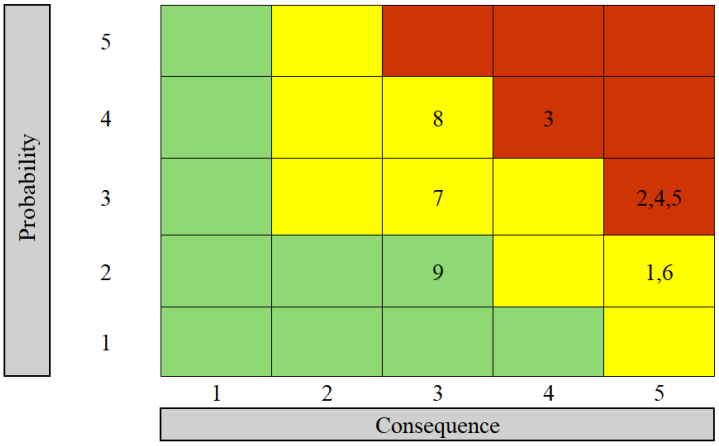
The test was used to evaluate structural stability rather than acoustic performance. The prototype held stable during the duration of the test, with the only issue being loose hex nuts after the vibration test. It was determined that lock-nuts would solve this issue; this change was reflected in our budget. Future testing should repeat the procedure using final flight material and include posttest inspection of loosening, deformation, actuator misalignment, membrane damage, and seal degradation.

Risks

NASA's upcoming sustained lunar operations include mission durations growing from about 7 days to 30-45 days as well as missions to Mars lasting 1200 days [15]. This long-term stay increases the importance of noise protection for astronauts and the need to ensure all possible risks are mitigated [8, 11]. Possible causes and solutions regarding maintenance measurand longevity of the design have been considered to enable reliable noise suppression. To analyze any risks posed by integrating MASS into

the human lander, a risk management table, which can be seen in Appendix C, and matrix shown in Table 1 were developed. Through this analysis, six critical risks were identified.

Table 1. MASS Risk Matrix.



The greatest risks for this system are related to functionality in terms of acoustic damping, with a high-pressure differential (Risk 3) being the most critical risk based on the matrix. The system's ability to resonate with target frequencies is highly dependent on maintaining an internal pressure that is equal to the surroundings. If the pressure does not equalize then the device resonant frequency will not match the dominant frequency in the surroundings. To mitigate this, the current design incorporates an orifice at the base of the reverberation chamber. This will allow air to escape and return as the system actuates, adjusting the volume of air inside as the height changes and keeping the pressure stable. Extensive analysis and testing will be done in future design phases to ensure that the hole size is suitable in terms of recovery time and equalization. Maintenance will be required at regular intervals that includes ensuring the orifice is still opening and is not blocked.

The next risk with a high consequence rating is electrical system failure (Risk 2). Frequent or long-term actuator motion could lead to wire fatigue, loose connections, or actuator motor wear. Additionally, vibration from launch loads could damage these same components. Strategies to mitigate electrical failure include using space-rated wiring and cables, providing strain relief on all moving cables, and mounting the PCB in a way that is damped from vibration. Additionally, current limits should be implemented in the actuators and a fail-safe integrated into the system that allows the device to remain at a default constant chamber depth that corresponds to the dominant, most common ECLSS frequency on the human landers.

There may also be further failures relating to the positioning and accuracy of the identified frequencies. Actuator positioning error (Risk 4) presents a challenge since improper chamber depth would shift the

resonant frequency away from the targeted acoustic band. This could result uneven actuator motion, a drift in calibration accuracy, and inability to suppress the current dominant frequency. To mitigate this risk, MASS will have actuator positioning feedback incorporated into its architecture, periodic calibration checks, and software travel limits to ensure the operation remains within the allowable chamber range of depth. In addition to the mechanical positioning accuracy, the effectiveness of the control system is dependent on reliable acoustic sensing. Acoustic monitoring inaccuracy (Risk 5) could cause the system to tune toward an incorrect dominant frequency which in turn would reduce the suppression performance of the system or cause unnecessary actuator movement. This risk would be mitigated through careful sensor calibration and testing of FFT averaging techniques. The microphone monitoring the surrounding noise would go through extensive validation testing and will be revalidated periodically.

Excessive mass (Risk 7) and excessive volume (Risk 8) are also important risks to consider when implementing MASS into the HLS. Additional payload weight and intrusion into habitable crew space directly impacts mission feasibility so reduction of weight and volume is critical. This design mitigates these risks through lightweight material selection, reduction of unnecessary structural components, and the optimization of cavity depth to minimize protrusion into the space. The maximum cavity depth was reduced by 0.105 m which improved the compactness of the system while still maintaining the target suppression frequency range. Continued design refinement and structural optimization would further reduce system mass and occupied volume before it is implemented.

Implementation

While the MASS design shown in the Quad Chart consists of one panel, the full-scale implementation of this design on NASA's Artemis would utilize multiple panels with separate sensors to reduce frequency decibel levels from separate sources. The first step towards implementation is a budget assessment to ensure that the cost of this project will fall within a reasonable budget.

Budget

When developing Table 2. which represents a lifecycle budget for developing one flight ready MASS unit using a 10-person engineering team over a three-year period. The largest cost driver is the personnel, with annual labor held constant at $1.26 million based on ten full time members at $126,00 per year. Direct costs were separated from personnel to demonstrate the expected expense for prototype hardware, testing facilities, and final flight hardware. A 30% cost margin is included in account of design changes, added testing, procurement uncertainty, and integration risk. Overall, the projected lifecycle cost is approximately $5.63 million, which provide a realistic estimate for a finalize MASS from prototype to flight ready.

Table 2. MASS Implementation Budget Summary.

Mission Phase	Year 1: Dev Phase (2026)	Year 2: Test Phase (2027)	Year 3: Fab & Impl. (2028)	Total ($)
PERSONNEL				
Engineering & Design	900,000	600,000	500,000	2,000,000
Fabrication & Assembly	250,000	160,000	360,000	77,0000
Testing & Inspection	110,000	500,000	400,000	1,010,000
Total Personnel(12)	1,260,000	1,260,000	1,260,000	3,780,000
DIRECT COSTS				
Prototyping Hardware	100,000	50,000	0	150,000
Testing Facilities & Equip.	0	150,000	100,000	250,000
Final Hardware (1 Unit)	0	0	150,000	150,000
Total Direct Cost	100,000	200,000	250,000	550,000
FINAL COST CALC.				
Total Projected Cost	1,360,000	1,460,000	1,510,000	4,330,000
Total Cost Margin(-30%)	408,000	438,000	453,000	1,299,000
Total Project Cost	**1,768,000**	**1,898,000**	**1,963,000**	**5,629,000**

The annual operating cost was determined using the estimated power input, mission time, system run rate, and the current power market rate. It is assumed that the MASS system would be actuated and run for one minute of every hour of the mission, the power usage would be 1.36 kW, and the current market rate was $0.12 per kilowatt-hour. Two values were determined: one for a 30-day long mission to the moon, and one for a 1200-day long mission to Mars.

The cost for a 30-day long mission is $58.75, and the cost for a 1200-day long mission is $2,350.08. This is a reasonable operating cost for the benefit of astronaut comfort and focus during Artemis Human Lander missions to both the moon and Mars.

Path-to-Flight Project Timeline

A three-year plan is proposed below to successfully implement the MASS system into NASA's ECLSS, with the assumption that a mission will happen in the next 5-8 years. This plan includes the development and testing of the product as well as evaluation of the final system.

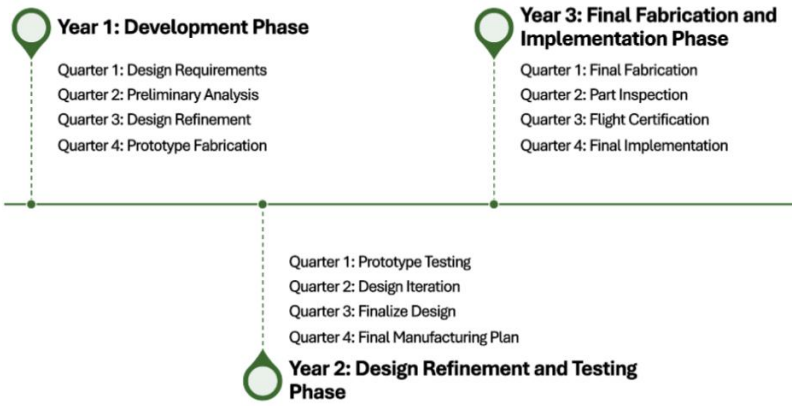


Figure 8. Project Timeline

Year 1, the development phase, focuses on the design requirements of the system. This will be CAD refinement from the Acoustic Artemis design, as well as material testing and final material selection. Year 1 will conclude with prototype fabrication. Year 2, the design refinement and testing phase, focuses on testing of the functional prototype, especially testing focused on launch fatigue aerospace thermal dynamics. Year 2 concludes with a final plan for manufacturing. Year 3, the final fabrication and implementation phase, focuses on the final fabrication of the MASS units being utilized for long-duration space flights. Year 3 concludes with the final implementation of the system.

Conclusion and Next Steps

The MASS concept shows the feasibility of a tuned membrane absorber for mitigating low frequency noise generated by the ECLSS abord the Artemis Human Lander. Through simulation modeling, material selection, and prototype validation, the MASS system was shown to theoretically suppress frequencies in the 60-200 Hz range while remaining within NASA volume, mass, and power constraints.

The required cavity depth for MASS was calculated as a function of frequency and membrane surface mass density. Initial analysis resulted in a required cavity depth of up to 0.305 m for a 3 mm neoprene membrane. However, by increasing membrane surface density through a fluorosilicone membrane material selection, and by using two different material thicknesses with overlapping operating ranges to span the entire target range, the required cavity depth was reduced to 0.05 m. This significantly improved compactness and reduced protrusion into habitable cabin space.

Prototype testing verified the critical subsystems of MASS. The actuator and control verification test successfully demonstrated automatic adjustment of the reverberation chamber across the target frequency range. Pressure equalization testing showed that the chamber could stabilize internal pressure after a change in cavity depth with an average equalization time of 4.49 seconds. Vibration testing exposed the prototype to various vibration profiles ranging from 20-2000 Hz with RMS accelerations approaching 10.3 g, which validated structural integrity under simulated spacecraft launch conditions.

This project demonstrates that MASS is a feasible solution for reducing low frequency ECLSS noise onboard future Artemis missions. Future research should focus on experimental acoustic chamber testing to quantify insertion loss across the target frequency range. Additionally, future development to the design should include lightweight structural optimization, refinement of control architecture, and environmental testing under relevant thermal and vacuum conditions. With further refinement and validation, MASS has potential to improve comfort, communication, and habitability for astronauts during long-duration missions.

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For each appendix, include a cover page that explains the contents of that appendix and put them in ***order mentioned in the report****. Possible appendices:*

- *Final Budget – NEED TO PUT IN CHART*
- *Manufacturing pictures - COMPLETE*
- *User Manual – NEEDS EDITING*
- *DVPR and other test results*
- *Analysis, Code, etc.*
- *Gantt Chart - COMPLETE*

Appendix A: Final Budget

This appendix outlines the final budget of the verification prototype that was built for the final design expo. All components ranging from electronics to structural components are listed in Table A. The final total was estimated based on parts that would have been purchased then the actual cost based on reused parts is also outlined.

Table A.1. Updated Budget

Level	Number	Part Description	Part Number/Link	Vendor	Price Per Part	Quantity	Total Price
1	1.1	RPI 2gb	6007	adafruit	$75.00	1	$75.00
1	1.2	stm32 pill	STM32BPSL01	Amazon	$39.99	1	$39.99
1	1.3	Drivers w/HS	606	Amazon	$2.00	2	$4.00
1	1.4	Microphone	UMIK	MiniDSP	$79.99	1	$79.99
1	1.7	Pressure Sensor	4258	adafruit	$12.50	1	$12.50
1	1.8	Power Supply	PJ044	Amazon	$21.98	1	$21.98
1	1.9	Breadboard Wires	--	Cal Poly IEE Lab	$7.00	1	$7.00
1	1.1	Stepper Motors	MS17HD6P4150	MOONS	$42.10	2	$84.20
2	2.1	Mass Loaded Vinyl - 2 lb	B0DT4Y35W8	Amazon	$95.00	1	$95.00
2	2.2	Gasket - TPU		Bambu Labs	$41.99	1	$41.99
2	2.3	Upper Compressive Housing - PLA		Amazon	$12.99	1	$12.99
2	2.4	Aluminum Composite Support Plate	308670323	Home Depot	$21.75	1	$21.75
2	2.5	Middle Compressive Housing - PLA		Amazon	--	--	--
2	2.6	x16 M4 Screws		Amazon	$9.99	1	$9.99
3	3.1	Reverberation Chamber - TPU		Bambu Labs	--	--	--
3	3.2	Chamber Base - PLA		Amazon	$12.99	1	$12.99
3	3.3	Linear Rail Stock - 6 ft	89535K17	McMaster	$28.85	1	$28.85
3	3.4	Linear Rail Sliders - PLA		Amazon	--	--	--
3	3.5	x24 M4 Screws		Amazon	--	--	--
4	4.1	Vertical T-slot Framing	5537T101	McMaster	$45.60	1	$45.60
4	4.2	Short T-slot Framing	5537T102	McMaster	$22.80	1	$22.80
4	4.3	Long T-slot Framing	5537T103	McMaster	$22.80	1	$22.80
4	4.4	x8 Triangle Brackets		Amazon	$74.40	1	$74.40
4	4.5	x32 M4 Screws		Amazon	--	--	--

LEVELS	
Level 1	Electronics
Level 2	Membane Assembly
Level 3	Reverberation Housing Assembly
Level 4	Structural Assembly
Prototype Totals	
Total Material Cost	$713.82 ***Cost of materials with estimated material pricing for those not purchased
Total Purchased	$493.39 ***Total spent by team

Appendix B: Manufacturing Pictures

Outlined in this appendix are pictures of the manufacturing process. The manufacturing process included 3D printing with both TPU and PLA, turning linear rails to size, and cutting aluminum rails to size. When all parts were manufactured, they were assembled and checked for tolerancing and overall fit.



Figure B.1. Human Lander Noise Suppression Team Integrating Electronics Into the Mechanical System

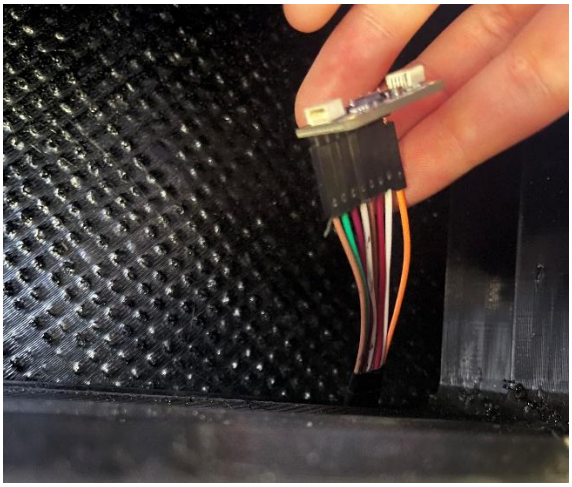


Figure B.2. Pressure Sensor Placement into Internal Reverberation Chamber



Figure B.3. Checking Bearing Fit After Turning Upper Shaft to Size



Figure B.4. Baseplate Fit Check and Placement



Figure B.5. Assembly Setup Including Membrane Placement and Fasteners

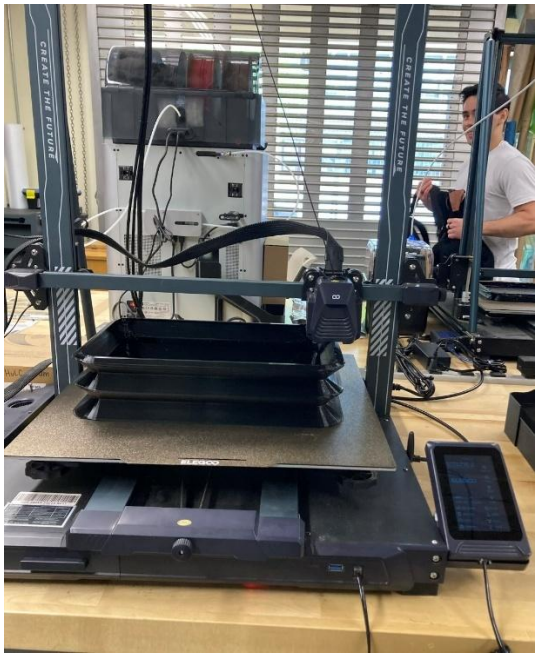


Figure B.6. Reverberation Chamber Being Manufactured on Elegoo Neptune 4 with TPU

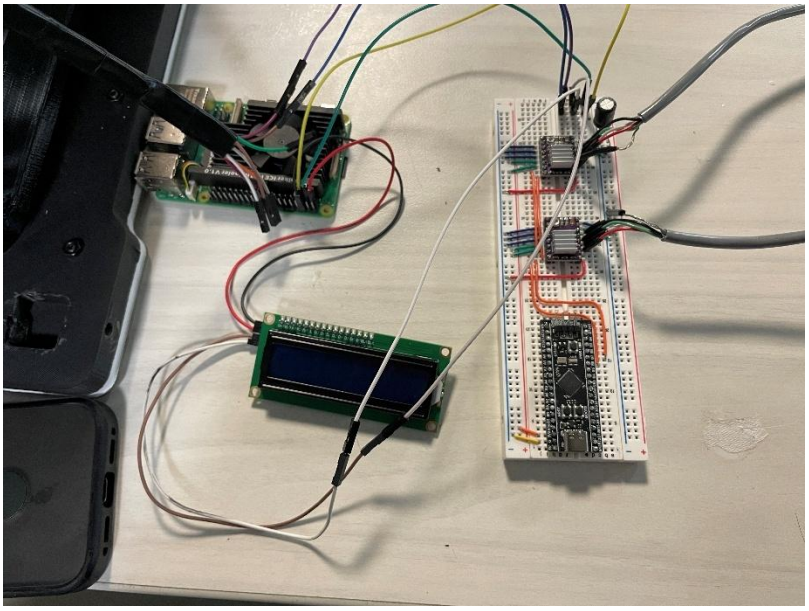


Figure B.7. Breadboard Configuration and Raspberry Pi Connections

Appendix C: User Manual

The MASS User Manual describes how to assembly, maintain, and utilize the MASS system. This includes required tools, mechanical components, hardware, electronics, assembly, operation, and maintenance.

Modular Acoustic Suppression System (MASS) User Manual

Produced by Cal Poly Senior Project Team:

NASA HuLC Noise Suppression



For the project sponsor:

Dr. John Chen

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REQUIRED TOOLS

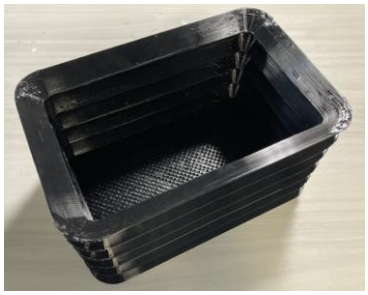
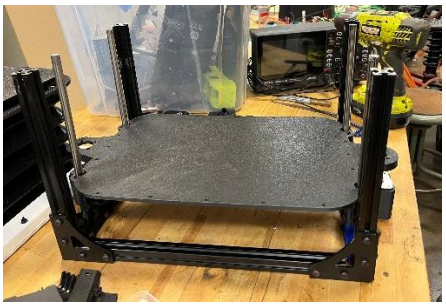
Hex Key

2.5 mm



COMPONENTS

4x 9.925” Aluminum Extrusion 	**2x** 13.5” Aluminum Extrusion 
2x 8.9” Aluminum Extrusion 	**2x** Lead Screws 

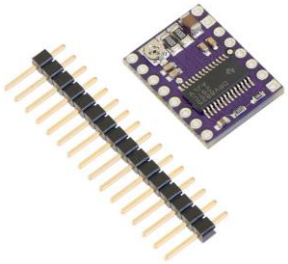
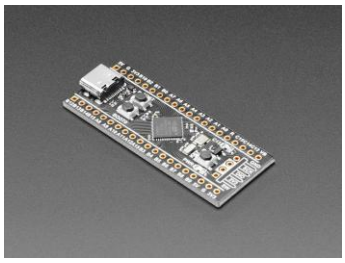
1x Reverberation Chamber 	**1x** Chamber Base 
1x Acoustic Membrane 	**4x** Linear Rails 
1x Gasket Assembly 	

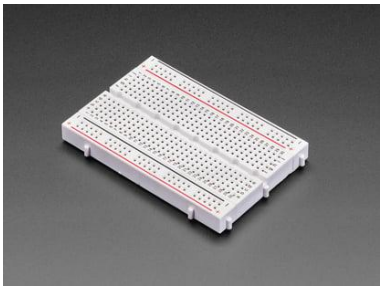
HARDWARE

32x ¼-20 Screws 	**32x** ¼-20 Tee Nuts 	**32x** ¼-20 Washers 

NOTE:
This is a breakdown of the electronic components that make up the electronic package. A single electronic package will come preassembled to be inserted into the base of the modular acoustic suppression system.

ELECTRONICS

1x Raspberry Pi 2gB 	**2x** DRV825 Motor Driver 	**1x** STM32F411CEU6 “Blackpill” 
1x Arcity 5V 12V 24V Output Switching Power Supply Unit 	**1x** 12 C Decibel Sound Level Meter Module 	**1x** LPS35HW Water Resistant Pressure Sensor 

2x Moons MS17HD6P4150 Stepper Motor Unit 	**1x** MINIDSP microphone 	**1x** Breadboard- 400 Tie Points 

ASSEMBLY

CAUTION

- **This assembly contains sharp- edged aluminum components.** All parts have been deburred and sanded, but care should be taken during handling and transport.
- **This product does not mitigate sound generated by frequencies over 250 Hz.**
- **Failure to properly assemble, inspect, and maintain the device may result in injury or equipment failure.** Noise will not be mitigated should the equipment fail.

NOTE:

- All parts are outlined in the component and hardware sections.

****NOTE:** *This a rough draft of the assembly process – pictures will be updated as they are manufactured****

1. Insert four tee nuts into the two 13.5” aluminum rails.



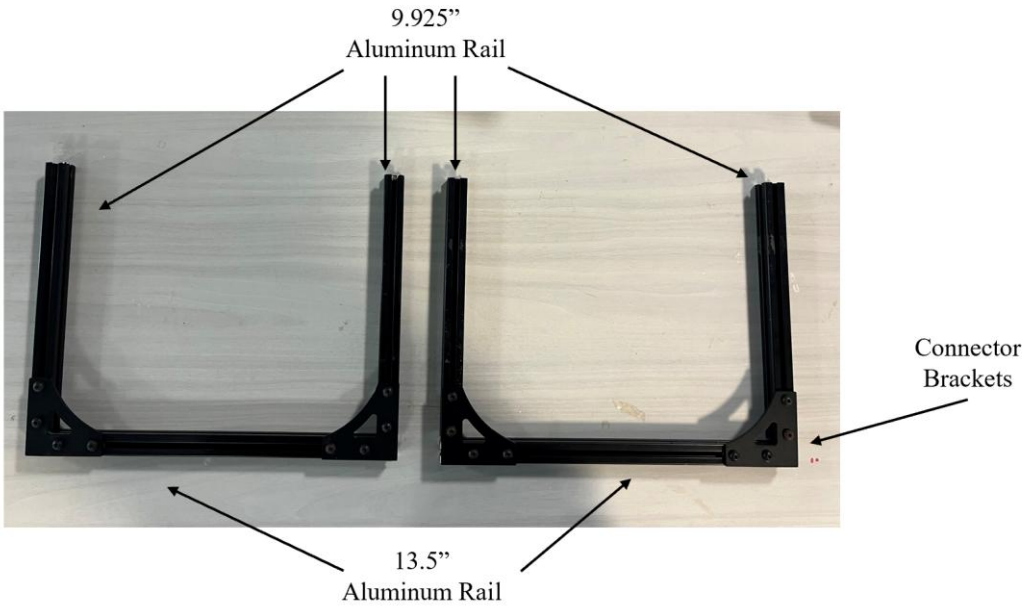
2. Insert four tee nuts into the two 8.9” aluminum rails.



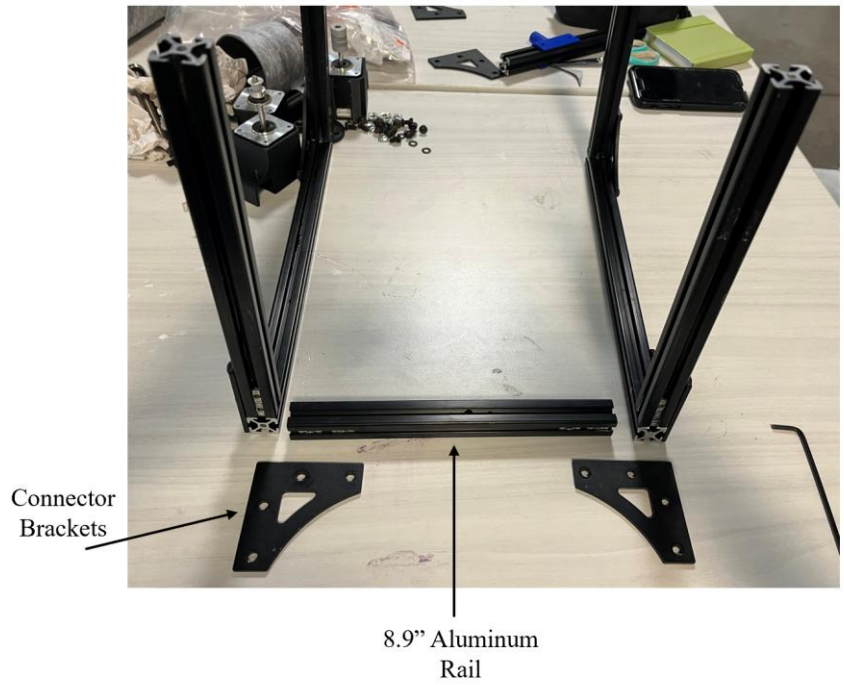
3. Insert two tee nuts into each of the four 9.925” aluminum rails.



4. Connect the 13.5” aluminum rails to the 9.925” rails at a ninety-degree angle using the ¼-20 screws and the eight connector brackets.



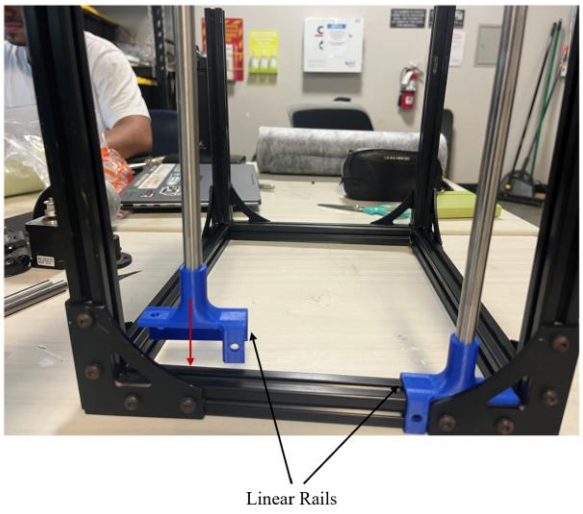
5. Connect the assembled 9.925” and 13.5” frame to the two 8.9” rails with four connector brackets and 16 ¼-20 screws and washers.



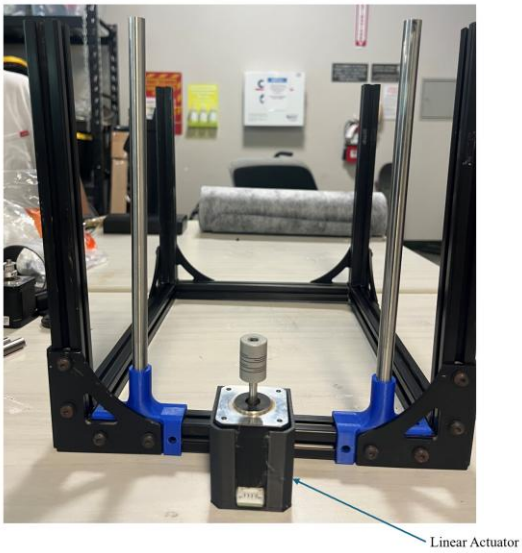
6. Press the four linear rails into the linear rail connectors.



7. Connect the linear rails to the frame base using four ¼-20 screws



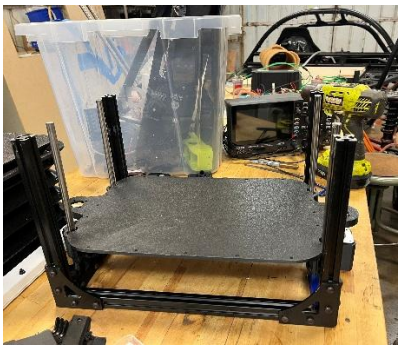
8. Connect the linear actuators to the frame base using one ¼-20 screw



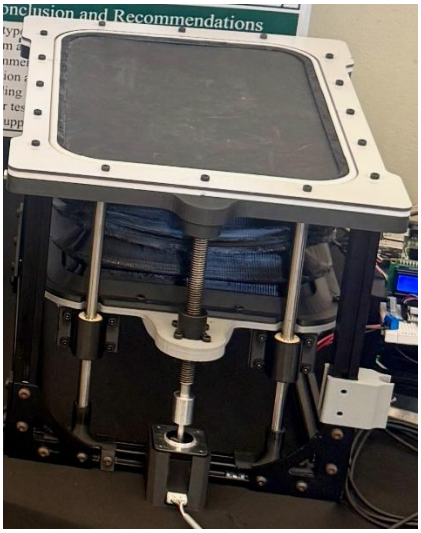
9. Place the electronic package next to the frame.



10. Connect the base plate to the frame using fasteners.



11. Connect the reverberation chamber to the base plate using fasteners.
12. Place the membrane and gasket over the open side of the chamber and place between the bottom and top mounting plates.
13. Fasten the plates together and create pressure seal with the membrane and reverberation chamber.
14. Attach frequency sensor and electronics to the frame and actuators.



OPERATION

CAUTION

- **This assembly contains sharp- edged aluminum components.** All parts have been deburred and sanded, but care should be taken during handling and transport.
- **This product does not mitigate sound generated by frequencies over 250 Hz.**
- **Failure to properly assemble, inspect, and maintain the device may result in injury or equipment failure.** Noise will not be mitigated should the equipment fail.
- **Keep fingers and other body parts clear of moving parts around linear actuation to avoid pinching or crushing hazards.**

1. **Pre-Use Inspection**
    a. Visually inspect the MASS assembly using the routine maintenance procedure below.
2. **Assembly**
    a. Follow the detailed assembly plan shown above.
    b. Ensure that all fasteners are tightened and in place.
    c. Ensure that all parts are correctly inserted.
3. **User Setup**
    a. Attach the MASS assembly to a flat surface using rubber mounting fasteners near the outlet of the highest decibel fan without obstructing the flow.
    b. Ensure that the system is properly in place and will not collide with other systems during operation.
4. **Operation**
    a. When not in use, store the accordion in an unloaded state.
    b. Ensure that the frequency sensor is properly in place.
    c. Switch on the system to begin use, check LED display to ensure microphone is operational.
    d. Ensure that the incoming frequency matches the displayed frequency.
5. **Post-Use Maintenance**
    a. Follow the detailed routine maintenance plan shown below.

MAINTENANCE

Recommended Routine Maintenance

- Visual Inspection (Before and After Use/Testing)
    - Inspect all structural components including the linear rails, membrane, accordion housing, and frame for cracks and punctures.
- Fastener Retention
    - Inspect for missing fasteners and retighten as needed.

Annual or Seasonal Maintenance

- Full System Check (Before Launch)

- Disassemble full assembly and inspect for any wear on linear rails and metal parts.
- Replace any visually damaged parts.
  Replace the membrane and accordion housing.

Storage and Handling

- Storage
  - The system should be stored in a box protected from the sun to prevent damage.
  - Parts should initially be stored separately and should only be assembled before the first testing and use of the unit.
- Handling and Transport
  - The final assembly should be transported in a cool, dry space to prevent damage.

Appendix D: DVPR and Testing

Appendix D includes the Design Verification Plan and Report, which outlines the tests that were planned, enacted, and the test procedures that were followed. Results are also detailed in the tables below.

Testing Plan: Table D.1 outlines the specific tests that will be performed to verify MASS' design. The specifications being tested are pressure, range of motion, frequency response, and vibration tests. Note that Test 1 was not completed due to lack of proper facilities.

Table D.1. Testing Plan

Test #	Spec #	Test Description	Measurement	Acceptance Criteria	Facilities & Equipment	Start Date	Planned Completi-on
1	1	Low Frequency Reduction	dB, Hz	NC-40: dB<60, 60-200 Hz	Quiet room, testing fan setup, SLM, speaker	5/21	Data 5/21, processed by 5/22
2	1,2,3	Pressure Equali-zation Verification	(kilo)Pascal	Pressure differential between atm and cavity <0.1 kPa	Quiet Room, digital media gauge,	5/19	Data 5/19, processed by 5/22
3		Mechanical Durability	Temperature, cycles, dB, Hz	Consistent bolt pretension, membrane assembly shifts < 1 1 mm during frequency sweep	Vibration laboratory, speaker	5/19	Data 5/19, processed by 5/22
4		Mounting Safety	Cycles	No detachment, <5 degree rotation of fasteners, <1 mm shift in	Vibration laboratory, speaker	5/19	Data 5/19, processed by 5/22

				any direction, no cracks			

Testing Procedures:

Low Frequency Reduction – This is a theoretical testing procedure that would be used to verify the reduction of noise in future tests.

Test Name: Low Frequency Reduction Verification

Purpose: *The purpose of this test is to verify that the design responds to frequencies within the range of 63-250 Hz and the surrounding area does not contain noise above the NC 40 rating.*

Scope: This test focuses on the surrounding space around the prototype.

Equipment:

- Sound level meter
- Speaker that can produce frequencies ranging from 63-250 Hz

Hazards: Standard operating hazards still apply (see user manual for further information). No additional hazards from the testing setup.

PPE Requirements:

- Ear protection
- Safety glasses

Facility:

- Any quiet room on the Cal Poly campus. The team will ensure that no sounds louder than the 70 dB infiltrate the room.

Procedure:

Setup:

1) Place speaker 2 feet from the top membrane of the prototype.

2) Place sound level meter directly to one sound of the prototype.

Process 1:

1. Begin by playing 63 Hz tone through the speaker.
2. Wait for the system to respond to the tone (30 seconds)
3. Measure the height of the reverberation chamber and record value.
4. Read the decibel level from the sound level meter and record value.
5. Increase frequency tone by 17 Hz.
6. Repeat steps 1-5 until 250 Hz is reached.

Process 2:

1. Repeat process 1 but with a fan as the sound input. Test all fan settings and record device response.

Results:

Pass Criteria: System actively responds to frequency and sounds do not exceed NC40 criteria

Fail Criteria: Sound exceeds NC-40 criteria

Number of samples to test: 1

Test Date(s): 5/21/2026

Test Results: *insert link*

Performed By: Cal Poly HuLC Noise Suppression Team

<u>Pressure Difference</u>

Test Name: Pressure Difference

Purpose: *The purpose of the pressure difference test is to ensure that there is no pressure difference between the atmospheric pressure and the pressure inside the MASS system.*

Scope: This test focuses on only the inside cavity of the MASS system.

Equipment: One pressure sensor is necessary for this test to see the gage pressure with the atmospheric pressure as the datum.

Hazards: There are no hazards associated with this test. Standard operating hazards still apply (see user manual for further information).

PPE Requirements: No PPE is required for this test.

Facility: The test will take place in the Cal Poly Machine Shops.

Procedure:

1) Insert pressure sensor inside of the MASS chamber.

2) Actuate the system to change the volume of the chamber by 1 inch.

3) Record any significant changes in pressure inside the chamber.

4) Repeat steps 2 and 3 until the system reaches its maximum actuation height.

Results:

Pass Criteria: Pressure does not exceed 0.1 kPa.

Fail Criteria: Pressure exceeds 0.1 kPa.

Number of samples to test: 1

Test Date(s): 5/19/2026

Test Results: See Table D.2

Performed By: Cal Poly HuLC Noise Suppression Team

MASS Mechanical Durability Test

Purpose: To ensure extended lifetime of the mechanical system through cyclic testing of durability from movement and frequency input. This test will allow an appropriate lifespan to be determined for MASS and whether any material changes are necessary to increase this lifespan.

Scope:

- This test involves the TPU housing for the membrane absorber, as well as the membrane itself.

Equipment:

- Shake Table in the Vibrations Laboratory
- Speaker producing critical frequencies

Hazards:

- Structure breaks apart, possible flying debris
- Standard operating hazards (see user manual for further information)

PPE Requirements:

- Safety glasses
- Closed toe shoes, pants

Facility:

- Vibrations laboratory (for shake test)
- Any room with table to clamp to (for frequency durability test)

Procedure:

Frequency Duration:

1) MASS is placed on inside a box

2) Speaker playing frequencies within our target range begins a frequency sweep

3) Response and amplitude of vibration is observed

Vibration Test:

1) MASS is clamped to the Shake Table, making sure that the housing is fully secured

2) Shake table is activated at different amplitudes of vibration

3) Behavior of the system structure is observed

Results:

Number of Samples: 1 Prototype (Analyze with FEA prior to test to verify we will not break theoretically during the test, physical tests will be used to verify manufacturing durability)

Number of Verifications:

- A shake test will be done once at the resonant frequencies determined through FEA simulation for a duration of 60 seconds
- For the frequency verification, a sweep test will be performed once across the 60 to 200 Hz range to ensure mechanical stability at all absorbed frequencies

Pass Criteria:

- For the vibration test:
  - Assembly must stay together
  - Bolt pretension will be remeasured to ensure continued structural integrity
- For frequency test:
  - Membrane must stay intact with sweep of frequencies
  - Tension of membrane must remain constant

Fail Criteria:

- For the frequency durability test:
  - The membrane breaks, or cannot absorb vibrations of the target frequencies
- For the vibration test:
  - The frame of the design breaks, or the accordion housing structure cannot handle the transferred vibration and breaks

Test Date(s):

Test Results: *Insert Link*

Performed By: Cal Poly HuLC Noise Suppression Team

Mounting Safety

Test Name: MASS Mounting Safety

Purpose: Verify that the MASS mounting method does not loosen, slip, crack, or detach under repeated loading representative of handing, vibration, and actuation cycling.

Scope: Mounting interfaces will be set up with the frame connected to the mounting plate, rubber mechanical fasteners, and any brackets used for the mounting of the module.

Equipment:

- Fully assemble MASS module with intended mounting hardware

- Mounting surface/fixture (rigid plate, wall mockup, or test stand)
- Torque wrench or driver with torque setting
- Marker/ paint pen (witness marks)
- Ruler/calipers (for displacement)

Hazards:

- Crush points at brackets, rails, and fasteners during installation
- Falling objects if the module loosens or detaches during cycling
- Sharp edges from metal framing, brackets, and hardware
- Electrical hazard if actuator wiring is powered during the test

PPE Requirements:

- Safety glasses
- Closed toe shoes
- Work gloves

Facility:

A machine shop area with sturdy bench stands with a rigid mounting fixture, with clear space around the setup to keep people out of the drop zone.

Procedure:

1) Pre-test inspection

2) Mount the unit

3) Mark baseline

- Apply witness marks across each critical fasteners interface

4) The cycling method

- Vibration input cycling: mount fixture to a vibrating tool

5) Cycling the mounting

- Perform N Cycles
- Every 100 cycles check witness marks and no visible damages.

7)Post-test Inspection

8) Pass/Fail determination

9) Document result

Results: Pass Criteria, Fail Criteria, Number of samples to test

Pass Criteria

- No detachment at any points
- Fasten loosening: With rotation of less than 5 degree
- Modules shift: Maximum permanent shift of 1mm in any direction
- Wall detachment: Only shift a maximum of 0.5 mm from the mounted edges
- No cracks and hardware deformation

Fail Criteria

- Any partial or full detachment
- Fasten loosening: With rotation greater than 10 degrees
- Modules shift: Greater permanent shift 2mm
- Wall detachment: Greater shift of 1 mm from the mounted edges
- Any cracks and hardware deformation

Number of samples to test:

Minimum: 1

Test Date(s): 5/19/2026

Test Results:

Performed By: Cal Poly HuLC Noise Suppression Team

Test Results:

Table D.2. Chamber Pressure Equalization Results

Actuation Distance [in]	**Initial Pressure Difference [kPa]**	**Time To Equalize [seconds]**
3	1.1	3.89
3	1	4.86
3	1.2	4.68
3	1.3	5.34
3	1	3.66

Table D.3. Mechanical Durability Throughout Formal Shake Test

Requirement	**Pass/Fail**
Bolts and nuts remain secure	Pass
Press fits remain in place	Pass
No cracks are propagated through structure	Pass

Vibration Test Data:

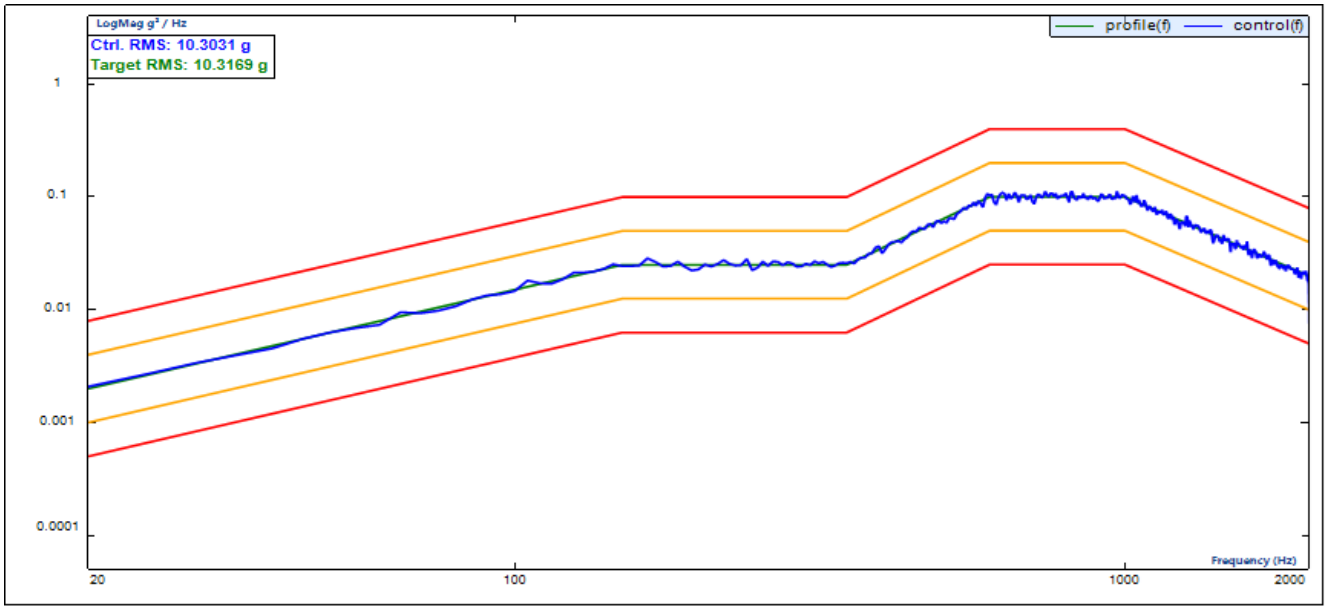
Test Report

Report time: 5/27/2026 3:43:46 PM **Data measured at:** 5/27/2026 3:43:27 PM

Test name: nanoracks_IDD4 **Test type:** Random

Test status: Test Stopped (Schedule Finished) **Run folder name:** Run6 5-27-2026 3-37-44 PM

Control Composite



Remaining time: 00:00:00 **Total elapsed time:** 00:05:59**Full level elapsed time:** 00:05:00

Lines: 400 **DOF:** 120 **Frequency range (fa):** Calculated by profile

Average: 60 **Run start time:** 5/27/2026 3:37:46 PM **Data measured at:** 5/27/2026 3:43:27 PM

Run Log

Absolute time	Test Time	Event type	Ctrl./Target RMS	Event Data	Event Source
Random test run log 5/27/2026 3:37:31 PM					
nanoracks_IDD4: No description					
Run description: Untitled Test Note					
5/27/2026 3:37:34 PM	00:00:00	Measuring noise	0.030/0.030 g	0	System
3:37:35 PM	00:00:00	Running in pre-test	0.066/1.032 g	0	System
3:37:44 PM	00:00:00	Pretest finished	1.029/1.032 g	0	System
3:37:47 PM	00:00:00	Start schedule		0	User Cmd
3:37:56 PM	00:00:06	Schedule level	2.579/2.579 g	25.00%	Schedule
3:38:07 PM	00:00:17	Schedule level	5.130/5.158 g	50.00%	Schedule
3:38:17 PM	00:00:27	Schedule level	7.730/7.738 g	75.00%	Schedule
3:38:28 PM	00:00:37	Schedule level	10.310/10.317 g	100.00%	Schedule
3:43:27 PM	00:05:37	Create report		4096	Action
3:43:27 PM	00:05:37	Stop the test	10.303/10.317 g	FLE: 300.00 (sec)	Schedule
3:43:27 PM	00:05:37	Schedule Finished		2	System

Scheduled tests finished successfully: full level elapsed: 00:05:00; total elapsed: 00:05:59	

Test Parameters

Lines: 400 **DOF:** 120

Delta frequency: 5 Hz **Frequency range (fa):** Calculated by profile

Drive limit (V Pk): 2 **Sigma clipping:** 5

Abort sensitivity: 50% **Overlap Ratio:** 50%

Average: 60 **Control strategy:** Weighted average

Level change rate: 20dB/s **Abort ramp down rate:** 20dB/s

Adjust level step: 10% **Non-linear control:** Enabled

Input Channel Table

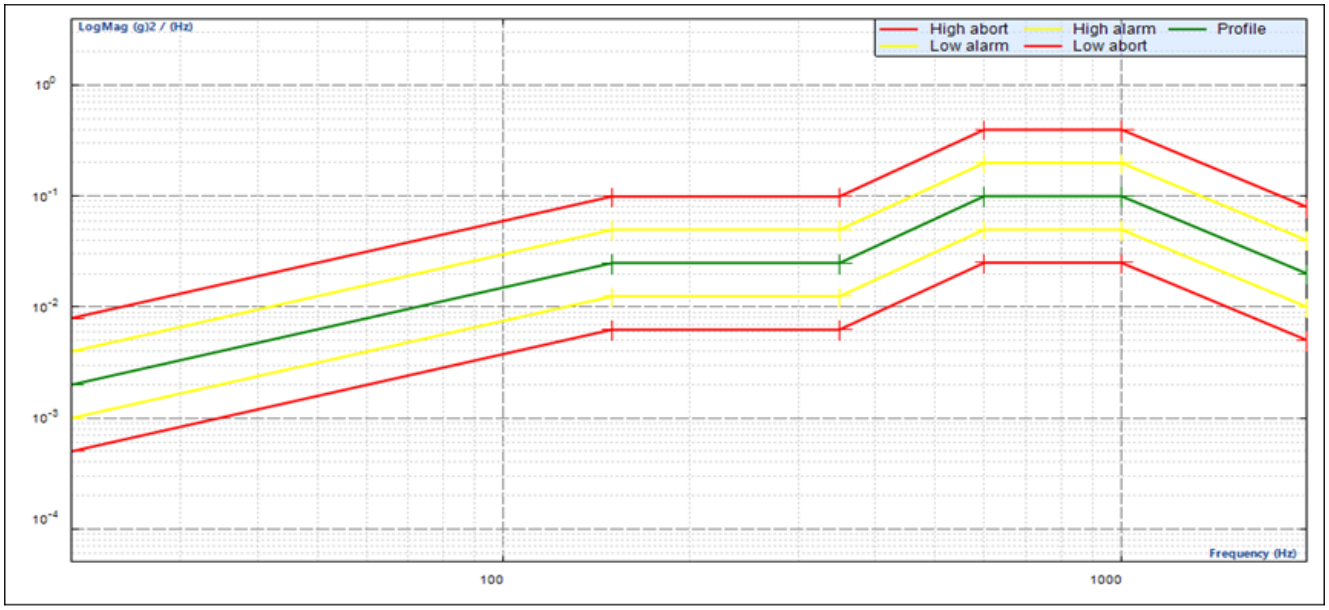
Location ID	On/Off	Measurement quantity	Unit	Sensitivity	Input mode	High-Pass Filter Fc	Description	Sensor
Ch2	On	Acceleration	g	98.00000 (mV/g)	AC-Single End	2.0000 Hz		N/A
Ch3	On	Acceleration	g	100.00000 (mV/g)	AC-Single End	2.0000 Hz		N/A

Location ID	Channel type	Sensor S/N	Max sensor range	Input range	Control weighting	Integration / Differentiation	DOFs	Meas. point	Coordinate

			(V)						
Ch2	Control		20.000 0 (V)	Auto	100.00%(1.0000)	No Integration	+2Z	2	+Z
Ch3	Monitor		20.000 0 (V)	Auto	N/A	No Integration	+3Z	3	+Z

Ch1,Ch4 are turned off.

Testing Profile



Frequency	**Acceleration**	**Slope**	**High abort**	**High alarm**	**Low alarm**	**Low abort**
20 Hz	0.002 $(g)^2/Hz$		6 dB	3 dB	-3 dB	-6 dB
		3.77348 dB/oct				
150 Hz	0.025 $(g)^2/Hz$		6 dB	3 dB	-3 dB	-6 dB
		0 dB/oct				
350 Hz	0.025 $(g)^2/Hz$		6 dB	3 dB	-3 dB	-6 dB

		7.74247 dB/oct				
600 Hz	0.1 (g)²/Hz		6 dB	3 dB	-3 dB	-6 dB
		0 dB/oct				
1000 Hz	0.1 (g)²/Hz		6 dB	3 dB	-3 dB	-6 dB
		-6.9897 dB/oct				
2000 Hz	0.02 (g)²/Hz		6 dB	3 dB	-3 dB	-6 dB

Run Schedule

Loop number: 1

Level 25.00%, duration 00:00:10

Level 50.00%, duration 00:00:10

Level 75.00%, duration 00:00:10

Level 100.00%, duration 00:05:00

Shaker Check Limits

Physical Quantity	**Random profile RMS**	**Profile Expected Values**	**Shaker Parameters**	**Expected/Shaker Limits**
Acceleration(g)	10.32	30.95 Pk	36.71 Pk	84.3%
Velocity(in/s)	1.391	4.173 Pk	2756 Pk	0.2%
Displacement(in)	0.00368	0.02208 Pk-Pk	39.37 Pk-Pk	0.1%
Force(LBF)	980.3	2941 Pk	1349 RMS	72.7%

Control RMS Limits during Test

Calculate based on the table: Yes

Random profile RMS	**10.3169 g**			
High abort	20.5849 g	(+)	6.00(dB)	199.5(%)
High alarm	14.573 g	(+)	3.00(dB)	141.3(%)
Low alarm	7.3038 g	(-)	-3.00(dB)	70.8(%)
Low abort	5.1707 g	(-)	-6.00(dB)	50.1(%)

Operation Information

Product model: Spider-81B **Product serial number:** 1322880

EDM version: 12.1.0.5 **Operator:** X.LIANG

Meter model: FLUKE 8845A **Meter serial number:** 3177003

Last calibrated date: 5/1/2018 **Description:**

Due for next calibration: 5/1/2019 **Adjustment date:** 5/1/2018

BIT version: 30309 **Hardware version:** 7.6.5

DSP application version: 12.1.0.5 **Firmware version:** 12.1.0.5

PCB ID: 40715

Validated Gain and Offset Errors after Calibration

Location ID	Offset	Gain Error (% full range)	Results
Output 1	0.00000 V	0.031 %	Pass
Input 1	-0.00041 V	0.002 %	Pass
Input 2	-0.00022 V	0.001 %	Pass
Input 3	-0.00012 V	0.001 %	Pass
Input 4	-0.00070 V	0.002 %	Pass

Charge Calibration Overview

Product model: Spider-81B **Product serial number:** 1322880

Last calibrated date: 5/1/2018 **Charge Amplifier Calibration Unit Serial Number:**

Before Charge Calibration (Calibration mode: 1,000 pC)

Location ID	**Gain Error (% full range)**
Channel 1	> 2%
Channel 2	> 2%
Channel 3	> 2%
Channel 4	> 2%

Before Charge Calibration (Calibration mode: 10,100 pC)

Location ID	**Gain Error (% full range)**
Channel 1	< 2%
Channel 2	< 2%
Channel 3	< 2%
Channel 4	< 2%

After Charge Calibration (Calibration mode: 1,000 pC)

Location ID	**Gain Error (% full range)**	**Results**
Channel 1	< 2%	Pass
Channel 2	< 2%	Pass
Channel 3	< 2%	Pass
Channel 4	< 2%	Pass

After Charge Calibration (Calibration mode: 10,100 pC)

Location ID	**Gain Error (% full range)**	**Results**
Channel 1	< 2%	Pass

Channel 2	< 2%	Pass
Channel 3	< 2%	Pass
Channel 4	< 2%	Pass

Safety Verification

Table D.4. Design Hazard Checklist

Description of Hazard	**Planned Corrective Action**	**Corrective Action Taken**	**Planned Date**	**Actual Date**
Mounted system has potential to fall and cause injury	Design mounting for longer life using materials with higher modulus of elasticity. Suggest safer mounting arrangements.	Stainless steel bolts were used in manufacturing, however this verification prototype will not be mounted but will be clamped for showcase	2/15	5/19
Mounted system protrudes from wall creating overhang	Moldable mounting to allow for out-of-the-way mounting arrangements.	Stainless steel bolts used, no mounting system designed for prototype. Device will also work clamped on a table	2/19	5/19
Sharp edges present on aluminum frame	Sand extrusions and provide covering around module	All cut extrusions were thoroughly sanded and checked for sharp edges. All cut pieces are covered.	2/19	5/19
Stored energy in battery powering system	Sleeve and route wiring to prevent accidental breakages or expose live wires.	All wirings have sleeves and are checked for breakages prior to operating. Proper grounding has been implemented.	2/21	5/21
Mounted system actuates along linear rails creating possible pinch points	Coverings and blocks will be implemented into the design so this part is not accessible.	This is still a hazard as we would like to see the actuation during operation of the prototype. A fabric covering may be used for	2/21	N/A

		final operation.		
Silicone manufacturing/flurosilicone for NASA	Proper PPE is required during the manufacturing process including mask, gloves, and eye protection.	Silicone was not used so this did not need to be implemented but will be noted in the manufacturing plan for NASA.	2/21	N/A

Images From Testing:

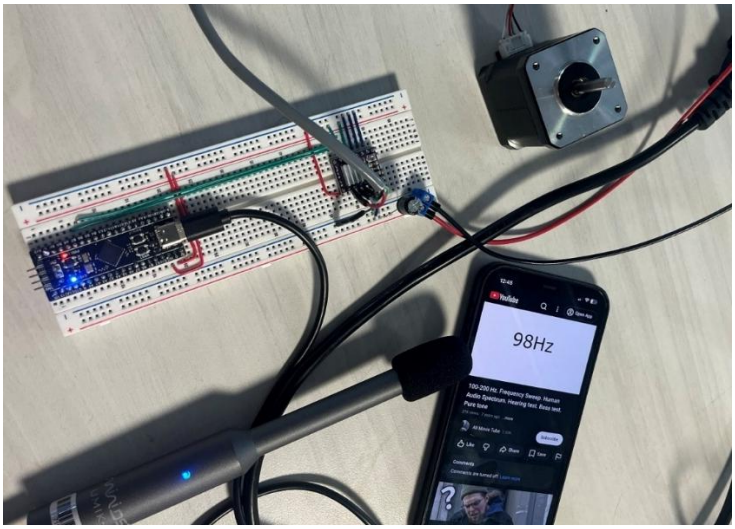


Figure D.1. Frequency Sweep with Microphone Connected to STM32 Board to Control Stepper Motor Function.

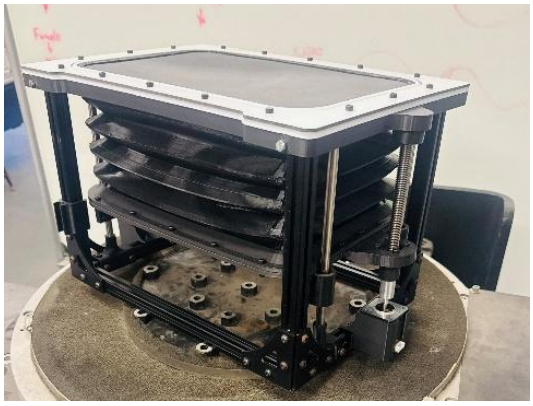


Figure D.2 Vibration Testing Setup

Appendix E: Code and Analysis

Shown below in Appendix E is the code utilized for the MASS system, which includes a control loop to convert frequency meausrements to a voltage to actuate the linear actuators to the correct height value for the corresponding freuqncy.

```
#!/usr/bin/env python3 """ Task-based microphone FFT limp-mass absorber height
controller for Raspberry Pi. """

from future import annotations

import argparse import queue import signal import threading import time from
dataclasses import dataclass from typing import Optional

import numpy as np import sounddevice as sd import serial from serial.tools
import list_ports

Default audio / FFT settings

DEFAULT_SAMPLE_RATE = 48000 DEFAULT_BLOCK_SIZE = 8196

Application target frequency range

DEFAULT_FREQ_MIN = 117.0 DEFAULT_FREQ_MAX = 203.0

Limp-mass absorber model settings from the MATLAB script

DEFAULT_LMA_K = 143.0 DEFAULT_SURFACE_MASS_DENSITY = 9.75 # kg/m^2
DEFAULT_DEPTH_MIN = 0.0508 # 2 inches in meters DEFAULT_DEPTH_MAX = 0.1524 # 6
inches in meters

Calibrated actuator step range (Adjusted for half-stepping)

DEFAULT_ACTUATOR_STEPS_MIN = 0 DEFAULT_ACTUATOR_STEPS_MAX = 12000

Higher = smoother, slower response

DEFAULT_SMOOTHING = 0.85

Ignore weak/noisy FFT peaks

DEFAULT_CONFIDENCE_THRESHOLD = 5.0

Do not spam serial with tiny position changes

DEFAULT_MIN_POSITION_CHANGE = 0.005 DEFAULT_MIN_DEPTH_CHANGE_MM = 1.0
DEFAULT_MIN_STEP_CHANGE = 10
```

```
@dataclass(frozen=True) class MotorCommand: detected_freq_hz: float
smoothed_freq_hz: float confidence: float required_depth_m: float
clamped_depth_m: float actuator_position: float actuator_steps: int
should_stop: bool

class AudioCaptureTask(threading.Thread): def init(self, stop_event:
threading.Event, audio_queue: queue.Queue[np.ndarray], audio_device:
Optional[int | str], sample_rate: int, block_size: int) -> None:
super().init(name="AudioCaptureTask", daemon=True) self.stop_event =
stop_event self.audio_queue = audio_queue self.audio_device = audio_device
self.sample_rate = sample_rate self.block_size = block_size self.stream:
Optional[sd.InputStream] = None

def _callback(self, indata, frames, time_info, status) -> None:
    if status:
        print(f"Audio status: {status}", flush=True)

    samples = indata[:, 0].astype(np.float32, copy=True)

    try:
        self.audio_queue.put_nowait(samples)
    except queue.Full:
        try:
            self.audio_queue.get_nowait()
        except queue.Empty:
            pass
        try:
            self.audio_queue.put_nowait(samples)
        except queue.Full:
            pass

def run(self) -> None:
    self.stream = sd.InputStream(
        device=self.audio_device,
        samplerate=self.sample_rate,
        blocksize=self.block_size,
        channels=1,
        dtype="float32",
        callback=self._callback,
    )

    print("Audio task started.")
    with self.stream:
        while not self.stop_event.is_set():
            time.sleep(0.05)
    print("Audio task stopped.")


class FFTAnalysisTask(threading.Thread): def init(self, stop_event:
threading.Event, audio_queue: queue.Queue[np.ndarray], command_queue:
queue.Queue[MotorCommand], sample_rate: int, freq_min: float, freq_max: float,
lma_k: float, surface_mass_density: float, depth_min: float, depth_max: float,
actuator_steps_min: int, actuator_steps_max: int, smoothing: float,
confidence_threshold: float, min_position_change: float, min_depth_change_mm:
float, min_step_change: int, serial_command_mode: str) -> None:
```

```
super().init(name="FFTAnalysisTask", daemon=True) self.stop_event = stop_event
self.audio_queue = audio_queue self.command_queue = command_queue
self.sample_rate = sample_rate self.freq_min = freq_min self.freq_max =
freq_max self.lma_k = lma_k self.surface_mass_density = surface_mass_density
self.depth_min = depth_min self.depth_max = depth_max self.actuator_steps_min
= actuator_steps_min self.actuator_steps_max = actuator_steps_max
self.smoothing = smoothing self.confidence_threshold = confidence_threshold
self.min_position_change = min_position_change self.min_depth_change_mm =
min_depth_change_mm self.min_step_change = min_step_change
self.serial_command_mode = serial_command_mode self.smoothed_freq:
Optional[float] = None self.last_queued_command: Optional[MotorCommand] = None

def run(self) -> None:
    print("FFT task started.")

    while not self.stop_event.is_set():
        try:
            samples = self.audio_queue.get(timeout=0.1)
        except queue.Empty:
            continue

        detected_freq, confidence = get_dominant_frequency(
            samples=samples, sample_rate=self.sample_rate,
freq_min=self.freq_min, freq_max=self.freq_max
        )

        if confidence < self.confidence_threshold:
            command = MotorCommand(
                detected_freq_hz=detected_freq, smoothed_freq_hz=0.0,
confidence=confidence,
                required_depth_m=0.0, clamped_depth_m=0.0,
actuator_position=0.0, actuator_steps=0, should_stop=True
            )
        else:
            if self.smoothed_freq is None:
                self.smoothed_freq = detected_freq
            else:
                self.smoothed_freq = (self.smoothing * self.smoothed_freq +
(1.0 - self.smoothing) * detected_freq)

            required_depth_m = lma_depth_from_frequency(self.smoothed_freq,
self.lma_k, self.surface_mass_density)
            clamped_depth_m = clamp(required_depth_m, self.depth_min,
self.depth_max)
            depth_fraction = depth_to_normalized_position(
                clamped_depth_m,
                self.depth_min,
                self.depth_max
            )

            # depth_fraction:
            #   0.0 = 2 inch / top / shallow chamber
            #   1.0 = 6 inch / bottom / deep chamber
            #
            # STM32 position convention:
```

```
            #   0 steps     = 6 inch / bottom
            #   48000 steps = 2 inch / top
            actuator_position = 1.0 - depth_fraction

            actuator_steps = normalized_position_to_steps(
                actuator_position,
                self.actuator_steps_min,
                self.actuator_steps_max
            )

            command = MotorCommand(
                detected_freq_hz=detected_freq,
smoothed_freq_hz=self.smoothed_freq, confidence=confidence,
                required_depth_m=required_depth_m,
clamped_depth_m=clamped_depth_m, actuator_position=actuator_position,
                actuator_steps=actuator_steps, should_stop=False
            )

        if self._should_queue_command(command):
            self._put_latest_command(command)
            self.last_queued_command = command

    print("FFT task stopped.")

def _should_queue_command(self, command: MotorCommand) -> bool:
    if self.last_queued_command is None:
        return True
    if self.last_queued_command.should_stop != command.should_stop:
        return True
    if command.should_stop:
        return False
    if self.serial_command_mode == "height-mm":
        depth_change_mm = abs(command.clamped_depth_m -
self.last_queued_command.clamped_depth_m) * 1000.0
        return depth_change_mm >= self.min_depth_change_mm
    if self.serial_command_mode in {"position-steps", "speed-legacy"}:
        return abs(command.actuator_steps -
self.last_queued_command.actuator_steps) >= self.min_step_change
    return abs(command.actuator_position -
self.last_queued_command.actuator_position) >= self.min_position_change

def _put_latest_command(self, command: MotorCommand) -> None:
    try:
        self.command_queue.put_nowait(command)
    except queue.Full:
        try:
            self.command_queue.get_nowait()
        except queue.Empty:
            pass
        try:
            self.command_queue.put_nowait(command)
        except queue.Full:
            pass
```

```
class SerialOutputTask(threading.Thread): def init(self, stop_event:
threading.Event, command_queue: queue.Queue[MotorCommand], ser: serial.Serial,
serial_command_mode: str) -> None: super().init(name="SerialOutputTask",
daemon=True) self.stop_event = stop_event self.command_queue = command_queue
self.ser = ser self.serial_command_mode = serial_command_mode

def run(self) -> None:
    print("Serial task started.")
    while not self.stop_event.is_set():
        try:
            command = self.command_queue.get(timeout=0.1)
        except queue.Empty:
            continue
        self._send_command(command)
    print("Serial task stopped.")

def _send_command(self, command: MotorCommand) -> None:
    line, display = self._format_serial_command(command)
    self.ser.write(line.encode("ascii"))
    self.ser.flush()

    if command.should_stop:
        pass # Suppressed stop logging to avoid terminal spam
    else:
        print(f"Detected: {command.detected_freq_hz:7.2f} Hz | Smoothed:
{command.smoothed_freq_hz:7.2f} Hz | Pos Steps: {command.actuator_steps} |
Command: {display}", flush=True)

def _format_serial_command(self, command: MotorCommand) -> tuple[str, str]:
    if command.should_stop:
        return "STOP\n", "STOP"
    if self.serial_command_mode == "position-steps":
        return f"POSITION {command.actuator_steps}\n", f"POSITION
{command.actuator_steps}"
    if self.serial_command_mode == "height-mm":
        height_mm = command.clamped_depth_m * 1000.0
        return f"HEIGHT_MM {height_mm:.2f}\n", f"HEIGHT_MM {height_mm:.2f}"
    if self.serial_command_mode == "normalized":
        return f"POSITION_NORM {command.actuator_position:.4f}\n",
f"POSITION_NORM {command.actuator_position:.4f}"
    if self.serial_command_mode == "speed-legacy":
        return f"SPEED {command.actuator_steps}\n", f"SPEED
{command.actuator_steps}"
    raise ValueError(f"Unsupported serial command mode:
{self.serial_command_mode}")


def clamp(x: float, x_min: float, x_max: float) -> float: return max(x_min,
min(x_max, x))

def lma_depth_from_frequency(freq_hz: float, lma_k: float,
surface_mass_density: float) -> float: if freq_hz <= 0.0: return float("inf")
return (lma_k / freq_hz) ** 2 / surface_mass_density

def depth_to_normalized_position(depth_m: float, depth_min: float, depth_max:
```

```
float) -> float: # 0 = min depth (top of chamber), 1 = max depth (bottom of
chamber) if depth_max <= depth_min: raise ValueError("depth_max must be
greater than depth_min") return clamp((depth_m - depth_min) / (depth_max -
depth_min), 0.0, 1.0)

def normalized_position_to_steps(position: float, steps_min: int, steps_max:
int) -> int: pos = clamp(position, 0.0, 1.0) return int(round(steps_min + pos
* (steps_max - steps_min)))

def get_dominant_frequency(samples: np.ndarray, sample_rate: int, freq_min:
float, freq_max: float) -> tuple[float, float]: samples =
samples.astype(np.float32) samples = samples - np.mean(samples) window =
np.hanning(len(samples)).astype(np.float32) windowed = samples * window

spectrum = np.fft.rfft(windowed)
magnitudes = np.abs(spectrum)
freqs = np.fft.rfftfreq(len(samples), d=1.0 / sample_rate)

min_index = np.searchsorted(freqs, freq_min)
max_index = np.searchsorted(freqs, freq_max)

search_mags = magnitudes[min_index:max_index]
search_freqs = freqs[min_index:max_index]

if len(search_mags) == 0:
    return 0.0, 0.0

local_peak_index = int(np.argmax(search_mags))
peak_freq = float(search_freqs[local_peak_index])
peak_mag = float(search_mags[local_peak_index])
avg_mag = float(np.mean(search_mags) + 1e-9)

confidence = peak_mag / avg_mag
return peak_freq, confidence


def list_serial_ports() -> None: ports = list(list_ports.comports()) if not
ports: print("No serial ports found.") return print("Available serial ports:")
for p in ports: print(f" {p.device} - {p.description}")

def list_audio_devices() -> None: print("Available audio devices:")
print(sd.query_devices())

def resolve_audio_device(name_or_index: Optional[str]) -> Optional[int | str]:
if name_or_index is None: return None try: return int(name_or_index) except
ValueError: pass devices = sd.query_devices() for index, device in
enumerate(devices): if device["max_input_channels"] <= 0: continue if
name_or_index.lower() in device["name"].lower(): return index raise
RuntimeError(f"Could not find audio input device containing: {name_or_index}")

def prompt_for_serial_port_if_needed(port: Optional[str]) -> str: if port:
return port list_serial_ports() selected = input("\nEnter microcontroller
serial port, example /dev/ttyACM0: ").strip() if not selected: raise
RuntimeError("No serial port selected.") return selected
```

```
def build_arg_parser() -> argparse.ArgumentParser: parser =
argparse.ArgumentParser(description="Task-based microphone FFT limp-mass
absorber height controller.") parser.add_argument("--list-audio",
action="store_true", help="List audio devices and exit.")
parser.add_argument("--list-serial", action="store_true", help="List serial
ports and exit.") parser.add_argument("--audio-device", default="UMIK",
help="Audio input index or name substring. Default: UMIK")
parser.add_argument("--serial-port", default=None, help="Microcontroller
serial port") parser.add_argument("--baudrate", type=int, default=115200)

parser.add_argument("--sample-rate", type=int, default=DEFAULT_SAMPLE_RATE)
parser.add_argument("--block-size", type=int, default=DEFAULT_BLOCK_SIZE)
parser.add_argument("--freq-min", type=float, default=DEFAULT_FREQ_MIN)
parser.add_argument("--freq-max", type=float, default=DEFAULT_FREQ_MAX)

parser.add_argument("--lma-k", type=float, default=DEFAULT_LMA_K)
parser.add_argument("--surface-mass-density", type=float,
default=DEFAULT_SURFACE_MASS_DENSITY)
parser.add_argument("--depth-min", type=float, default=DEFAULT_DEPTH_MIN)
parser.add_argument("--depth-max", type=float, default=DEFAULT_DEPTH_MAX)

parser.add_argument(
    "--serial-command-mode",
    choices=["height-mm", "position-steps", "normalized", "speed-legacy"],
    default="position-steps",
    help="Command protocol sent to the microcontroller. Default: position-
steps",
)
parser.add_argument("--actuator-steps-min", type=int,
default=DEFAULT_ACTUATOR_STEPS_MIN)
parser.add_argument("--actuator-steps-max", type=int,
default=DEFAULT_ACTUATOR_STEPS_MAX)

parser.add_argument("--smoothing", type=float, default=DEFAULT_SMOOTHING)
parser.add_argument("--confidence-threshold", type=float,
default=DEFAULT_CONFIDENCE_THRESHOLD)
parser.add_argument("--min-position-change", type=float,
default=DEFAULT_MIN_POSITION_CHANGE)
parser.add_argument("--min-depth-change-mm", type=float,
default=DEFAULT_MIN_DEPTH_CHANGE_MM)
parser.add_argument("--min-step-change", type=int,
default=DEFAULT_MIN_STEP_CHANGE)

return parser


def main() -> None: args = build_arg_parser().parse_args()

if args.list_audio:
    list_audio_devices()
    return

if args.list_serial:
    list_serial_ports()
    return
```

```
audio_device = resolve_audio_device(args.audio_device)
serial_port = prompt_for_serial_port_if_needed(args.serial_port)

sd.check_input_settings(device=audio_device, samplerate=args.sample_rate,
channels=1, dtype="float32")

print(f"\nOpening serial port {serial_port} at {args.baudrate} baud...")
ser = serial.Serial(port=serial_port, baudrate=args.baudrate, timeout=1,
write_timeout=1)
time.sleep(2.0) # Wait for MCU to reset

# STARTUP HOMING ROUTINE
print("\n--- STARTUP HOMING ---")
print("Driving baseplate downwards to the 6-inch mark.")
ser.write(b"HOME_DOWN\n")
input("Press ENTER when you hear/see the motors stall at the bottom...")
ser.write(b"ZERO\n")
print("System zeroed at 6 inches (0 steps).\n")

stop_event = threading.Event()
audio_queue: queue.Queue[np.ndarray] = queue.Queue(maxsize=16)
command_queue: queue.Queue[MotorCommand] = queue.Queue(maxsize=8)

audio_task = AudioCaptureTask(stop_event, audio_queue, audio_device,
args.sample_rate, args.block_size)
fft_task = FFTAnalysisTask(
    stop_event, audio_queue, command_queue, args.sample_rate, args.freq_min,
args.freq_max,
    args.lma_k, args.surface_mass_density, args.depth_min, args.depth_max,
    args.actuator_steps_min, args.actuator_steps_max, args.smoothing,
args.confidence_threshold,
    args.min_position_change, args.min_depth_change_mm, args.min_step_change,
args.serial_command_mode
)
serial_task = SerialOutputTask(stop_event, command_queue, ser,
args.serial_command_mode)

print("Starting task-based microphone FFT LMA height control.")
print("Press Ctrl+C to stop.\n")

serial_task.start()
audio_task.start()
fft_task.start()

try:
    while not stop_event.is_set():
        time.sleep(0.2)
except KeyboardInterrupt:
    pass

print("\nShutdown requested. Stopping background tasks...")
stop_event.set()
audio_task.join(timeout=2.0)
```

```
fft_task.join(timeout=2.0)
serial_task.join(timeout=2.0)

#  SHUTDOWN HOMING ROUTINE
print("\n--- SHUTDOWN HOMING ---")
print("Driving baseplate downwards to the 6-inch mark for safe power-off.")
try:
    ser.write(b"HOME_DOWN\n")
    input("Press ENTER when the motors stall at the bottom to terminate...")
    ser.write(b"ZERO\n")
except Exception as e:
    print(f"Could not complete shutdown homing: {e}")
finally:
    ser.write(b"STOP\n")
    ser.close()
    print("Serial closed. Program stopped.")


if name == "main": main()
```

Appendix F: Gantt Chart and Project Timeline

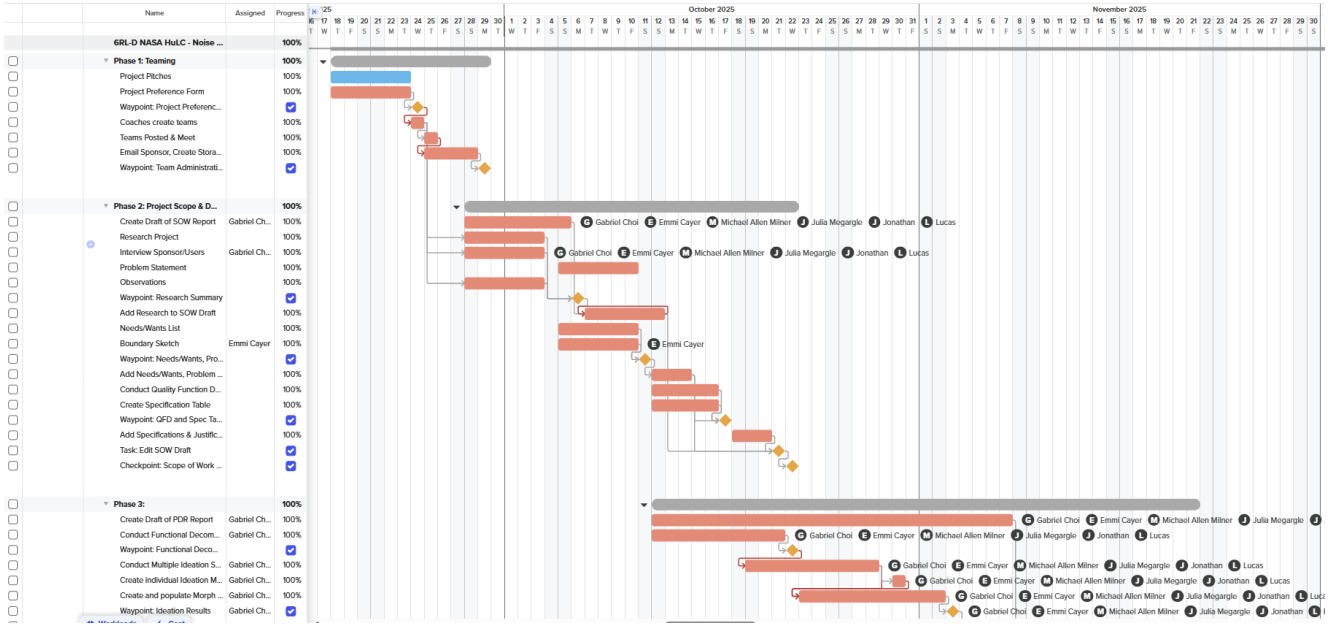


Figure F.1. Gantt Chart Section 1.

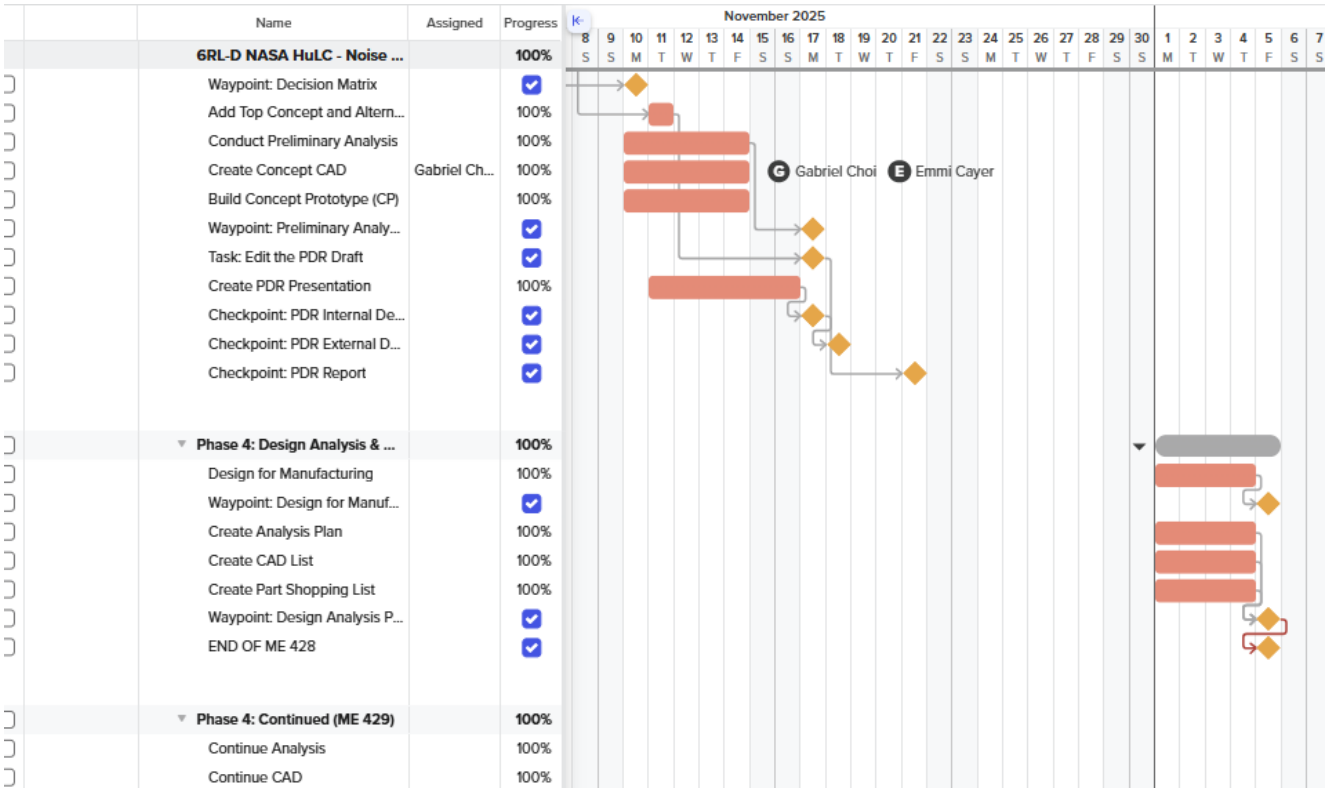


Figure F.2. Gantt Chart Section 2.

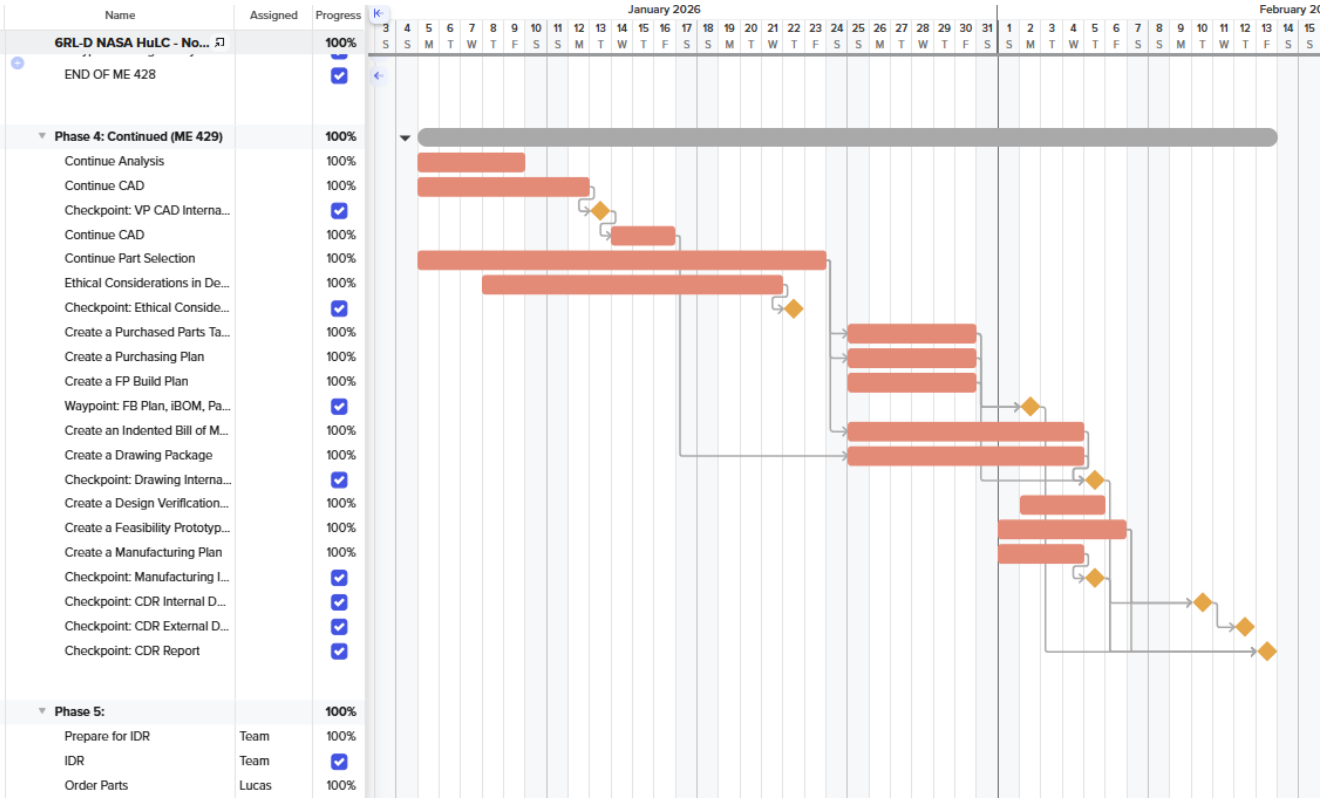


Figure F.3. Gantt Chart Section 3.

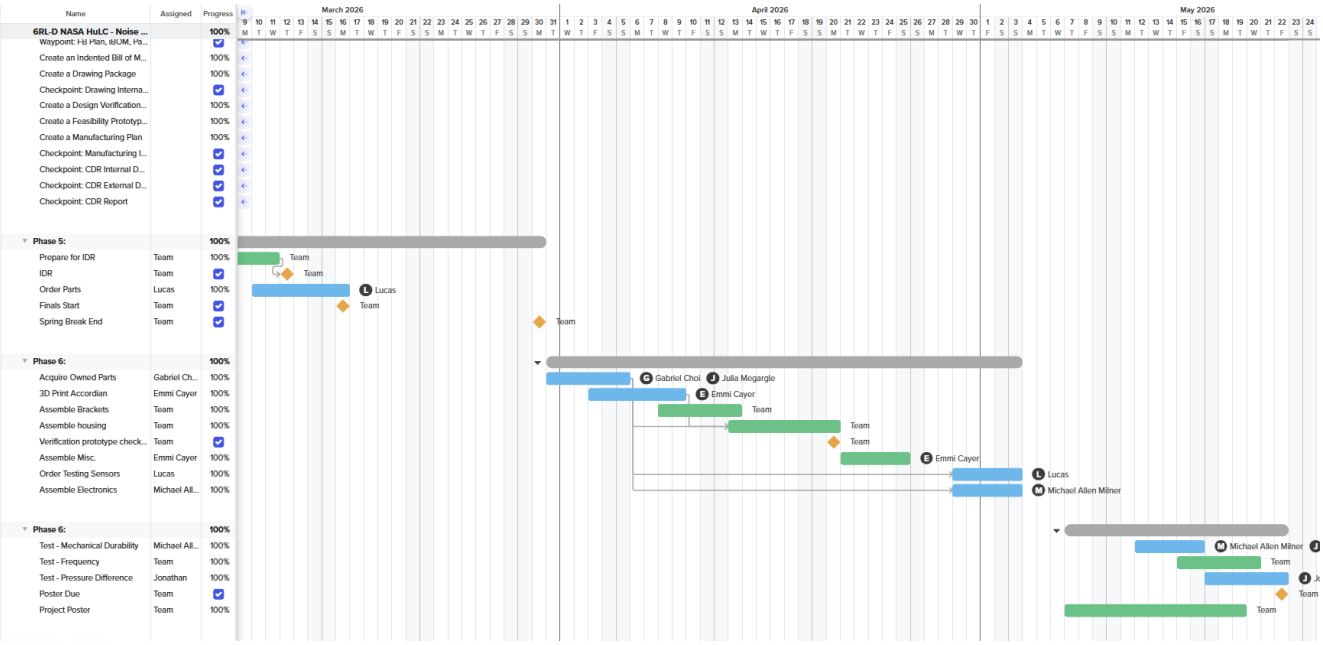


Figure F.4. Gantt Chart Section 4.

Modular Acoustic Suppression System (MASS)

Emmi Cayer, Gabriel Choi, Jonathan Corvera, Lucas Kaemmerer, Julia Megargle, Michael Milner

Advisor: Rick Lasko

Sponsor: Dr. John Chen

Background

Currently, the Environmental Control and Life Support Systems (ECLSS) in the human lander produces noise outside of the specified Noise Criterion levels affecting astronauts' sleep, concentration, and communication.

Objective: Design an innovative solution to reduce noise generated by ECLSS systems in long-duration space habitats.

Design Specifications

System Constraints:

- Minimal barriers to NASA adoption (low mass, low power consumption)
- No additional risks to crew
- Mission operational life of 30-days for lunar missions, 1200-days for Mars missions

Design Specifications:

Specification	Required Value	Design Value
Acoustic	63-250 Hz, 50-64 dB	---
Weight	<100 lb	14 lb
Power	<0.95 kW	0.136 kW
Volume	<1.5 cubic feet	< 1.0 cubic feet

Manufacturing and Assembly

3D Printing Operations:

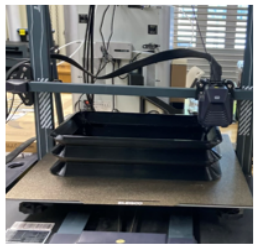
- Reverberation chamber and gasket - TPU filament
- Top/bottom mounting plates, linear rail mounts - PLA filament

Lathe Operations:

- Power screw turned down to size

Waterjet Operations:

- Top mounting plate support

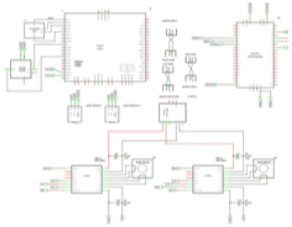


3D Printing Fabrication

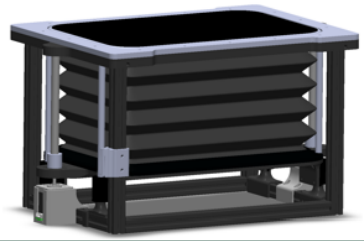
Final Design

Key Features:

- Noise Reduction – Flexible membrane reduces sound by achieving resonance with incoming sound waves.
- Adjustable – Chamber height is adjusted by linear actuators that respond to varying frequency inputs, changing membrane's resonant frequency.
- Compact – Components contained in 18in x 12in x 12in box, making implementation and transportation easy.



Electrical System Design

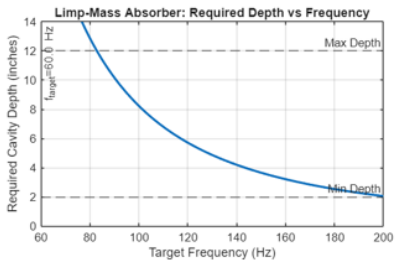


Mechanical System Design

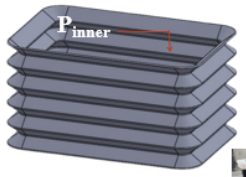
Analysis

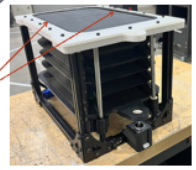
Membrane Selection:

- Analyzed as oscillating mass-acoustic spring-damper
- Required depth plotted and analyzed in MATLAB



Plot of Required Depth for Target Frequencies





Pressure Equalization:

- Functions by equalizing to surrounding pressure during operation ($P_{inner} = P_{outer}$)
- Air released through small gaps between membrane and gasket allowing equalization while maintaining constant air volume during operation

Testing and Results

Test 1: Linear Actuator and Electronic Verification

- Sweep of frequencies ranging from 60-250 Hz
- Ensured actuators adjusted to changing frequencies, switching directions and height (6" max) dependent on frequency spectrum
- Smoothed frequency enables consistent transition despite highly variable sound conditions

Detected Frequency [Hz]	Smoothed Frequency [Hz]	Required Depth [in]	Confidence (>5 is good)
128.91	131.04	4.80	11.33
175.78	139.28	4.26	6.57
234.34	188.32	2.32	5.55

Test 2: Pressure Equalization Verification

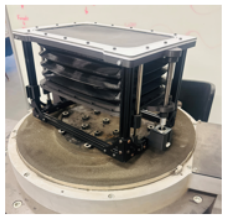
- Chamber depth shifted 3 inches to verify pressure equalization at varying depths throughout 5 tests
- Successfully equalized after 4.49 seconds on average

Test 3: Vibration Test

- Verified mechanical system's structural integrity at critical resonant frequencies



Stepper Motor Testing Setup



Vibration Testing Setup

Conclusion and Recommendations

- Prototype successfully verifies acoustic control
- system and pressure equalization
- Recommend material switch to silicone for weight reduction and easier height adjustment as well as extending frequency range
- Further testing required in acoustic chamber to verify noise suppression